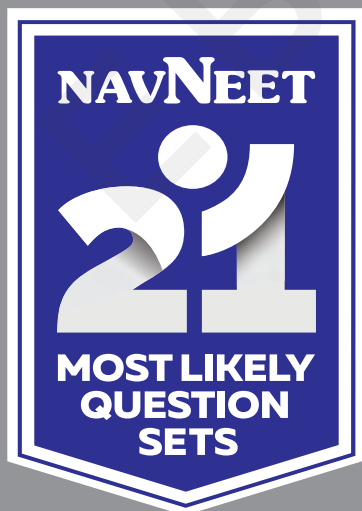


For
2026
Examinations



MATHEMATICS
(PART – I)

Board's
Question Paper
of July 2025
Included

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2026

**An excellent book for the
Preparation of the Board's
Examination**

MATHEMATICS

(PART – I)

STANDARD X

This book contains ...

1. Format (Outline) of the Question Paper at the beginning of the book
2. Study Tips for Students
3. **Section 1** : Board's July 2024 Question Paper with full Solution
4. **Section 2** : Question Sets for chapter-wise study and Revision :
 - (a) Guidance on how to attempt the question
 - (b) Model answers with solution of each question type
 - (c) Questions for Practice
5. Creative type question (open ended questions, heuristic approach, approximation, etc.) and challenging questions included in each set.
6. **Section 3** : Board's July 2025 Question Paper for Practice (with Solution on QR Code)

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CTP
(21-9-25)

Outline of Question Paper

MATHEMATICS (PART – I) AND MATHEMATICS (PART – II)

Q. No.	Types of Questions	Marks	Marks with options
Q. 1	(A) Solve Multiple choice questions <i>(1 mark each)</i>	4	4
	(B) Solve 4 out of 4 <i>(1 mark each)</i>	4	4
Q. 2	(A) Complete 2 activities out of 3 <i>(2 marks each)</i>	4	6
	(B) Solve any 4 out of 5 subquestions <i>(2 marks each)</i>	8	10
Q. 3	(A) Complete 1 activity out of 2 <i>(3 marks each)</i>	3	6
	(B) Solve any 2 out of 4 subquestions <i>(3 marks each)</i>	6	12
Q. 4	Solve any 2 out of 3 subquestions <i>(4 marks each)</i>	8	12
Q. 5	Solve any 1 out of 2 subquestions <i>(3 marks each)</i>	3	6
	Total Marks	40	60

Q. 3 (B) One of the subquestions should be asked on prove the given statement.

Q. 4 These questions should be out of the textbooks but based on prescribed syllabus and also challenging.

Q. 5 These questions should be creative for which the responses should be as follows :

For Example :

- ☛ Expecting the student to express his/her own views
- ☛ To draw a figure as per the given information
- ☛ To complete an incomplete construction
- ☛ To complete the given flow chart
- ☛ To construct a problem from given information or situation
- ☛ Open ended questions
- ☛ Estimation and Approximation
- ☛ Comprehension of mathematical passage.

Study Tips for Students

Dear Students,

Your Board Examination is approaching near. The coming days are for self-study and to make an effective study plan. Navneet 21 Most Likely Question Sets provides a deeper understanding of important topics and how questions are asked from these topics.

Navneet 21 Most Likely Question Sets are prepared according to the latest syllabus and the format of the Question Paper.

Important questions and answers from the examination point of view have been incorporated in this book.

1. The questions are based on the syllabus of Mathematics : Part-I
2. MCQ in **Q. 1 (A)** is objective type. This is based on your knowledge and understanding of the subject. **Q. 1 (B)** Short answer questions.
3. The subquestions in **Q. 2 (A)** are activity based questions. To solve this, you have to fill in the blanks/boxes. These are easy questions.
4. The subquestions in **Q. 2 (B)** are of **2 marks** and based on knowledge and understanding. They usually would be from the textbook.
5. The subquestions in **Q. 3 (A)** is activity based questions of **3 marks** and **Q. 3 (B)** of **3 marks** each based on knowledge, understanding and application.
6. **Q. 4** of **4 marks** is very important. It is a challenging question based on the syllabus but the question will not be from the textbook.
7. **Q. 5** is of constructive nature. This question is to evaluate your capacity to express your views regarding the given question.

The question may be of various types based on the syllabus.

In this book, each set is divided into 1 mark, 2 marks, 3 marks, 4 marks questions with solutions. Assignments in each case have been given for practice.

We have included various types of questions as expected by the Board.

While solving a question :

- * Write proper steps giving reasons wherever necessary.
- * Draw proper figure wherever necessary.
- * Underline the final answer in blue or black or put a box around the answer.
- * Always start the answer to a new question on a new page.
- * All the subquestions of a particular question should be answered in a continuous manner. If you don't know the answer to a particular subquestion, keep proper space for writing it afterwards.
- * Be neat and clean in writing.
- * Practice solving questions methodically. This will help you to solve the questions in the examination in time.
- * Solve the questions given in **(1) Navneet Practice Papers and (2) Navneet 21 Most Likely Question Sets** on your own and then compare your solutions with the solutions given in the Sets and correct them if necessary.
- * Refer to **Navneet Board Papers Smart Solutions : Mathematics – Std. X (Part – I & Part – II)** for past papers and their solutions. This will be useful for how to write the solutions in exam.
- * Don't be afraid of **HOTS** problems. They are, after all, based on the topics you have learnt.

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Section 1**BOARD'S QUESTION PAPER : JULY 2024**
(With Full Solution and Marking Scheme)**Time : 2 Hours]****[Max. Marks : 40**

- Note :**
- (i) All questions are compulsory.
 - (ii) Use of calculator is **not** allowed.
 - (iii) The numbers to the right of the questions indicate full marks.
 - (iv) In case of MCQs [Q. No. 1 (A)], only the first attempt will be evaluated and will be given credit.
 - (v) For every MCQ, four alternatives (A), (B), (C), (D) of answers are given. Alternative of correct answer is to be written in front of the subquestion number.

Q. 1. (A) Choose the correct alternative and write its alphabet in front of the subquestion number : 4

In this question, 4 multiple choice subquestions each of 1 mark will be asked. Four alternatives, for each subquestion will be given. Only one of them will be correct. Write the alphabet of the correct alternative against the subquestion number.

Note :

- (1) Here, all the subquestions are compulsory.
- (2) Method of solution is not expected. Write the correct alternative with the alphabet given in the brackets.

(i) For simultaneous equations in variables x and y , $D_x = 49$, $D_y = -63$, $D = 7$, then what is the value of x ?

- (A) 7 (B) -7 (C) $\frac{1}{7}$ (D) $-\frac{1}{7}$

(ii) Which of the following is a quadratic equation?

- (A) $\frac{5}{x} - 3 = x^2$ (B) $x(x+5) = 2$
(C) $n - 1 = 2n$ (D) $\frac{1}{x^2}(x+2) = x$

- (iii) The sequence $-10, -6, -2, 2, \dots$ is
 (A) an A.P., reason $d = -16$ (B) an A.P., reason $d = 4$
 (C) an A.P., reason $d = -4$ (D) not an A.P.
- (iv) Which number from the following cannot represent probability?
 (A) $\frac{2}{3}$ (B) 1.5 (C) 15% (D) 0.7

Solutions

- (i) (A) (ii) (B) (iii) (B) (iv) (B)

(1 mark for each correct answer)

Explanation : Students are not expected to write the explanation. Here, it is given for the better understanding of the students.

(i) $x = \frac{D_x}{D}$

(ii) The remaining are not quadratic equations.

(iii) $d = -6 - (-10) = -6 + 10 = 4.$

(iv) The probability can never be greater than 1.

Q. 1. (B) Solve the following subquestions :

4

In this question, 4 subquestions each of 1 mark will be asked. All questions are compulsory.

- (i) Find the value of the following determinant :

$$\begin{vmatrix} 5 & -2 \\ -3 & 1 \end{vmatrix}.$$

- (ii) Find the first term and common difference for the following A.P. :

$5, 1, -3, -7, \dots$

- (iii) Face value of a share is ₹ 100 and premium is ₹ 65, then what is the market value of that share?

- (iv) Write sample space, if one die is thrown.

Solutions

$$(i) \begin{vmatrix} 5 & -2 \\ -3 & 1 \end{vmatrix} = 5 \times 1 - (-2) \times (-3) \quad \left(\frac{1}{2} \text{ mark}\right)$$

$$= 5 - 6$$

$$= -1 \quad \left(\frac{1}{2} \text{ mark}\right)$$

(ii) A.P. : 5, 1, -3, -7, ...

First term = $a = 5$ $\left(\frac{1}{2} \text{ mark}\right)$

Common difference $d = (1 - 5) = -4$ $\left(\frac{1}{2} \text{ mark}\right)$

(iii) FV = ₹ 100, Premium = ₹ 65

MV = FV + Premium $\left(\frac{1}{2} \text{ mark}\right)$

$= ₹(100 + 65) = ₹ 165$

MV = ₹ 165 $\left(\frac{1}{2} \text{ mark}\right)$

(iv) The sample space

$S = \{1, 2, 3, 4, 5, 6\}.$ (1 mark)

Q. 2. (A) Complete the following activities and rewrite it :

(Any two) 4

In this question, 3 activities each of 2 marks will be given. You have to attempt any two of them. Various types of activities can be asked.

(i) Complete the following table for drawing the graph of the equation $x + 2y = 4$.

x	-2	0	
y		2	1
(x, y)		(0, 2)	

- (ii) Determine the nature of roots of the quadratic equation

$$m^2 + 2m + 9 = 0.$$

Activity :

$$m^2 + 2m + 9 = 0$$

$$a = \boxed{}, b = 2, c = 9$$

$$b^2 - 4ac = 2^2 - 4 \times 1 \times \boxed{}$$

$$= \boxed{} - 36$$

$$\therefore b^2 - 4ac = \boxed{}$$

$$\therefore b^2 - 4ac < 0$$

$\therefore$ the roots of the quadratic equation are not real.

- (iii) Smita invested ₹ 12,000 and purchased shares of FV ₹ 10 at a premium of ₹ 2. Find the number of shares she purchased. Complete the given activity to get the answer.

Activity :

$$FV = ₹ 10, \text{ Premium} = ₹ 2$$

$$MV = FV + \boxed{} = \boxed{} + 2 = ₹ 12$$

$$\text{Number of shares} = \frac{\text{Total investment}}{MV}$$

$$= \frac{12000}{\boxed{}}$$

$$= \boxed{}$$

$$= \boxed{} \text{ shares.}$$

Solutions

(i) **Activity :**

x	-2	0	$\boxed{2}$
y	$\boxed{3}$	2	1
(x, y)	$\boxed{(-2, 3)}$	$(0, 2)$	$\boxed{(2, 1)}$

[Substitute $x = -2$ in the given equation.

$$x + 2y = 4 \quad \therefore -2 + 2y = 4 \quad \therefore 2y = 4 + 2 \quad \therefore y = 3.$$

Substitute $y = 1$ in the given equation.

$$x + 2y = 4 \quad \therefore x + 2(1) = 4$$

$$\therefore x + 2 = 4 \quad \therefore x = 4 - 2 \quad \therefore x = 2.]$$

(ii) **Activity :**

$$m^2 + 2m + 9 = 0$$

$$a = \boxed{1}, b = 2, c = 9$$

$$b^2 - 4ac = 2^2 - 4 \times 1 \times \boxed{9}$$

$$= \boxed{4} - 36$$

$$\therefore b^2 - 4ac = \boxed{-32}$$

$$\therefore b^2 - 4ac < 0$$

$\therefore$ the roots of the quadratic equation are not real.

(iii) **Activity :**

$$FV = ₹ 10, \text{ Premium} = ₹ 2$$

$$MV = FV + \boxed{\text{Premium}}$$

$$= \boxed{10} + 2$$

$$= ₹ 12$$

$$\text{Number of shares} = \frac{\text{Total investment}}{MV}$$

$$= \frac{12000}{\boxed{12}}$$

$$= \boxed{1000} \text{ shares.}$$

[**Scheme of marking for activities :** $\frac{1}{2}$ mark for each correct answer in the boxes.]

Q. 2. (B) Solve the following subquestions : (Any four)**8**

In this question, 5 subquestions each of 2 marks will be given. You have to attempt any four of them.

Note : Answers with solutions are expected.

- (i) Solve the following simultaneous equations :

$$x + y = 5; \quad x - y = 3.$$

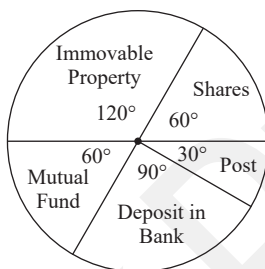
- (ii) Find k , if $x = 3$ is a root of the quadratic equation $kx^2 - 10x + 3 = 0$.

- (iii) Find the 19th term of the following A.P. :

$$7, 13, 19, 25, \dots$$

- (iv) The taxable value of a wristwatch belt is ₹ 586. Rate of GST is 18%. What is the price of the belt for the customer?

- (v) The annual investments of a family are shown in the following pie diagram. Answer the following questions based on it.



- (a) If the investment in the shares is ₹ 2000, find the total investment.
- (b) How much amount is deposited in the post?

Solutions

(i) $x + y = 5 \quad \dots (1) \quad x - y = 3 \quad \dots (2)$

Adding equations (1) and (2),

$$x + y = 5 \quad \dots (1)$$

$$x - y = 3 \quad \dots (2)$$

$$\hline 2x = 8$$

(½ mark)



$$\therefore x = \frac{8}{2}$$

$$\therefore x = 4 \quad \quad \quad (1/2 \text{ mark})$$

Substituting $x = 4$ in equation (1),

$$x + y = 5$$

$$\therefore 4 + y = 5 \quad \quad \quad (1/2 \text{ mark})$$

$$\therefore y = 5 - 4$$

$$\therefore y = 1 \quad \quad \quad (1/2 \text{ mark})$$

Ans. $(x, y) = (4, 1)$ is the solution of the given simultaneous equations.

(ii) $x = 3$ is the root of the equation $kx^2 - 10x + 3 = 0$. Substitute $x = 3$ in the given equation.

$$\therefore k(3)^2 - 10(3) + 3 = 0 \quad \quad \quad (1/2 \text{ mark})$$

$$\therefore 9k - 30 + 3 = 0 \quad \quad \quad (1/2 \text{ mark})$$

$$\therefore 9k - 27 = 0 \quad \quad \quad (1/2 \text{ mark})$$

$$\therefore 9k = 27$$

$$\therefore k = \frac{27}{9} \quad \quad \quad \therefore k = 3 \quad \quad \quad (1/2 \text{ mark})$$

Ans. The value of k is **3**.

(iii) Here, $a = t_1 = 7$, $t_2 = 13$, $t_3 = 19$, $t_4 = 25$, ...

$$d = t_2 - t_1 = 13 - 7 = 6. \quad \quad \quad (1/2 \text{ mark})$$

We have to find 19th term. $t_{19} = ?$

$$t_n = a + (n - 1)d \quad \quad \quad \dots \text{ (Formula) } (1/2 \text{ mark})$$

$$\therefore t_{19} = 7 + (19 - 1) \times 6 \quad \quad \dots \text{ (Substituting the values) } (1/2 \text{ mark})$$

$$= 7 + 18 \times 6$$

$$= 7 + 108$$

$$\therefore t_{19} = 115 \quad \quad \quad (1/2 \text{ mark})$$

Ans. The 19th term of the given A.P. is **115**.

(iv) The rate of GST is 18%.

The taxable value of the belt is ₹ 586

$$\therefore \text{GST on the belt} = \frac{18}{100} \times 586 \quad (1/2 \text{ mark})$$

$$= ₹ 105.48 \quad (1/2 \text{ mark})$$

The price of the belt for the customer

$$= \text{Taxable value} + \text{GST} \quad (1/2 \text{ mark})$$

$$= ₹(586 + 105.48)$$

$$= ₹ 691.48$$

Ans. The price of the belt for the customer is ₹ 691.48. (1/2 mark)

(v) (a) Let the total investment be ₹ x .

The central angle for the investment in shares is given to be 60° .

The central angle for the investment in shares

$$= \frac{\text{Investment in shares}}{\text{Total investment}} \times 360^\circ \quad (1/2 \text{ mark})$$

$$\therefore 60^\circ = \frac{2000}{x} \times 360^\circ$$

$$\therefore x = \frac{2000 \times 360^\circ}{60^\circ}$$

$$\therefore x = 12000 \quad (1/2 \text{ mark})$$

(b) Let the amount deposited in post be ₹ y .

The central angle for the amount deposited in post

$$= \frac{\text{The amount deposited in post}}{\text{Total investment}} \times 360^\circ$$

$$\therefore 30^\circ = \frac{y}{12000} \times 360^\circ$$

$$\therefore y = \frac{30^\circ \times 12000}{360^\circ} \quad (1/2 \text{ mark})$$

$$= \frac{12000}{12}$$

$$= 1000$$

$$\therefore y = 1000$$

(1/2 mark)

Ans. (a) The total investment is ₹ 12,000

(b) The amount deposited in the post is ₹ 1000.

Q. 3. (A) Complete the following activities and rewrite it :

(Any one) 3

In this question, 2 activities each of 3 marks will be given. You have to attempt any one of them. Various types of activities can be asked.

- (i) A share is sold for the market value of ₹ 1000. Brokerage is paid at the rate of 0.1%. Find the amount received after sale by completing the activity.

Activity :

Brokerage = × the rate of brokerage.

$$= 1000 \times \frac{0.1}{\text{}}$$

$$= 10 \times 0.1$$

$$= \text{}$$

Amount received after sale = MV –

$$= \text{} - 1$$

$$= ₹ \text{}$$

- (ii) The numbers 2, 4, 6, 8, 10, 12 are on the surface of one die. If this die is rolled once, then complete the activity to find the probability of getting a perfect square number on the upper face.

Activity :

$$S = \left\{ \text{} \right\} \quad \therefore n(S) = \text{}$$

Condition for event B : Getting a perfect square number on the upper face of that die.

$$B = \left\{ \boxed{} \right\} \quad \therefore n(B) = \boxed{}$$

$$P(B) = \frac{n(B)}{n(S)} = \frac{\boxed{}}{6}$$

$$= \boxed{}.$$

Solutions

(i) **Activity :**

Brokerage = $\boxed{\text{MV}}$ $\times$ the rate of brokerage.

$$= 1000 \times \frac{0.1}{\boxed{100}} = 10 \times 0.1$$

$$= ₹ \boxed{1}$$

Amount received after sale = MV – $\boxed{\text{Brokerage}}$

$$= \boxed{1000} - 1$$

$$= ₹ \boxed{999}$$

(ii) **Activity :**

$$S = \left\{ \boxed{2, 4, 6, 8, 10, 12} \right\} \quad \therefore n(S) = \boxed{6}$$

Condition for event B : Getting a perfect square number on the upper face of that die.

$$B = \left\{ \boxed{4} \right\} \quad \therefore n(B) = \boxed{1}$$

$$P(B) = \frac{n(B)}{n(S)} = \frac{\boxed{1}}{6} = \frac{\boxed{1}}{\boxed{6}}$$

[**Scheme of marking for activities :** $\frac{1}{2}$ mark for each correct answer in the boxes.]

Q. 3. (B) Solve the following subquestions : (Any two)**6**

In this question, 4 subquestions each of 3 marks will be asked. You have to solve any two of them.

Note : Answers with solutions are expected.

- (i) Solve the following simultaneous equations by Cramer's method :

$$4m - 2n = -4; 4m + 3n = 16.$$

- (ii) Solve the following quadratic equation by formula method :

$$y^2 + \frac{1}{3}y = 2.$$

- (iii) Two dice are rolled. Write the sample space S and number of sample points $n(S)$. Also write the events in the set form and number of sample points in the event according to the given condition.

- (a) Condition for event A : The sum of the digits on the upper face is a multiple of 5.

- (b) Condition for event B : The sum of the digits on the upper face is 25.

- (iv) In the following table, the toll (in ₹) paid from 6 am to 6 pm by drivers and the number of vehicles is shown. Find the mean of the toll by 'assumed mean' method.

Toll (₹)	300–400	400–500	500–600	600–700	700–800
Number of Vehicles	80	110	120	70	40

Solutions

(i) $4m - 2n = -4$. Here, $a_1 = 4, b_1 = -2, c_1 = -4$

$$4m + 3n = 16 \quad a_2 = 4, b_2 = 3, c_2 = 16$$

$$D = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} = \begin{vmatrix} 4 & -2 \\ 4 & 3 \end{vmatrix} = 4 \times 3 - (-2) \times 4$$

$$= 12 + 8 = 20 \quad \left(\frac{1}{2} \text{ mark}\right)$$

$$D_m = \begin{vmatrix} c_1 & b_1 \\ c_2 & b_2 \end{vmatrix} = \begin{vmatrix} -4 & -2 \\ 16 & 3 \end{vmatrix} = (-4) \times 3 - (-2) \times 16$$

$$= -12 + 32 = 20 \quad \left(\frac{1}{2} \text{ mark}\right)$$

$$D_n = \begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix} = \begin{vmatrix} 4 & -4 \\ 4 & 16 \end{vmatrix} = 4 \times 16 - (-4) \times 4$$

$$= 64 + 16 = 80 \quad \left(\frac{1}{2} \text{ mark}\right)$$

By Cramer's rule,

$$m = \frac{D_m}{D} = \frac{20}{20} = 1; \quad \left(\frac{1}{2} \text{ mark}\right)$$

$$n = \frac{D_n}{D} = \frac{80}{20} = 4. \quad \left(\frac{1}{2} \text{ mark}\right)$$

Ans. $m = 1$ and $n = 4$ is the solution of the given simultaneous equations. $\left(\frac{1}{2} \text{ mark}\right)$

(ii) $y^2 + \frac{1}{3}y = 2$.

$$\therefore y^2 + \frac{1}{3}y - 2 = 0$$

Comparing with the standard form $ax^2 + bx + c = 0$,

$$a = 1, b = \frac{1}{3}, c = -2. \quad \left(\frac{1}{2} \text{ mark}\right)$$

$$b^2 - 4ac = \left(\frac{1}{3}\right)^2 - 4(1)(-2) \quad \left(\frac{1}{2} \text{ mark}\right)$$

$$= \frac{1}{9} + 8 = \frac{1+72}{9} = \frac{73}{9}$$

$$y = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-\left(\frac{1}{3}\right) \pm \sqrt{\frac{73}{9}}}{2 \times 1}$$

($\frac{1}{2}$ mark)

$$= \frac{-\frac{1}{3} \pm \frac{1}{3}\sqrt{73}}{2}$$

($\frac{1}{2}$ mark)

$$= \frac{-1 \pm \sqrt{73}}{6}$$

$$\therefore y = \frac{-1 + \sqrt{73}}{6} \quad \text{or} \quad y = \frac{-1 - \sqrt{73}}{6}. \quad (\frac{1}{2} + \frac{1}{2} \text{ marks})$$

Ans. $\frac{-1 + \sqrt{73}}{6}, \frac{-1 - \sqrt{73}}{6}$ are the roots of the given quadratic equation.

(iii) Two dice are rolled.

$\therefore$ the sample space

$S = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6), (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6), (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6), (5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6), (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)\}.$

($\frac{1}{2}$ mark)

$$\therefore n(S) = 36.$$

($\frac{1}{2}$ mark)

(a) Event A : The sum of the digits on the upper face is a multiple of 5.

$$\therefore A = \{(1, 4), (2, 3), (3, 2), (4, 1), (4, 6), (5, 5), (6, 4)\}.$$

($\frac{1}{2}$ mark)

$$\therefore n(A) = 7.$$

($\frac{1}{2}$ mark)

(b) Event B : The sum of the digits on the upper face is 25.

$B = \{ \}$. Because, the maximum sum of the digits on the upper face is 12. (1/2 mark)

$\therefore n(B) = 0$. (1/2 mark)

(iv) Let us take 550 as the assumed mean.

Then $A = 550$ and deviation $d_i = x_i - A = x_i - 550$.

Class Toll (in ₹)	Class mark x_i	Deviations $d_i = x_i - A$ $d_i = x_i - 550$	Frequency (Number of Vehicles) f_i	Frequency $\times$ deviation $f_i \times d_i$
300–400	350	– 200	80	– 16000
400–500	450	– 100	110	– 11000
500–600	550 $\rightarrow A$	0	120	0
600–700	650	100	70	7000
700–800	750	200	40	8000
Total			$\Sigma f_i = 420$	$\Sigma f_i d_i = -12000$

(1½ marks for tabulation)

Here, $\Sigma f_i d_i = -12000$, $\Sigma f_i = 420$

$$\bar{d} = \frac{\Sigma f_i d_i}{\Sigma f_i} = \frac{-12000}{420} \approx -28.57 \quad (1/2 \text{ mark})$$

$$\begin{aligned} \text{Mean} = \bar{X} &= A + \bar{d} \\ &= 550 + (-28.57) = 550 - 28.57 = 521.43. \end{aligned} \quad (1/2 \text{ mark})$$

Ans. The mean of the toll is ₹ **521.43**. (1/2 mark)

3 subquestions each of 4 marks will be asked. You have to solve any two of them.

Note : (1) Answers with solutions are expected.

(2) These will be challenging questions.

(3) These questions will not be from the textbook, but based on the syllabus.

- (i) Manisha wants to distribute 540 bananas among some students. If 30 students were more, each would get 3 bananas less. Find the number of students.
- (ii) ₹ 2000 is invested at 10 p.c.p.a. simple interest. Check at the end of the every year whether the interest amount is in A.P. or not. If this is an A.P., then find interest amount after 10 years.
- (iii) In one group of students; 10% students got 0 to 20 marks; 20% students got 20 to 40 marks; 35% students got 40 to 60 marks; 20% students got 60 to 80 marks and remaining 30 students got 80 to 100 marks, then
- (a) prepare a grouped frequency distribution table.
- (b) find the mode of the marks scored.

Solutions

- (i) Let the number of students be x .

Then each will get $\frac{540}{x}$ bananas. ($\frac{1}{2}$ mark)

30 students were more.

$\therefore$ the number of students is $(x + 30)$.

$\therefore$ each will get $\frac{540}{(x + 30)}$ bananas. ($\frac{1}{2}$ mark)

From the given condition,

$$\frac{540}{x} - \frac{540}{x + 30} = 3 \quad \text{($\frac{1}{2}$ mark)}$$

$$\therefore 540 \left(\frac{1}{x} - \frac{1}{x + 30} \right) = 3$$

$$\therefore 180 \left[\frac{x+30-x}{x(x+30)} \right] = 1 \dots \text{(Dividing both the sides by 3)} \quad (1/2 \text{ mark})$$

$$\therefore 180 \times 30 = x(x+30)$$

$$\therefore x^2 + 30x = 5400$$

$$\therefore x^2 + 30x - 5400 = 0$$

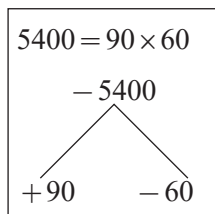
$$\therefore x^2 + 90x - 60x - 5400 = 0$$

$$\therefore x(x+90) - 60(x+90) = 0$$

$$\therefore (x+90)(x-60) = 0$$

$$\therefore x+90=0 \text{ or } x-60=0$$

$$\therefore x = -90 \text{ or } x = 60$$



(1/2 mark)

(1/2 mark)

(1/2 mark)

But the number of students cannot be negative.

$\therefore x = -90$ is unacceptable.

$\therefore x = 60$

Ans. The number of students is **60**.

(1/2 mark)

$$(ii) SI = \frac{P \times N \times R}{100}$$

(1/2 mark)

Here, $P = ₹ 2000$, $R = 10$ p.c.p.a.

Interest at the end of 1st year

$$= \frac{2000 \times 1 \times 10}{100} = ₹ 200 \quad (1/2 \text{ mark})$$

Interest at the end of 2nd year

$$= \frac{2000 \times 2 \times 10}{100} = ₹ 400 \quad (1/2 \text{ mark})$$

Interest at the end of 3rd year

$$= \frac{2000 \times 3 \times 10}{100} = ₹ 600 \quad (1/2 \text{ mark})$$

₹ 200, ₹ 400, ₹ 600, ... forms an A.P. (1/2 mark)

Here, $a = t_1 = 200$, $d = 400 - 200 = 200$, $t_{10} = ?$ (1/2 mark)

$$t_n = a + (n-1)d \quad \dots \text{(Formula)}$$



$$\begin{aligned}
 \therefore t_{10} &= 200 + (10 - 1) \times 200 \dots \text{(Substituting the values)} \text{ (}\frac{1}{2}\text{ mark)} \\
 &= 200 + 9 \times 200 \\
 &= 200 + 1800 \\
 &= 2000.
 \end{aligned}$$

Ans. The interest amount after 10 years will be ₹ 2000. ($\frac{1}{2}$ mark)

(iii) Let the total number of students be x .

We convert the percentages into the number of students.

$$10\% \text{ of } x = \frac{10}{100} \times x = \frac{x}{10} \quad \dots (1)$$

$$20\% \text{ of } x = \frac{20}{100} \times x = \frac{x}{5} \quad \dots (2)$$

$$35\% \text{ of } x = \frac{35}{100} \times x = \frac{7x}{20} \quad \dots (3)$$

$$\text{Now, } \frac{x}{10} + \frac{x}{5} + \frac{7x}{20} + \frac{x}{5} + 30 = x$$

$$\therefore 2x + 4x + 7x + 4x + 600 = 20x$$

... (Multiplying both the sides by 20) ($\frac{1}{2}$ mark)

$$\therefore 17x + 600 = 20x$$

$$\therefore 20x = 17x + 600$$

$$\therefore 20x - 17x = 600$$

$$\therefore 3x = 600 \quad \therefore x = \frac{600}{3} \quad \therefore x = 200 \quad \text{(\frac{1}{2} mark)}$$

Substituting $x = 200$ in equations (1), (2) and (3), we get,

$$\therefore \frac{x}{10} = \frac{200}{10} = 20, \quad \frac{x}{5} = \frac{200}{5} = 40, \quad \frac{7x}{20} = \frac{7 \times 200}{20} = 70. \quad \text{(\frac{1}{2} mark)}$$

(a) We make a tabulation :

Marks	0–20	20–40	40–60	60–80	80–100	Total
Number of students	20	40	70	40	30	200

(1 mark for tabulation)

Here, the maximum frequency 70 is in the class 40–60.

$\therefore$ the modal class is 40–60.

(b) L = Lower class limit of the modal class = 40

h = Class interval of the modal class = 20

f_1 = Frequency of the modal class = 70

f_0 = Frequency of the class preceding the modal class = 40

f_2 = Frequency of the class succeeding the modal class = 40.

$$\text{Mode} = L + \left[\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right] \times h \quad \dots \text{ (Formula) } \left(\frac{1}{2} \text{ mark} \right)$$

$$= 40 + \left[\frac{70 - 40}{2(70) - 40 - 40} \right] \times 20$$

... (Substituting the values) $\left(\frac{1}{2} \text{ mark} \right)$

$$= 40 + \left(\frac{30}{140 - 80} \right) \times 20$$

$$= 40 + \frac{30}{60} \times 20 = 40 + 10 = 50$$

Ans. The mode of the marks scored is **50**. $\left(\frac{1}{2} \text{ mark} \right)$

Q. 5. Solve the following subquestions : (Any one)

3

In this question, 2 subquestions each of 3 marks will be asked. You have to solve any one of them.

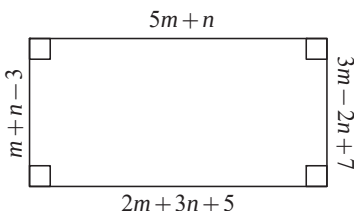
Note : (1) Answers with solutions are expected.

(2) The subquestions will be of constructive nature.

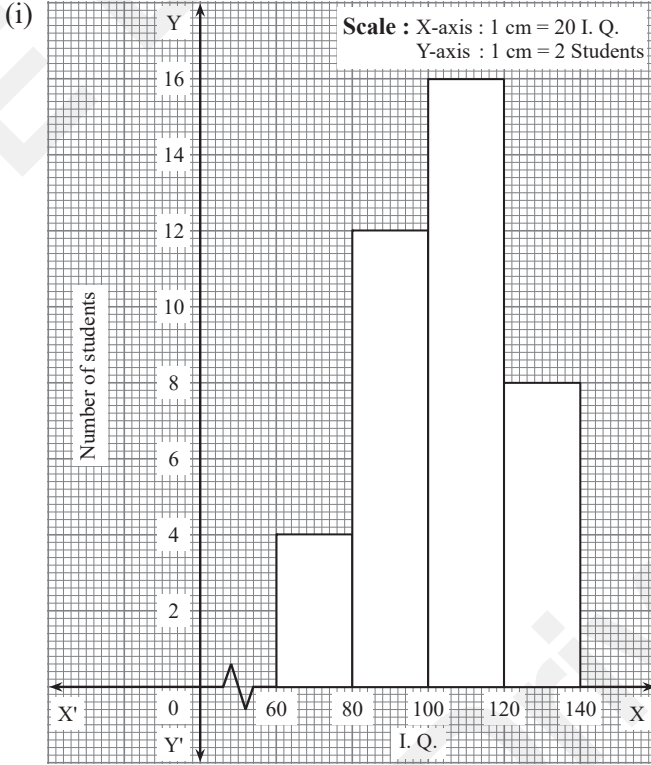
(i) Draw a histogram of the following data :

Students I.Q.	60–80	80–100	100–120	120–140
Number of Students	4	12	16	8

(ii) Using information given in the following figure, find the length and breadth of this rectangle in cm.



Solutions



[**Scheme of marking** : $\frac{1}{2}$ mark for taking proper scale and drawing both the axes, $\frac{1}{2}$ mark for kink mark, $\frac{1}{2}$ mark for each correct bar.]

(ii) The opposite sides of a rectangle are of equal lengths.

$$\therefore 5m + n = 2m + 3n + 5$$

$$\therefore 5m - 2m + n - 3n = 5$$

$$\therefore 3m - 2n = 5$$

... (1) ($\frac{1}{2}$ mark)

$$\text{Also, } 3m - 2n + 7 = m + n - 3$$

$$\therefore 3m - m - 2n - n = -3 - 7$$

$$\therefore 2m - 3n = -10$$

... (2) ($\frac{1}{2}$ mark)

Multiplying equation (1) by 3 and equation (2) by 2,

$$9m - 6n = 15 \quad \dots (3) \quad \text{and} \quad 4m - 6n = -20 \quad \dots (4)$$

Subtracting equation (4) from equation (3),

$$\begin{array}{rcl} 9m - 6n & = & 15 \quad \dots (3) \\ 4m - 6n & = & -20 \quad \dots (4) \\ - & + & + \\ \hline 5m & = & 35 \end{array} \quad \therefore m = \frac{35}{5} \quad \therefore m = 7 \quad (\frac{1}{2} \text{ mark})$$

Substituting $m = 7$ in equation (1),

$$3m - 2n = 5$$

$$\therefore 3(7) - 2n = 5$$

$$\therefore 21 - 2n = 5$$

$$\therefore -2n = 5 - 21$$

$$\therefore -2n = -16$$

$$\therefore n = \frac{-16}{-2} \quad \therefore n = 8 \quad (\frac{1}{2} \text{ mark})$$

Substituting the values of m and n .

$$\text{The length of the rectangle} = 5m + n$$

$$= 5 \times 7 + 8 = 35 + 8 = 43$$

$$\text{The breadth of the rectangle} = m + n - 3$$

$$= 7 + 8 - 3 = 15 - 3 = 12$$

Ans. The length of the rectangle is **43 cm** and its breadth is **12 cm**.

($\frac{1}{2} + \frac{1}{2}$ marks)

Question
Set
1LINEAR EQUATIONS IN
TWO VARIABLESPoints To Remember :

- **Linear Equation in Two Variables :**

An equation in two variables with the degree of the variable 1 is called a linear equation in two variables.

- **General Form of a Linear Equation in Two Variables :**

$ax + by + c = 0$ is the general form of a linear equation in two variables.

Here, a , b and c are real numbers and $a \neq 0$, $b \neq 0$.

- **Determinant Method of solving Simultaneous Equations (Cramer's Rule) :**

The equations are written as $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$.

$$D = \begin{array}{cc} \text{Coefficients of } x & \text{Coefficients of } y \\ \left| \begin{array}{c} a_1 \\ a_2 \end{array} \right| & \left| \begin{array}{c} b_1 \\ b_2 \end{array} \right| \end{array} \quad \text{i.e. } D = \left| \begin{array}{cc} a_1 & b_1 \\ a_2 & b_2 \end{array} \right|$$

$$D_x = \begin{array}{cc} \text{Constant terms} & \text{Coefficients of } y \\ \left| \begin{array}{c} c_1 \\ c_2 \end{array} \right| & \left| \begin{array}{c} b_1 \\ b_2 \end{array} \right| \end{array} \quad \text{i.e. } D_x = \left| \begin{array}{cc} c_1 & b_1 \\ c_2 & b_2 \end{array} \right|$$

$$D_y = \begin{array}{cc} \text{Coefficients of } x & \text{Constant terms} \\ \left| \begin{array}{c} a_1 \\ a_2 \end{array} \right| & \left| \begin{array}{c} c_1 \\ c_2 \end{array} \right| \end{array} \quad \text{i.e. } D_y = \left| \begin{array}{cc} a_1 & c_1 \\ a_2 & c_2 \end{array} \right|$$

$$x = \frac{D_x}{D} \quad \text{and} \quad y = \frac{D_y}{D}.$$

• **For Simultaneous Equations :**

Comparison of ratios	Graphical interpretation	Algebraic interpretation
(i) $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$	Intersecting lines	Unique solution
(ii) $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$	Coincident lines	Infinitely many solutions
(iii) $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$	Parallel lines	No solution

• **Determinant :**

$\begin{vmatrix} a & b \\ c & d \end{vmatrix}$ is a determinant. The degree of this determinant is 2. The value of this determinant is $ad - bc$.

1.1 1 mark each

Q. Choose the correct alternative :

1. What is the value of k , if $\left(\frac{1}{9}, k\right)$ is the solution of the simultaneous

equations $\frac{2}{x} - \frac{3}{y} = 15$ and $\frac{8}{x} + \frac{5}{y} = 77$?

- (A) 11 (B) -11 (C) 1 (D) -1

Ans. (C) [Substitute $x = \frac{1}{9}$ and $y = k$ in any one equation.]

2. For drawing the graph of $x + 2y = 4$, if $y = 1$, what is the value of x ? (March '24)

- (A) 1 (B) 2 (C) -2 (D) 6

Ans. (B) [Substitute $y = 1$ in the given equation.]

3. $ax + by = c$; $mx + ny = d$ and $an \neq bm$, then these simultaneous equations have

- (A) only one solution (B) no solution
(C) infinite number of solutions (D) only two solutions



Ans. (A) [If $an \neq bm$, i.e. $\frac{a}{m} \neq \frac{b}{n}$, then the lines representing the equations intersect each other. So the equations have only one solution.]

4. Find the value of $\begin{vmatrix} 5 & 3 \\ -7 & -4 \end{vmatrix}$. **(July '19–'22–'25)**

(A) -1 (B) -41 (C) 41 (D) 1

Ans. (D) $[5 \times (-4) - 3 \times (-7) = -20 + 21 = 1.]$

5. $\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix}$. Write the degree of the given determinant. **(March '25)**

(A) 1 (B) 2 (C) 3 (D) 4

Ans. (B) [There are 2 elements in each column and 2 elements in each row.]

6. What is the value of D for solving simultaneous equations $x + 2y = -7$ and $2x - 3y = 12$ by Cramer's rule?

(A) -7 (B) -21 (C) 14 (D) -2

Ans. (A) $\left[D = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \right]$

7. If $2x - 3y = 22$; $3x - 2y = 23$, what is the value of $(x + y)$?

(A) 9 (B) -9 (C) 1 (D) -1

Ans. (C) [Subtract equation (1) from equation (2).]

8. To draw the graph of $4x + 5y = 19$, find y , when $x = 1$.

(Nov. '20; March '23)

(A) 4 (B) 3 (C) 2 (D) -3

Ans. (B) [Substituting $x = 1$, $4(1) + 5y = 19$
 $\therefore 4 + 5y = 19 \quad \therefore 5y = 15 \quad \therefore y = 3.$]

Q. Choose the correct alternative :

1. What are the coordinates of the point of intersection of the lines $3x - y = 2$; $2x - y = 3$.
(A) $(-1, 5)$ (B) $(-1, -5)$ (C) $(1, 5)$ (D) $(1, -5)$
2. What is the value of k , if $(-2, k)$ is the solution of the simultaneous equations $4x + 3y = 1$ and $3x + 4y = 6$?
(A) 3 (B) -3 (C) 5 (D) 7
3. If $7x + 3y = 15$ and $3x + 7y = 25$, then what is the value of $(x - y)$?
(A) 10 (B) -10 (C) $\frac{5}{2}$ (D) $-\frac{5}{2}$
4. What are the coordinates of the point of intersection of the graph of the equation $2x - 3y = 12$ and the Y-axis?
(A) $(0, 6)$ (B) $(0, -6)$ (C) $(0, -4)$ (D) $(0, 4)$
5. Find the value of $\begin{vmatrix} 2 & 3 \\ 4 & 5 \end{vmatrix}$. (Sept. '21)
(A) 2 (B) -2 (C) 22 (D) -22
6. For simultaneous equations in variables x and y , $D_x = 49$, $D_y = -63$, $D = 7$, then what is the value of y ? (March '22)
(A) 9 (B) 7 (C) -7 (D) -9
7. Find the value of the determinant $\begin{vmatrix} 3 & 2 \\ 4 & 5 \end{vmatrix}$. (July '23)
(A) 2 (B) 7 (C) -7 (D) 23
8. What is the value of k , for which the simultaneous equations $x + 2y = 4$ and $3x - ky = 12$ have infinitely many solutions?
(A) 6 (B) 3 (C) -3 (D) -6

Answers :

- (B) [Subtract $2x - y = 3$ from $3x - y = 2$, $x = -1$, substituting $x = -1$, $y = -5$.]
- (A) [Substitute $x = -2$ and $y = k$ in any one equation.]
- (D) [Subtract $3x + 7y = 25$ from $7x + 3y = 15$.]
- (C) [x -coordinate of any point on the Y-axis is zero. Substitute $x = 0$ in the given equation.]
- (B) [$2 \times 5 - 3 \times 4 = 10 - 12 = -2$.]
- (D) $\left[y = \frac{D_y}{D} = \frac{-63}{7} = -9. \right]$
- (B) [$3 \times 5 - 2 \times 4$.]
- (D) $\left[\text{For infinite solutions : } \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} \right]$

[**Note :** The matter given in the brackets after answers in MCQ are explanations. Students are not expected to write the explanation.]

1.2 1 mark each**Q. Answer the following questions :**

- Find the value of $(x + y)$, if $3x + 5y = 9$; $5x + 3y = 7$. (March '19)

Ans. $3x + 5y = 9 \dots (1)$

$5x + 3y = 7 \dots (2)$

$8x + 8y = 16 \dots$ [Adding equations (1) and (2)]

$\therefore x + y = 2 \dots$ (Dividing both the sides by 8)

- Find the value of $(x - y)$, if $17x + 15y = 11$ and $15x + 17y = 21$.

(July '23; March '24)

Ans.

$17x + 15y = 11 \dots (1)$

$15x + 17y = 21 \dots (2)$

$2x - 2y = -10 \dots$ [Subtracting equation (2) from equation (1)]

$\therefore x - y = -5 \dots$ (Dividing both the sides by 2)

3. To draw the graph of $5x - 6y = 13$, find the value of x , when $y = 2$.

Ans. Substitute $y = 2$ in the given equation $5x - 6y = 13$.

$$\therefore 5x - 6(2) = 13 \quad \therefore 5x - 12 = 13$$

$$\therefore 5x = 13 + 12 \quad \therefore 5x = 25 \quad \therefore x = \frac{25}{5}$$

$$\therefore x = 5.$$

-
4. Find the value of the determinant $\begin{vmatrix} 4 & 3 \\ 2 & 7 \end{vmatrix}$. **(March '23)**

Ans. $\begin{vmatrix} 4 & 3 \\ 2 & 7 \end{vmatrix} = 4 \times 7 - 3 \times 2$
 $= 28 - 6 = 22.$

-
5. For simultaneous equations in variables x and y , if $D_x = 25$, $D_y = 40$, $D = 5$, then what is the value of x ? **(Nov. '20)**

Ans. $x = \frac{D_x}{D} = \frac{25}{5} \quad \therefore x = 5.$

Assignment

1.2

Q. Answer the following questions :

1. Find the value of $(x + y)$
 - (a) If $x + 2y = 5$; $2x + y = 4$. **(Sept. '21)**
 - (b) $15x + 17y = 21$; $17x + 15y = 11$. **(March '20; July '23)**
2. Find the value of $(x - y)$, if $2x - 3y = 2$; $9x - 10y = 23$.
3. To draw the graph of $4x + 5y = 19$, find y , when $x = 1$. **(March '22)**
4. The sum of the father's age and twice the age of his son is 70. Use the given information to form a linear equation in two variables. **(July '22)**
5. The cost of two tables and five chairs is ₹ 6600. Use the given information to form a linear equation in two variables.

Answers :

1. The value of $(x + y)$ is (a) 3; (b) 1. 2. The value of $(x - y)$ is 3.
3. $y = 3$. 4. $x + 2y = 70$. 5. $2x + 5y = 6600$.



1.3 2 marks each

1. Complete the following activity to find the value of x . (July '22)

$$3x - y = 2 \quad \dots (1)$$

$$2x + y = 8 \quad \dots (2)$$

$$\boxed{}x = \boxed{}$$

$$\therefore x = \frac{\boxed{}}{5}$$

$$\therefore x = \boxed{}.$$

Activity :

$$3x - y = 2 \quad \dots (1)$$

$$2x + y = 8 \quad \dots (2)$$

$$\boxed{5}x = \boxed{10}$$

$$\therefore x = \frac{\boxed{10}}{5}$$

$$\therefore x = \boxed{2}.$$

2. If $(0, 2)$ is the solution of $2x + 3y = k$, then complete the following activity to find the value of k . (March '24)

$(0, 2)$ is the solution of the equation $2x + 3y = k$.

Put $x = \boxed{}$ and $y = \boxed{}$ in the given equation.

$$\therefore 2 \times \boxed{} + 3 \times 2 = k$$

$$\therefore 0 + 6 = k \quad \therefore k = \boxed{}.$$

Activity :

$(0, 2)$ is the solution of the equation $2x + 3y = k$.

Put $x = \boxed{0}$ and $y = \boxed{2}$ in the given equation.

$$\therefore 2 \times \boxed{0} + 3 \times 2 = k$$

$$\therefore 0 + 6 = k \quad \therefore k = \boxed{6}.$$

3. To draw the graph of $4x + 5y = 19$, complete the following activity to find y , when $x = 1$. (Sept. '21)

$$4x + 5y = 19$$

$$\therefore 4 \times \boxed{} + 5y = 19$$

$$\therefore 5y = 19 - \boxed{}$$

$$\therefore y = \frac{\boxed{}}{5} \quad \therefore y = \boxed{}.$$

Activity : $4x + 5y = 19$

$$\therefore 4 \times \boxed{1} + 5y = 19$$

$$\therefore 5y = 19 - \boxed{4}$$

$$\therefore y = \frac{\boxed{15}}{5} \quad \therefore y = \boxed{3}.$$

4. Complete the activity to find the value of the determinant

$$\begin{vmatrix} 2\sqrt{3} & 9 \\ 2 & 3\sqrt{3} \end{vmatrix}.$$

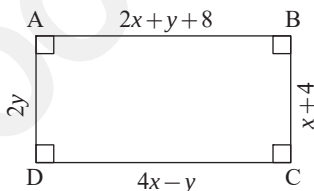
(March '22)

$$\begin{vmatrix} 2\sqrt{3} & 9 \\ 2 & 3\sqrt{3} \end{vmatrix} = 2\sqrt{3} \times \boxed{} - 9 \times \boxed{} \\ = \boxed{} - 18 = \boxed{}$$

Activity : $\begin{vmatrix} 2\sqrt{3} & 9 \\ 2 & 3\sqrt{3} \end{vmatrix} = 2\sqrt{3} \times \boxed{3\sqrt{3}} - 9 \times \boxed{2} \\ = \boxed{18} - 18 = \boxed{0}.$

5. $\square ABCD$ is a rectangle. Write two simultaneous equations, using the information given in the diagram, in the form of $ax + by = c$.

(March '24)



Solution : Opposite sides of a rectangle are equal.

$$\therefore DC = AB \quad \dots (1) \quad \text{and} \quad BC = AD \quad \dots (2)$$

$$\therefore 4x - y = 2x + y + 8 \quad \text{and} \quad x + 4 = 2y$$

$$\therefore 4x - y - 2x - y = 8 \quad \quad \quad x - 2y = -4$$



$$\therefore 2x - 2y = 8$$

$$\therefore x - y = 4$$

Ans. The equations are $x - y = 4$; $x - 2y = -4$.

- 6. Solve the simultaneous equations $x + y = 3$; $3x - 2y = 4$.**

(March '25)

Solution : $x + y = 3$... (1) $3x - 2y = 4$... (2)

Multiplying equation (1) by 2,

$$2x + 2y = 6 \quad \dots (3)$$

Adding equations (3) and (2),

$$2x + 2y = 6 \quad \dots (3)$$

$$3x - 2y = 4 \quad \dots (2)$$

$$\begin{array}{r} 2x + 2y = 6 \\ 3x - 2y = 4 \\ \hline 5x = 10 \end{array} \quad \therefore x = \frac{10}{5} \quad \therefore x = 2.$$

Substituting $x = 2$ in equation (1),

$$x + y = 3$$

$$\therefore 2 + y = 3 \quad \therefore y = 3 - 2 \quad \therefore y = 1$$

Ans. $(x, y) = (2, 1)$ is the solution.

- 7. If $\begin{vmatrix} 4 & 5 \\ m & 3 \end{vmatrix} = 22$, then find the value of m .**

(March '19)

Solution : $\begin{vmatrix} 4 & 5 \\ m & 3 \end{vmatrix} = 22$

$$\therefore 4 \times 3 - 5 \times m = 22$$

$$\therefore 12 - 5m = 22$$

$$\therefore -5m = 22 - 12$$

$$\therefore -5m = 10$$

$$\therefore 5m = -10 \quad \dots \text{[Multiplying both the sides by } (-1)\text{]}$$

$$\therefore m = -2$$

Ans. The value of m is -2 .

- 8. Find the values of $(x + y)$ and $(x - y)$ for the simultaneous equations $49x - 57y = 172$; $57x - 49y = 252$.**

(March '22)

Solution : $49x - 57y = 172 \quad \dots (1) \quad 57x - 49y = 252 \quad \dots (2)$

Subtracting equation (1) from equation (2),

$$\begin{array}{r} 57x - 49y = 252 \quad \dots (2) \\ - 49x + 57y = 172 \quad \dots (1) \\ \hline 8x + 8y = 80 \end{array}$$

$\therefore x + y = 10 \quad \dots (\text{Dividing both the sides by } 8)$

Adding equations (1) and (2),

$$\begin{array}{r} 49x - 57y = 172 \quad \dots (1) \\ 57x - 49y = 252 \quad \dots (2) \\ \hline 106x - 106y = 424 \end{array}$$

$\therefore x - y = 4 \quad \dots (\text{Dividing both the sides by } 106)$

Ans. The value of $(x + y)$ is **10**; the value of $(x - y)$ is **4**.

9. Find the values of D_x and D_y to solve the simultaneous equations

$x + 2y = -1$; $2x - 3y = 12$ by Cramer's rule. (July '22)

Solution : $x + 2y = -1$ Here, $a_1 = 1, b_1 = 2, c_1 = -1$
 $2x - 3y = 12$ Here, $a_2 = 2, b_2 = -3, c_2 = 12$

$$D_x = \begin{vmatrix} c_1 & b_1 \\ c_2 & b_2 \end{vmatrix} = \begin{vmatrix} -1 & 2 \\ 12 & -3 \end{vmatrix} = (-1) \times (-3) - 2 \times 12 = 3 - 24 = -21$$

$$D_y = \begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix} = \begin{vmatrix} 1 & -1 \\ 2 & 12 \end{vmatrix} = 1 \times 12 - (-1) \times 2 = 12 + 2 = 14$$

Ans. The value of D_x is **-21** and the value of D_y is **14**.

10. If the value of the determinant $\begin{vmatrix} 3\sqrt{6} & -4\sqrt{2} \\ 5\sqrt{3} & x \end{vmatrix}$ is $14\sqrt{6}$, find the value of x .

Solution : $\begin{vmatrix} 3\sqrt{6} & -4\sqrt{2} \\ 5\sqrt{3} & x \end{vmatrix} = 14\sqrt{6}$

$\therefore 3\sqrt{6} \times x - (-4\sqrt{2}) \times 5\sqrt{3} = 14\sqrt{6}$

$\therefore 3\sqrt{6}x + 20\sqrt{6} = 14\sqrt{6}$



$$\therefore 3x + 20 = 14$$

... (Dividing both the sides by $\sqrt{6}$)

$$\therefore 3x = 14 - 20$$

$$\therefore 3x = -6$$

$$\therefore x = \frac{-6}{3}$$

$$\therefore x = -2.$$

Ans. The value of x is -2 .

11. Two numbers differ by 3. The sum of the greater number and twice the smaller number is 15. Find the numbers. (March '19)

Solution : Let the greater number be x and the smaller number be y .

The two numbers differ by 3.

$$\therefore x - y = 3 \quad \dots (1)$$

Twice the smaller number = $2y$.

From the second condition,

$$x + 2y = 15 \quad \dots (2)$$

Subtracting equation (1) from equation (2),

$$x + 2y = 15 \quad \dots (2)$$

$$\underline{-x - y = -3} \quad \dots (1)$$

$$3y = 12 \quad \therefore y = \frac{12}{3} \quad \therefore y = 4$$

Substituting $y = 4$ in equation (1),

$$\begin{aligned} x - y &= 3 \\ \therefore x - 4 &= 3 \quad \therefore x = 3 + 4 \quad \therefore x = 7 \end{aligned}$$

Ans. The required numbers are **7** and **4**.

Assignment

1.3

1. Complete the following activity to find the value of x . (March '23)

$$5x + 3y = 9 \quad \dots (1)$$

$$2x - 3y = 12 \quad \dots (2)$$

Adding equations (1) and (2),

$$5x + 3y = 9 \quad \dots (1)$$

$$2x - 3y = 12 \quad \dots (2)$$

$$\underline{7x} = \square$$

$$\therefore x = \frac{\square}{\square}$$

$$\therefore x = \square$$

2. Complete the following table to draw the graph of $2x - 6y = 3$:
(March '20)

x	-5	
y		0
(x, y)		

3. Complete the following table to draw the graph of $x + y = 3$:
(July '23)

x	3		
y		5	3
(x, y)	$(3, 0)$		$(0, 3)$

4. Solve the simultaneous equations :
 (i) $4x + 3y = 11$; $3x + 4y = 10$. (July '19)
 (ii) $5x + 4y = 17$; $4x + 5y = 10$. (Sept. '21)
 (iii) $x + y = 4$; $2x - y = 2$. (March '23)
 (iv) $x + y = 6$; $x - y = 4$. (July '23)
5. Find the values of $(x + y)$ and $(x - y)$ for simultaneous equations
 $15x + 17y = 21$; $17x + 15y = 11$. (July '22; March '25)

6. Find the value of the determinant $\begin{vmatrix} \frac{7}{3} & \frac{5}{3} \\ \frac{3}{2} & \frac{1}{2} \end{vmatrix}$. (March '20)

7. To solve the simultaneous equations $3x + 5y = 26$; $x + 5y = 22$ by Cramer's rule, find the values of D_x and D_y . (March '22)
8. The sum of two numbers is 7 and their difference is 5. Find the numbers. (Nov. '20)



Answers :

1. $\boxed{21}$, $\frac{\boxed{21}}{\boxed{7}}$, $\boxed{3}$.
2. $\boxed{\frac{3}{2}}$, $\boxed{-\frac{13}{6}}$, $\boxed{\left(-5, -\frac{13}{6}\right)}$, $\boxed{\left(\frac{3}{2}, 0\right)}$
3. $\boxed{-2}$, $\boxed{0}$, $\boxed{0}$, $\boxed{(-2, 5)}$
4. (i) $x=2, y=1$ (ii) $x=5, y=-2$ (iii) $x=2, y=2$
(iv) $x=5, y=1$.
5. The value of $(x+y)$ is 1; the value of $(x-y)$ is -5 .
6. The value of the determinant is $\frac{-4}{3}$.
7. The value of D_x is 20 and the value of D_y is 40.
8. The required numbers are 6 and 1.

1.4 3 marks each

1. Complete the activity to solve the following simultaneous equations :

$$7x + 3y = 33 \quad \dots (1) \qquad 2x - 3y = -6 \quad \dots (2)$$

Adding equations (1) and (2),

$$7x + 3y = 33 \quad \dots (1)$$

$$2x - 3y = -6 \quad \dots (2)$$

$$\boxed{} = \boxed{}$$

$$\therefore x = \boxed{}$$

Substituting the value of x in equation (1),

$$7x + 3y = 33$$

$$\therefore \boxed{} + 3y = 33 \qquad \therefore 3y = \boxed{} \qquad \therefore y = \boxed{}$$

Activity :

Adding equations (1) and (2),

$$7x + 3y = 33 \quad \dots (1)$$

$$2x - 3y = -6 \quad \dots (2)$$

$$\boxed{9x} = \boxed{27}$$

$$\therefore x = \boxed{3}$$

Substituting the value of x in equation (1),

$$7x + 3y = 33$$

$$\therefore \boxed{21} + 3y = 33 \quad \therefore 3y = \boxed{12} \quad \therefore y = \boxed{4}$$

2. Complete the following activity to solve the simultaneous equations **$3x - 2y = 3$; $2x + y = 16$ by Cramer's rule.**

$$D = \begin{vmatrix} 3 & -2 \\ 2 & 1 \end{vmatrix} = \boxed{}; D_x = \begin{vmatrix} & -2 \\ & 1 \end{vmatrix} = \boxed{}; D_y = \begin{vmatrix} 3 & \\ 2 & \end{vmatrix} = 42.$$

$$x = \frac{D_x}{D} = \boxed{}; \quad y = \frac{D_y}{D} = \boxed{}.$$

Activity :

$$D = \begin{vmatrix} 3 & -2 \\ 2 & 1 \end{vmatrix} = \boxed{7}; D_x = \begin{vmatrix} 3 & -2 \\ 16 & 1 \end{vmatrix} = \boxed{35}; D_y = \begin{vmatrix} 3 & 3 \\ 2 & 16 \end{vmatrix} = 42;$$

$$x = \frac{D_x}{D} = \boxed{5}; \quad y = \frac{D_y}{D} = \boxed{6}.$$

3. Solve the simultaneous equations $3x - 2y = \frac{5}{2}$; $\frac{1}{3}x + 3y = -\frac{4}{3}$.**(July '22)**

$$\text{Solution : } 3x - 2y = \frac{5}{2}$$

Multiplying both the sides by 2,

$$6x - 4y = 5 \quad \dots (1)$$

$$\frac{1}{3}x + 3y = -\frac{4}{3}$$

Multiplying both the sides by 3,

$$x + 9y = -4 \quad \dots (2)$$

Multiplying equation (2) by 6,

$$6x + 54y = -24 \quad \dots (3)$$

Subtracting equation (1) from equation (3),

$$\begin{array}{rcl} 6x + 54y & = & -24 \quad \dots (3) \\ \underline{-6x + 4y} & = & \underline{5} \quad \dots (1) \\ \hline 58y & = & -29 \quad \therefore y = \frac{-29}{58} \quad \therefore y = -\frac{1}{2} \end{array}$$

Substituting $y = -\frac{1}{2}$ in equation (1),

$$\begin{aligned} 6x - 4y &= 5 \\ \therefore 6x - 4\left(-\frac{1}{2}\right) &= 5 \\ \therefore 6x + 2 &= 5 \quad \therefore 6x = 5 - 2 \quad \therefore 6x = 3 \\ \therefore x &= \frac{3}{6} \quad \therefore x = \frac{1}{2} \end{aligned}$$

Ans. $(x, y) = \left(\frac{1}{2}, -\frac{1}{2}\right)$ is the solution.

4. Solve the simultaneous equations $x + y = 7$; $x - y = -1$ graphically.

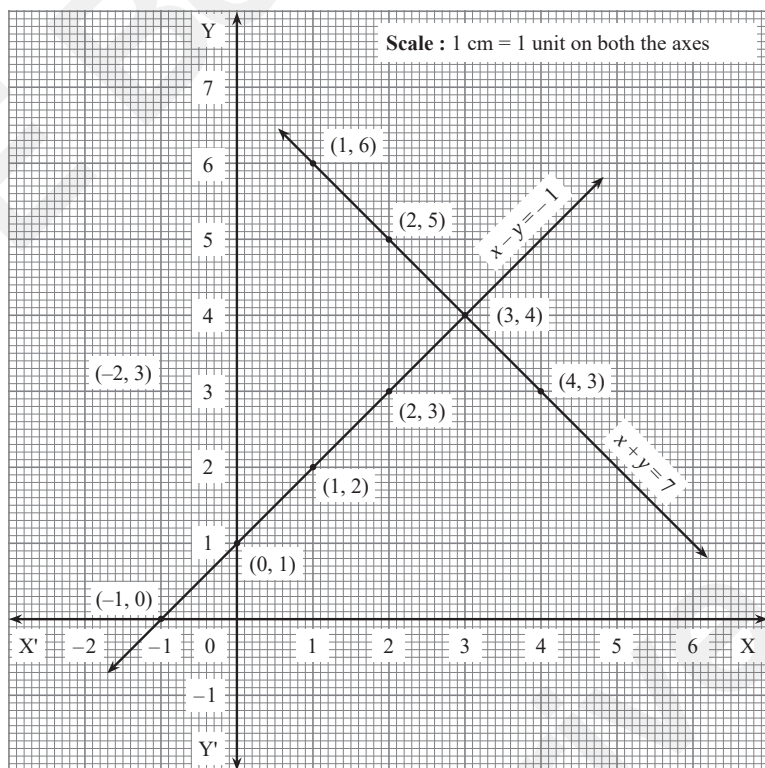
(Sept. '21)

Solution : $x + y = 7 \quad \therefore y = 7 - x$

x	1	2	3	4
y	6	5	4	3
(x, y)	(1, 6)	(2, 5)	(3, 4)	(4, 3)

$x - y = -1 \quad \therefore -y = -x - 1 \quad \therefore y = x + 1$

x	-1	0	1	2
y	0	1	2	3
(x, y)	(-1, 0)	(0, 1)	(1, 2)	(2, 3)



The coordinates of the point of intersection are $(3, 4)$.

Ans. $x = 3, y = 4$ is the solution.

5. Solve the simultaneous equations $4x + 3y = 18$; $3x - 2y = 5$ by Cramer's rule. (March '25)

Solution :

$$4x + 3y = 18 \quad \dots (1) \quad \text{Here, } a_1 = 4, b_1 = 3, c_1 = 18$$

$$3x - 2y = 5 \quad \dots (2) \quad \text{Here, } a_2 = 3, b_2 = -2, c_2 = 5$$

$$D = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} = \begin{vmatrix} 4 & 3 \\ 3 & -2 \end{vmatrix} = 4 \times (-2) - 3 \times 3 \\ = -8 - 9 = -17$$

$$D_x = \begin{vmatrix} c_1 & b_1 \\ c_2 & b_2 \end{vmatrix} = \begin{vmatrix} 18 & 3 \\ 5 & -2 \end{vmatrix} = 18 \times (-2) - 3 \times 5 \\ = -36 - 15 = -51$$

$$D_y = \begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix} = \begin{vmatrix} 4 & 18 \\ 3 & 5 \end{vmatrix} = 4 \times 5 - 3 \times 18 \\ = 20 - 54 = -34$$

By Cramer's rule,

$$x = \frac{D_x}{D} = \frac{-51}{-17} = 3, y = \frac{D_y}{D} = \frac{-34}{-17} = 2$$

Ans. $x = 3$ and $y = 2$ is the solution.

- 6. The sum of the present ages of Manish and Savita is 31 years. Manish's age 3 years ago was 4 times the age of Savita at that time. Find their present ages. (March '22)**

Solution : Let Manish's present age be x years and

Savita's present age be y years.

From the first condition,

$$x + y = 31 \quad \dots (1)$$

3 years ago, Manish's age was $(x - 3)$ years and Savita's age was $(y - 3)$ years.

From the second condition,

$$x - 3 = 4(y - 3)$$

$$\therefore x - 3 = 4y - 12 \quad \therefore x - 4y = -12 + 3$$

$$\therefore x - 4y = -9 \quad \dots (2)$$

Subtracting equation (2) from equation (1),

$$\begin{array}{rcl} x + y & = & 31 \quad \dots (1) \\ x - 4y & = & -9 \quad \dots (2) \\ \hline - & + & + \\ & 5y & = 40 \end{array}$$

$$\therefore y = \frac{40}{5} \quad \therefore y = 8$$

Substituting $y = 8$ in equation (1),

$$x + y = 31 \quad \therefore x + 8 = 31 \quad \therefore x = 31 - 8 \quad \therefore x = 23$$

Ans. The present ages of Manish and Savita are **23 years** and **8 years** respectively.

7. The sum of a two-digit number and the number obtained by reversing its digits is 143. Find the number, if the units place digit is greater than the tens place digit by 3. (July '22)

Solution : Let the digit at the tens place be x and that at the units place be y .

Then the number formed is $10 \times x + y = 10x + y$

The number obtained by reversing the digits $= 10y + x$

From the first condition,

$$10x + y + 10y + x = 143$$

$$\therefore 11x + 11y = 143$$

$$\therefore x + y = 13 \quad \dots \text{(Dividing both the sides by 11)} \quad \dots (1)$$

From the second condition, y is greater than x by 3.

$$\therefore y = x + 3 \quad \therefore x + 3 = y \quad \therefore x - y = -3 \quad \dots (2)$$

Adding equations (1) and (2),

$$x + y = 13 \quad \dots (1)$$

$$x - y = -3 \quad \dots (2)$$

$$\begin{array}{r} 2x = 10 \\ \hline \therefore x = \frac{10}{2} \end{array} \quad \therefore x = 5$$

Substituting $x = 5$ in equation (1),

$$x + y = 13$$

$$\therefore 5 + y = 13 \quad \therefore y = 13 - 5 \quad \therefore y = 8$$

The number is $10x + y = 10(5) + 8 = 50 + 8 = 58$

Ans. The number is **58**.

8. The denominator of a fraction is 4 more than twice its numerator. If 6 is subtracted from both the numerator and the denominator, the denominator becomes 12 times the numerator. Find the fraction. (March '20)

Solution : Let the numerator of the fraction be x and its denominator be y .

Twice the numerator $= 2x$.

From the first condition,

$$y = 2x + 4 \quad \therefore 2x + 4 = y \quad \therefore 2x - y = -4 \quad \dots (1)$$

If 6 is subtracted both from the numerator and the denominator, then the numerator becomes $(x - 6)$ and the denominator becomes $(y - 6)$.



From the second condition,

$$y - 6 = 12(x - 6)$$

$$\therefore y - 6 = 12x - 72$$

$$\therefore 12x - 72 = y - 6$$

$$\therefore 12x - y = -6 + 72$$

$$\therefore 12x - y = 66$$

... (2)

Subtracting equation (1) from equation (2),

$$12x - y = 66 \quad \dots (2)$$

$$2x - y = -4 \quad \dots (1)$$

$$\begin{array}{r} - \\ + \quad + \\ \hline 10x \quad = \quad 70 \end{array}$$

$$\therefore x = 7$$

Substituting $x = 7$ in equation (1),

$$2x - y = -4$$

$$\therefore 2(7) - y = -4$$

$$\therefore 14 - y = -4$$

$$\therefore -y = -4 - 14$$

$$\therefore -y = -18$$

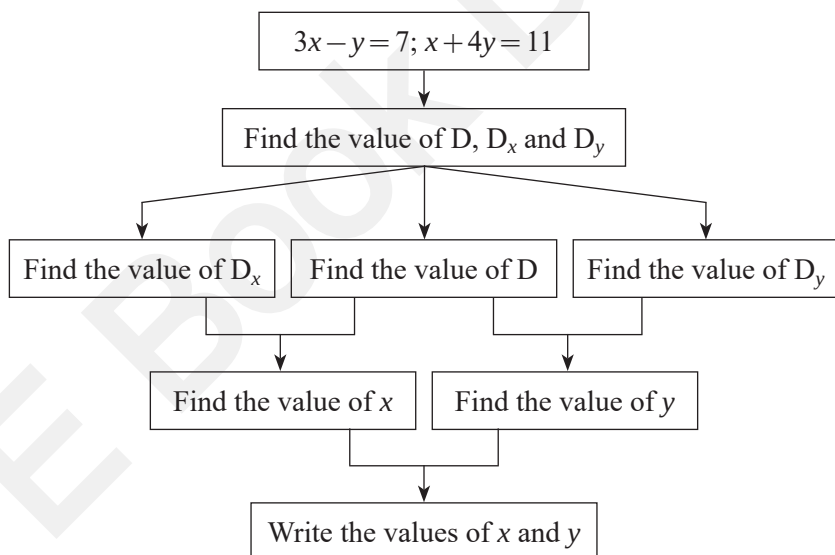
$$\therefore y = 18$$

The numerator x is 7 and the denominator y is 18.

Ans. The required fraction is $\frac{7}{18}$.

9. Observe the following flow chart and solve it :

(July '23)



Solution :

$$3x - y = 7 \quad \dots (1)$$

$$\text{Here, } a_1 = 3, b_1 = -1, c_1 = 7.$$

$$x + 4y = 11 \quad \dots (2)$$

$$\text{Here, } a_2 = 1, b_2 = 4, c_2 = 11.$$

$$D = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} = \begin{vmatrix} 3 & -1 \\ 1 & 4 \end{vmatrix} = 3 \times 4 - (-1) \times 1 \\ = 12 + 1 = 13$$

$$D_x = \begin{vmatrix} c_1 & b_1 \\ c_2 & b_2 \end{vmatrix} = \begin{vmatrix} 7 & -1 \\ 11 & 4 \end{vmatrix} = 7 \times 4 - (-1) \times 11 \\ = 28 + 11 = 39$$

$$D_y = \begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix} = \begin{vmatrix} 3 & 7 \\ 1 & 11 \end{vmatrix} = 3 \times 11 - 7 \times 1 \\ = 33 - 7 = 26$$

$$x = \frac{D_x}{D} = \frac{39}{13} = 3$$

$$y = \frac{D_y}{D} = \frac{26}{13} = 2$$

Ans. $(x, y) = (3, 2)$ is the solution.

10. Solve :

$$\sqrt{\frac{x}{y}} = 4, \frac{1}{x} + \frac{1}{y} = \frac{1}{xy}.$$

(March '20)

Solution :

$$\sqrt{\frac{x}{y}} = 4$$

Squaring both the sides,

$$\frac{x}{y} = 16 \quad \therefore x = 16y \quad \therefore x - 16y = 0 \quad \dots (1)$$

$$\frac{1}{x} + \frac{1}{y} = \frac{1}{xy}$$

Multiplying both the sides by xy ,

$$y + x = 1, \quad \text{i.e. } x + y = 1 \quad \dots (2)$$



Subtracting equation (1) from equation (2),

$$\begin{array}{rcl} x + y & = & 1 \quad \dots (2) \\ x - 16y & = & 0 \quad \dots (1) \\ \hline - & + & \\ 17y & = & 1 \end{array} \quad \therefore y = \frac{1}{17}$$

Substituting $y = \frac{1}{17}$ in equation (2),

$$\begin{aligned} x + y &= 1 \\ \therefore x + \frac{1}{17} &= 1 & \therefore x = 1 - \frac{1}{17} & \therefore x = \frac{17-1}{17} \\ \therefore x &= \frac{16}{17}. \end{aligned}$$

Ans. $(x, y) = \left(\frac{16}{17}, \frac{1}{17}\right)$ is the solution.

- 11. Draw the graphs representing the equation $2x = y + 2$ and $4x + 3y = 24$ on the same graph paper. Find the area of the triangle formed by these lines and the X-axis. (March '25)**

Solution :

$$2x = y + 2$$

$$\therefore y + 2 = 2x$$

$$\therefore y = 2x - 2$$

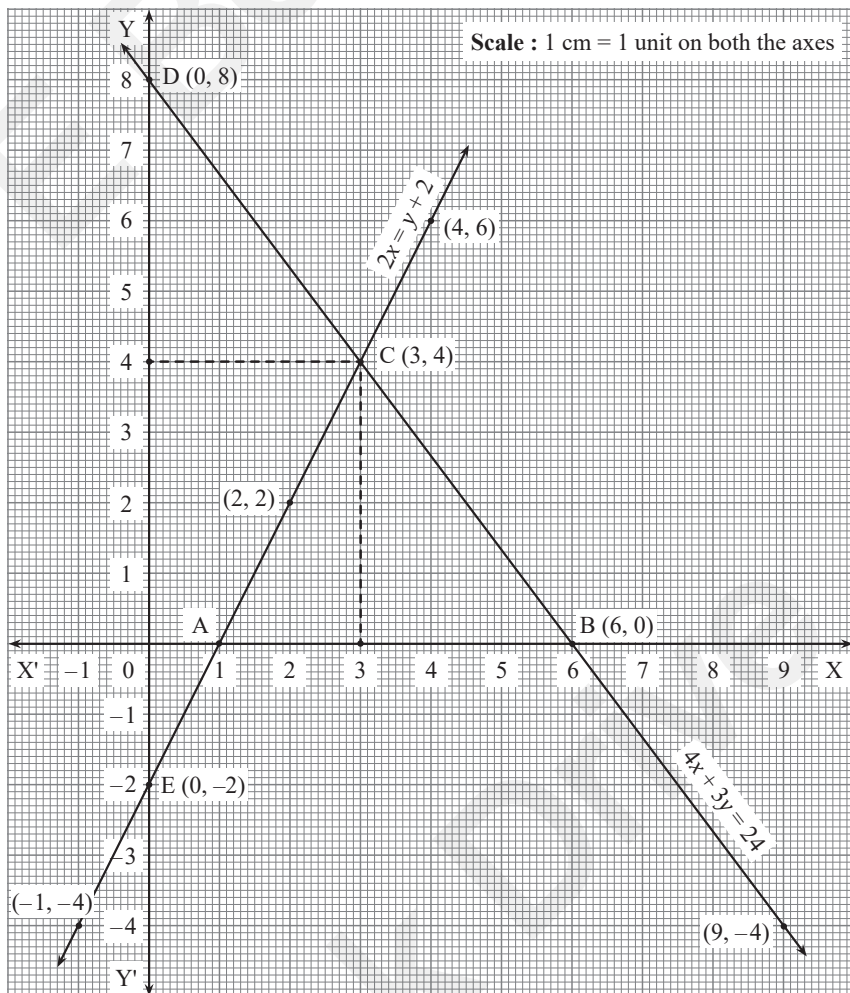
x	-1	0	2	4
y	-4	-2	2	6
(x, y)	$(-1, -4)$	$(0, -2)$	$(2, 2)$	$(4, 6)$

$$4x + 3y = 24$$

$$\therefore 3y = 24 - 4x$$

$$\therefore y = \frac{24 - 4x}{3}$$

x	0	3	6	9
y	8	4	0	-4
(x, y)	$(0, 8)$	$(3, 4)$	$(6, 0)$	$(9, -4)$



The lines intersect each other at the point C(3, 4).

From the graph $\triangle ABC$ is formed by the two lines and the X-axis.

For $\triangle ABC$, $d(A, B) = (6 - 1) = 5$

$\therefore AB = 5$ units and the perpendicular distance of point C on the X-axis is 4.

$$A(\triangle ABC) = \frac{1}{2} \times 5 \times 4 = 10 \text{ cm}^2.$$

1. Complete the activity to solve the following simultaneous equations :

$$5x - 3y = 19 \quad \dots (1) \qquad x - 6y = -7 \quad \dots (2)$$

Multiplying equation (1) by 2,

$$10x - 6y = \boxed{} \quad \dots (3)$$

Subtracting equation (2) from equation (3),

$$10x - 6y = 38 \quad \dots (3)$$

$$\begin{array}{r} x - 6y = -7 \quad \dots (2) \\ - \quad + \quad + \quad + \\ \hline \end{array}$$

$$9x = \boxed{} \qquad \therefore x = \boxed{}$$

Substituting the value of x in equation (1),

$$5x - 3y = 19$$

$$\therefore \boxed{} - 3y = 19 \qquad \therefore -3y = \boxed{} \qquad \therefore y = \boxed{}$$

2. Solve the following simultaneous equations graphically :

(i) $x + y = 0$; $2x - y = 9$. **(March '19)**

(ii) $x + y = 2$; $x - y = 4$. **(July '19)**

(iii) $x + y = 5$; $x - y = 1$. **(Nov. '20)**

(iv) $x + 3y = 7$; $2x + y = -1$. **(March '22)**

3. Solve the following simultaneous equations using Cramer's rule :

(i) $4m + 6n = 54$; $3m + 2n = 28$. **(Sept. '21; March '23)**

(ii) $3x - 4y = 10$; $4x + 3y = 5$. **(March '24)**

4. Two numbers differ by 3. The sum of twice the smaller number and thrice the greater number is 19. Find the numbers. **(July '23)**

5. The coordinates of the point of intersection of the lines $ax + by = 9$ and $bx + ay = 5$ are $(3, -1)$. Find the values of a and b . **(March '19)**

6. Construct any one linear equation in two variables. Obtain another equation by interchanging only the coefficients of the variables. Find the values of the variables. **(Nov. '20)**

7. Draw the graph of the equation $x + 2y = 4$. Find the area of the triangle formed by the line intersecting the X-axis and the Y-axis. **(March '23)**
8. Draw the graph of the equation $x + y = 4$ and answer the following questions : **(March '24)**
- (a) Which type of triangle is formed by the line with X and Y-axes based on its sides.
- (b) Find the area of that triangle.

Answers :

1. $\boxed{38}$, $\boxed{45}$, $\boxed{5}$, $\boxed{25}$, $\boxed{-6}$, $\boxed{2}$
2. (i) $(3, -3)$ (ii) $(3, -1)$ (iii) $(3, 2)$ (iv) $(-2, 3)$.
3. (i) $(6, 5)$ (ii) $(2, -1)$.
4. The required numbers are 5 and 2.
5. The values of a and b are 4 and 3 respectively.
6. This is an open ended question. Students should solve it.
 [For example, Let $3x + 5y = 12$... (1)
 Then $5x + 3y = 12$... (2) Solve.]
7. Area of the triangle is 4 cm^2 .
8. (a) An isosceles triangle is formed.
 (b) The area of the triangle is 8 cm^2 .

CHALLENGING QUESTIONS

1.5 4 marks each

1. Solve the simultaneous equations $x + y = 4$;
 $3x - 2y = 7$ graphically.

(July '22)

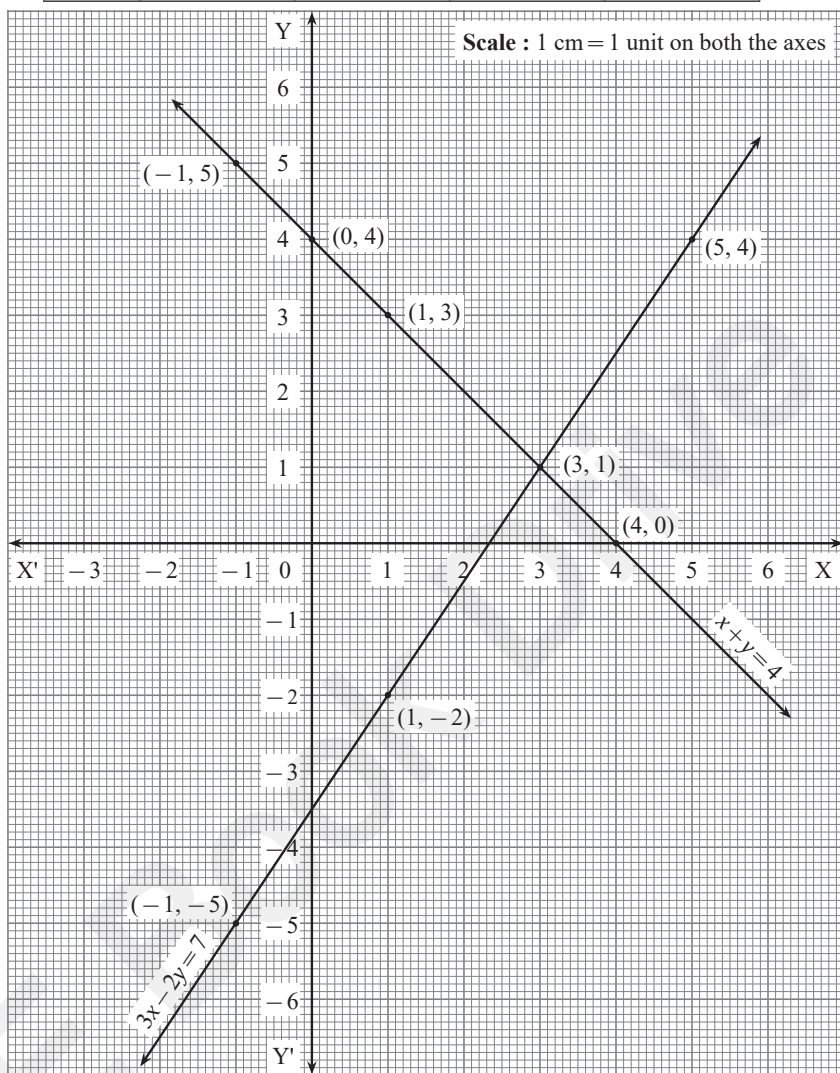
Solution : $x + y = 4 \quad \therefore y = 4 - x$

x	-1	0	1	4
y	5	4	3	0
(x, y)	$(-1, 5)$	$(0, 4)$	$(1, 3)$	$(4, 0)$

$$3x - 2y = 7 \quad \therefore -2y = 7 - 3x$$

$$\therefore 2y = 3x - 7 \quad \therefore y = \frac{3x - 7}{2}$$

x	-1	1	3	5
y	-5	-2	1	4
(x, y)	(-1, -5)	(1, -2)	(3, 1)	(5, 4)



The coordinates of the point of intersection are (3, 1).

Ans. The solution of the given simultaneous equations is $x = 3, y = 1$.

2. Solve : $\frac{12}{2x-y} + \frac{39}{x+2y} = 5$; $\frac{36}{2x-y} + \frac{26}{x+2y} = 8$.

Solution : $\frac{12}{2x-y} + \frac{39}{x+2y} = 5$... (1)

$\frac{36}{2x-y} + \frac{26}{x+2y} = 8$... (2)

Substituting a for $\frac{1}{2x-y}$ and b for $\frac{1}{x+2y}$, the given equations become

$12a + 39b = 5$... (3) $36a + 26b = 8$... (4)

Multiplying equation (3) by 3,

$36a + 117b = 15$... (5)

Subtracting equation (4) from equation (5),

$$\begin{array}{rcl} 36a + 117b & = & 15 \quad \dots (5) \\ 36a + 26b & = & 8 \quad \dots (4) \\ \hline 91b & = & 7 \end{array} \quad \therefore b = \frac{7}{91} \quad \therefore b = \frac{1}{13}$$

Substituting the value of b in equation (3),

$$\begin{aligned} 12a + 39b &= 5 \\ \therefore 12a + 39\left(\frac{1}{13}\right) &= 5 & \therefore 12a + 3 = 5 & \therefore 12a = 5 - 3 \\ \therefore 12a &= 2 & \therefore a = \frac{2}{12} & \therefore a = \frac{1}{6} \end{aligned}$$

Resubstituting the values of a and b ,

$a = \frac{1}{2x-y} = \frac{1}{6}$ $\therefore 2x - y = 6$... (6)

$b = \frac{1}{x+2y} = \frac{1}{13}$ $\therefore x + 2y = 13$... (7)

Multiplying equation (6) by 2,

$4x - 2y = 12$... (8)

Adding equations (8) and (7),

$$\begin{array}{rcl} 4x - 2y & = & 12 \quad \dots (8) \\ x + 2y & = & 13 \quad \dots (7) \\ \hline 5x & = & 25 \end{array} \quad \therefore x = \frac{25}{5} \quad \therefore x = 5$$



Substituting $x = 5$ in equation (7),

$$x + 2y = 13 \quad \therefore 5 + 2y = 13 \quad \therefore 2y = 13 - 5 \quad \therefore 2y = 8 \quad \therefore y = 4$$

Ans. $(x, y) = (5, 4)$ is the solution.

3. Out of 555 km, Vishal travelled certain distance by bus and the remaining distance by car. Bus travels at an average speed of 60 km/h and the average speed of the car is 75 km/h. He takes total 8 hours to complete the journey. Find the distance that Vishal travelled by bus. (July '19)

Solution : Let Vishal travelled x km by bus and y km by car.

$$\text{Time} = \frac{\text{Distance}}{\text{Speed}}$$

$$\text{Time taken by Vishal while travelling by bus} = \frac{x}{60} \text{ h.}$$

$$\text{Time taken by Vishal while travelling by car} = \frac{y}{75} \text{ h.}$$

From the given conditions,

$$x + y = 555 \quad \dots (1) \quad \frac{x}{60} + \frac{y}{75} = 8 \quad \dots (2)$$

Multiplying equation (2) by 300,

$$5x + 4y = 2400 \quad \dots (3)$$

We want to find the distance travelled by bus.

So we eliminate y .

Multiplying equation (1) by 4,

$$4x + 4y = 2220 \quad \dots (4)$$

Subtracting equation (4) from equation (3),

$$5x + 4y = 2400 \quad \dots (3)$$

$$4x + 4y = 2220 \quad \dots (4)$$

$$\begin{array}{r} - \\ - \\ - \\ \hline x \quad = \quad 180 \end{array}$$

Ans. Vishal travelled **180 km** by bus.

4. The fore wheel of a carriage makes 10 revolutions more than the rear wheel in covering 120 m. If the diameter of the fore wheel be increased by one-fourth, keeping the diameter of the rear wheel same, the fore wheel makes 4 revolutions more than the rear wheel in covering the same distance. Find the circumference of each wheel of the carriage.

Solution : Let the circumference of the fore wheel be x m and that of the rear wheel be y m.

For a wheel,

$$\text{Number of revolutions} = \frac{\text{Distance covered}}{\text{Circumference}}$$

From the first condition,

$$\frac{120}{x} - \frac{120}{y} = 10 \quad \dots (1)$$

The diameter of the fore wheel is increased by one-fourth.

$\therefore$ the new diameter of the fore wheel is $\frac{5}{4}$ times the previous diameter.

$\therefore$ the circumference of the fore wheel will now be $\frac{5}{4}$ times the previous circumference.

From the second condition,

$$\begin{aligned} \frac{120}{\frac{5}{4}x} - \frac{120}{y} &= 4 \\ \therefore \frac{120 \times 4}{5x} - \frac{120}{y} &= 4 \quad \therefore \frac{96}{x} - \frac{120}{y} = 4 \quad \dots (2) \end{aligned}$$

Substituting a for $\frac{1}{x}$ and b for $\frac{1}{y}$, the given equations become

$$120a - 120b = 10 \quad \dots (3) \quad 96a - 120b = 4 \quad \dots (4)$$

Subtracting equation (4) from equation (3),

$$\begin{aligned} 120a - 120b &= 10 & \dots (3) \\ 96a - 120b &= 4 & \dots (4) \end{aligned}$$

$$\begin{array}{r} - \quad + \quad - \\ 120a \quad \quad \quad = 10 \\ - 96a \quad \quad \quad = 4 \\ \hline 24a \quad \quad \quad = 6 \end{array}$$

$$\therefore a = \frac{6}{24} \quad \therefore a = \frac{1}{4}$$



Substituting $a = \frac{1}{4}$ in equation (3),

$$120a - 120b = 10$$

$$\therefore 120\left(\frac{1}{4}\right) - 120b = 10 \quad \therefore 30 - 120b = 10$$

$$\therefore -120b = 10 - 30 \quad \therefore -120b = -20$$

$$\therefore b = \frac{-20}{-120} = \frac{1}{6}$$

Resubstituting the values of a and b ,

$$a = \frac{1}{x} = \frac{1}{4} \quad \therefore x = 4$$

$$b = \frac{1}{y} = \frac{1}{6} \quad \therefore y = 6$$

Ans. The circumference of the fore wheel is **4 m** and
that of the rear wheel is **6 m**.

Assignment	1.5
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1. Solve :

$$(i) \frac{8}{x+2y} + \frac{3}{2x-y} = 3; \quad \frac{12}{x+2y} - \frac{6}{2x-y} = 1.$$

$$(ii) \frac{8}{x-y} + \frac{12}{x+y} = 3; \quad \frac{36}{x-y} + \frac{48}{x+y} = 13.$$

2. When one is added to the numerator as well as the denominator of a certain fraction, it becomes $\frac{1}{2}$. If one is subtracted from the numerator and the denominator both, the fraction becomes $\frac{1}{3}$. Find the original fraction. **(Sept. '21)**
3. The perimeter of an isosceles triangle is 24 cm. The length of its congruent sides is 13 cm less than twice the length of its base. Find the lengths of all sides of the triangle. **(March '22)**
4. A three-digit number is equal to 17 times the sum of its digits. If 198 is added to the number, the digits get reversed. The sum of the extreme digits is 1 less than the middle digit. Find the number.

5. A boat covers 32 km upstream and 36 km downstream in 7 hours. Also, it covers 40 km upstream and 48 km downstream in 9 hours. Find the speed of the boat in still water and that of the stream.
6. Anil is older than his friend Baiju by 3 years. Anil's father is thrice as old as Anil. Baiju is twice as old as his sister Chitra. The ages of Chitra and Anil's father differ by 49 years. Find the ages of Anil and Baiju.
7. Draw the graphs representing the equations $x + y = 5$ and $3x - y = 3$ on the same graph paper. Find the areas of the triangles formed by these lines, the X-axis and the Y-axis.

Answers :

1. (i) $x = 2, y = 1$ (ii) $x = 8, y = 4$.
2. The original fraction is $\frac{3}{7}$.
3. The lengths of the sides of the triangle are 7 cm, 7 cm and 10 cm.
4. The required number is 153.
5. The speed of the boat in still water is 10 km/h and the speed of the stream is 2 km/h.
6. Anil's age : 19 years; Baiju's age : 16 years.
7. The area of the triangle formed by the two lines and the X-axis is 6 cm^2 . The area of the triangle formed by the two lines and the Y-axis is 8 cm^2 .

Based on the Board's Evaluation Pattern
Vikas
MATHEMATICS PRACTICE BOOK
(Part I & II)
(Internal Evaluation Handbook)
With Solution on QR Code



**Question
Set
2**

QUADRATIC EQUATIONS

Points To Remember :

- $ax^2 + bx + c = 0$ ($a \neq 0$) is called the *standard form* (or *general form*) of the quadratic equation.
- The values of the variable which satisfy the given quadratic equation are called the *roots* of the quadratic equation.
- Solving quadratic equations :

(A) Factorisation Method :

- (i) Write the given equation in the form $ax^2 + bx + c = 0$.
- (ii) Find the two linear factors of the LHS of the equation.
- (iii) Equate each linear factor to zero.
- (iv) Solve each equation obtained in (iii) and write the roots of the given equation.

[**Note :** If the product of two numbers is zero, then at least one of them must be zero.]

(B) Completing Square Method :

- (i) Write the given equation in the form $ax^2 + bx + c = 0$.
- (ii) Consider the first two terms on LHS. Find the third suitable square term to make the polynomial a perfect square.
- (iii) Add the square term and subtract the same.
- (iv) Write the square of the first three terms and the last two terms.
- (v) Factorise and equate each factor to zero. Find the roots.

(C) Formula Method :

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ is the quadratic formula.

$b^2 - 4ac$ is called the *discriminant* of the quadratic equation and it is denoted by Δ (Delta).

If α and β are the roots of the quadratic equation,

$$\text{then } \alpha = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \text{ and } \beta = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

$$\therefore \alpha + \beta = -\frac{b}{a} \text{ and } \alpha\beta = \frac{c}{a}.$$

- Nature of the roots of a quadratic equation :
 - (i) $\Delta > 0$ Roots are real and unequal
 - (ii) $\Delta = 0$ Roots are real and equal
 - (iii) $\Delta < 0$ No real roots.
- If $b^2 - 4ac > 0$ and $b^2 - 4ac$ is not a complete square, then the roots of the quadratic equation are irrational and occur in pair.
e.g. if one root of a quadratic equation is $1 - 2\sqrt{3}$, then the other root is $1 + 2\sqrt{3}$. These roots are *conjugate of each other*.
- **Identities :**
 - (i) $a^2 + \frac{1}{a^2} = \left(a + \frac{1}{a}\right)^2 - 2.$
 - (ii) $a^2 + \frac{1}{a^2} = \left(a - \frac{1}{a}\right)^2 + 2.$
 - (iii) $(a + b)^2 = (a - b)^2 + 4ab.$
 - (iv) $(a - b)^2 = (a + b)^2 - 4ab.$
 - (v) $a^3 + b^3 = (a + b)^3 - 3ab(a + b).$

2.1	1 mark each
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Q. Choose the correct alternative :

1. Which of the following is not a quadratic equation?

(July '19; March '23)

- (A) $x^2 + 4x = 11 + x^2$ (B) $x^2 = 4x$
(C) $5x^2 = 90$ (D) $2x - x^2 = x^2 + 5$

Ans. (A) [x^2 from both the sides cancel out and the equation becomes a linear one.]



2. What is the value of k , if one root of the quadratic equation $kx^2 - 7x + 12 = 0$ is 3? (March '24)

(A) 1 (B) -1 (C) 3 (D) -3

Ans. (A) [Substitute $x = 3$ in the given equation.]

3. Which of the following equations has roots 3 and 5?

(Sept. '21; July '22-'23)

(A) $x^2 - 15x + 8 = 0$ (B) $x^2 - 8x + 15 = 0$

(C) $x^2 + 3x + 5 = 0$ (D) $x^2 + 8x - 15 = 0$

Ans. (B) [$\alpha = 3, \beta = 5$. the quadratic equation is $x^2 - (\alpha + \beta)x + \alpha\beta = 0$.]

4. One root of the quadratic equation $x^2 + mx - 5 = 0$ is 2. Find m .

(A) -2 (B) $-\frac{1}{2}$ (C) $\frac{1}{2}$ (D) 2

Ans. (C) [Substitute $x = 2$ in the given equation.]

5. What is the value of k , if the roots of $x^2 + kx + k = 0$ are real and equal?

(A) 0 (B) 4 (C) 0 or 4 (D) 2

Ans. (C) [The roots are real and equal. $\therefore \Delta = 0$.]

6. $\sqrt{5}m^2 - \sqrt{5}m + \sqrt{5} = 0$. Which of the following statements is true for this given equation?

(A) The roots are real and unequal (B) The roots are real and equal

(C) The roots are not real (D) Three roots

Ans. (C) [$\Delta = (-\sqrt{5})^2 - 4(\sqrt{5})(\sqrt{5})$
 $= 5 - 20 = -15$. $\therefore \Delta < 0$.]

7. What are the roots of the equation $\sqrt{3x^2 + 1} = 7$?

(A) 3, -3 (B) 4, -4 (C) 1, -1 (D) 7, -7

Ans. (B) [Square both the sides.]

8. Which of the following quadratic equations has sum of the roots -5 ?

- (A) $3x^2 - 15x + 3 = 0$ (B) $x^2 - 5x + 3 = 0$
(C) $x^2 + 3x - 5 = 0$ (D) $3x^2 + 15x + 3 = 0$

Ans. (D)

$$\left[\alpha + \beta = -\frac{b}{a}, -\frac{15}{3} = -5. \right]$$

Assignment

2.1

Q. Choose the correct alternative :

1. For which of the following quadratic equations is $\alpha + \beta = 11$ and $\alpha\beta = -33$?

- (A) $x^2 - 11x + 33 = 0$ (B) $x^2 + 11x - 33 = 0$
(C) $x^2 + 11x + 33 = 0$ (D) $x^2 - 11x - 33 = 0$

2. For the quadratic equation $x^2 + 10x - 7 = 0$, the values of a, b, c are
(March '19)

- (A) $a = -1, b = 10, c = 7$ (B) $a = 1, b = -10, c = 7$
(C) $a = 1, b = 10, c = -7$ (D) $a = 1, b = 10, c = 7$

3. Which of the following equations has roots -2 and 7 ?

- (A) $x^2 + 5x - 14 = 0$ (B) $x^2 - 5x - 14 = 0$
(C) $x^2 + 5x + 14 = 0$ (D) $x^2 - 5x + 14 = 0$

4. What is the nature of the roots of the quadratic equation $2x^2 + x - 10 = 0$?

- (A) Real and equal (B) Real and unequal
(C) Not real (D) Real

Answers :

1. (D) $\left[\alpha + \beta = -\frac{b}{a}; \alpha\beta = \frac{c}{a} \right]$ 2. (C)

3. (B) [Roots are -2 and 7 . $\therefore (x+2)(x-7) = 0$.]

4. (B) [$\Delta = b^2 - 4ac > 0$.]



2.2	1 mark each
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Q. Answer the following questions :

1. Find the values of a and b from the quadratic equation

$$2x^2 - 5x + 7 = 0.$$

(March '20)

Ans. $2x^2 - 5x + 7 = 0$

$\therefore a = 2, b = -5.$

-
2. Decide whether the equation $m^3 - 5m^2 + 4 = 0$ is a quadratic equation or not. Justify. **(Sept. '21)**

Ans. The given equation **is not** a quadratic equation because the maximum index of the variable m is 3 and not 2.

-
3. Determine whether 2 is a root of the equation $2m^2 - 5m = 0$ or not.

(March '22)

Ans. LHS = $2m^2 - 5m = 2(2)^2 - 5(2) = 8 - 10 = -2.$

RHS = 0

$\therefore$ LHS $\neq$ RHS. $\therefore$ 2 **is not** a root of the given equation.

-
4. Find the roots of the quadratic equation $(x + 5)(x - 4) = 0$.

(July '22)

Ans. $(x + 5)(x - 4) = 0$

$\therefore x + 5 = 0$ or $x - 4 = 0$

$\therefore x = -5$ or $x = 4.$

Assignment

2.2

Q. Answer the following questions :

1. Write the values of a, b, c for the following quadratic equations :

(i) $2x^2 - 3x + 7 = 0$ (ii) $5x^2 - 2 = -6x.$

2. Write the following quadratic equations in the standard form :

(i) $2y^2 = 7y - 3$ (ii) $3x^2 + 8 = 5x.$

3. Is $x = -1$ a root of the quadratic equation $x^2 - 3x - 4 = 0$?
4. Write the roots of the quadratic equation $x^2 + x - 2 = 0$.
5. Write the value of $b^2 - 4ac$, if $a = 1$, $b = -2$, $c = -4$.
6. If $\alpha + \beta = 7$ and $\alpha\beta = 10$, write the quadratic equation using variable x .

Answers :

1. (i) $a = 2$, $b = -3$, $c = 7$ (ii) $a = 5$, $b = 6$, $c = -2$.
2. (i) $2y^2 - 7y + 3 = 0$ (ii) $3x^2 - 5x + 8 = 0$.
3. $x = -1$ is the root of the given equation.
4. 1 and -2 are the roots.
5. The value of $b^2 - 4ac$ is 20.
6. $x^2 - 7x + 10 = 0$.

2.3 2 marks each

1. Complete the following activity to find the value of discriminant of the equation $x^2 + 10x - 7 = 0$. (July '23)

Comparing $x^2 + 10x - 7 = 0$ with $ax^2 + bx + c = 0$,

$$a = 1, b = 10, c = \boxed{}$$

$$\begin{aligned}\therefore b^2 - 4ac &= \boxed{} - 4 \times 1 \times (-7) \\ &= 100 + \boxed{} \\ &= \boxed{}.\end{aligned}$$

Activity :

Comparing $x^2 + 10x - 7 = 0$ with $ax^2 + bx + c = 0$,

$$a = 1, b = 10, c = \boxed{-7}$$

$$\begin{aligned}\therefore b^2 - 4ac &= \boxed{10^2} - 4 \times 1 \times (-7) \\ &= 100 + \boxed{28} \\ &= \boxed{128}.\end{aligned}$$



2. One of the roots of the quadratic equation $5m^2 + 2m + k = 0$ is $-\frac{7}{5}$.

Complete the following activity to find the value of k . (Nov. '20)

$-\frac{7}{5}$ is the root of the quadratic equation $5m^2 + 2m + k = 0$.

$\therefore$ substitute $m = -\frac{7}{5}$ in the equation.

$$5 \times \boxed{\quad}^2 + 2 \times \boxed{\quad} + k = 0$$

$$\therefore \frac{49}{5} - \frac{14}{5} + k = 0 \quad \therefore \boxed{\quad} + k = 0 \quad \therefore k = \boxed{\quad}.$$

Activity :

$-\frac{7}{5}$ is the root of the quadratic equation $5m^2 + 2m + k = 0$.

$\therefore$ substitute $m = -\frac{7}{5}$ in the equation.

$$5 \times \boxed{-\frac{7}{5}}^2 + 2 \times \boxed{-\frac{7}{5}} + k = 0$$

$$\therefore \frac{49}{5} - \frac{14}{5} + k = 0 \quad \therefore \boxed{7} + k = 0 \quad \therefore k = \boxed{-7}.$$

3. Complete the following activity to form a quadratic equation.

(March '25)

I am a quadratic equation.
↓
My standard form is .
↓
My roots are 5 and 12.
↓
Sum of my roots .
↓
Product of my roots .
↓
My quadratic equation is .

Activity :

I am a quadratic equation.
My standard form is $ax^2 + bx + c = 0$.
My roots are 5 and 12.
Sum of my roots 17 .
Product of my roots 60 .
My quadratic equation is $x^2 - 17x + 60 = 0$.

4. Complete the following activity to solve the quadratic equation

$\sqrt{2}x^2 + 7x + 5\sqrt{2} = 0$ by factorisation method.

$$\sqrt{2}x^2 + 7x + 5\sqrt{2} = 0$$

$$\therefore \sqrt{2}x^2 + \square + \square + 5\sqrt{2} = 0$$

$$\therefore x(\sqrt{2}x + 5) + \sqrt{2}(\sqrt{2}x + 5) = 0$$

$$\therefore (\sqrt{2}x + 5)(x + \sqrt{2}) = 0$$

$$\therefore \sqrt{2}x + 5 = 0 \quad \text{or} \quad \square = 0$$

$$\therefore x = \square \quad \text{or} \quad x = -\sqrt{2}.$$

Activity :

$$\sqrt{2}x^2 + 7x + 5\sqrt{2} = 0$$

$$\therefore \sqrt{2}x^2 + \square + \square + 5\sqrt{2} = 0$$

$$\therefore x(\sqrt{2}x + 5) + \sqrt{2}(\sqrt{2}x + 5) = 0$$

$$\therefore (\sqrt{2}x + 5)(x + \sqrt{2}) = 0$$

$$\therefore \sqrt{2}x + 5 = 0 \quad \text{or} \quad \square = 0$$

$$\therefore x = \square \quad \text{or} \quad x = -\sqrt{2}.$$



5. Solve the following quadratic equations by factorisation method :

(i) $m^2 + 14m + 13 = 0$. (March '25)

Solution : $m^2 + 14m + 13 = 0$

$$\therefore m^2 + 1m + 13m + 13 = 0$$

$$\therefore m(m+1) + 13(m+1) = 0$$

$$\therefore (m+1)(m+13) = 0$$

$$\therefore m+1 = 0 \text{ or } m+13 = 0$$

$$\therefore m = -1 \text{ or } m = -13$$

Ans. -1 and -13 are the roots of the given quadratic equation.

(ii) $x^2 + x - 20 = 0$. (Nov. '20; March '22-'24)

Solution : $x^2 + x - 20 = 0$

$$\therefore x^2 + 5x - 4x - 20 = 0$$

$$\therefore x(x+5) - 4(x+5) = 0$$

$$\therefore (x+5)(x-4) = 0$$

$$\therefore x+5 = 0 \text{ or } x-4 = 0$$

$$\therefore x = -5 \text{ or } x = 4$$

Ans. $-5, 4$ are the roots of the given quadratic equation.

6. Determine the nature of the roots of the quadratic equation $m^2 + 2m - 9 = 0$ using discriminant.

Solution : Comparing equation $m^2 + 2m - 9 = 0$ with the standard form $ax^2 + bx + c = 0$, $a = 1$, $b = 2$, $c = -9$.

$$\begin{aligned}\text{Discriminant } (\Delta) &= b^2 - 4ac \\ &= (2)^2 - 4(1)(-9) \\ &= 4 + 36 = 40\end{aligned}$$

Here, $\Delta > 0$

Ans. The roots are **real and unequal**.

7. Obtain a quadratic equation whose roots are -3 and -7 .

(March '19)

Solution : Let $\alpha = -3$ and $\beta = -7$.

Then $\alpha + \beta = -3 + (-7) = -10$ and $\alpha\beta = (-3) \times (-7) = 21$.

The required quadratic equation is

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

$$\therefore x^2 - (-10)x + 21 = 0 \quad \dots \text{(Substituting the values)}$$

$$\therefore x^2 + 10x + 21 = 0$$

Ans. The required quadratic equation is $x^2 + 10x + 21 = 0$.

Assignment

2.3

1. If 2 and 5 are the roots of the quadratic equation, then complete the following activity to form quadratic equation. **(March '24)**

Let $\alpha = 2$ and $\beta = 5$ be the roots of the quadratic equation.

Then the quadratic equation is

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

$$\therefore x^2 - \left(2 + \boxed{}\right)x + \boxed{} \times 5 = 0$$

$$\therefore x^2 - \boxed{}x + \boxed{} = 0.$$

2. Complete the following activity to form a quadratic equation.

(July '19)

I am a quadratic equation
↓
My standard form is
↓
My roots are 3 and 4
↓
Sum of my roots = Product of my roots =
↓
My equation is

3. Complete the following activity to determine the nature of the roots of the quadratic equation $x^2 + 2x - 9 = 0$. **(March '23)**



Compare $x^2 + 2x - 9 = 0$ with $ax^2 + bx + c = 0$.

$$a = 1, b = 2, c = \boxed{}.$$

$$\begin{aligned}\therefore \Delta &= b^2 - 4ac = (2)^2 - 4 \times \boxed{} \times \boxed{} \\ &= 4 + \boxed{} = 40.\end{aligned}$$

$$b^2 - 4ac > 0.$$

$\therefore$ the roots of the equation are real and unequal.

4. Find the value of k , if one of the roots of the quadratic equation :

(i) $6x^2 + 5x + k = 0$ is $-\frac{3}{2}$ (ii) $kx^2 - 11x - 6 = 0$ is 6.

5. Solve the following quadratic equations by factorisation method :

(i) $x^2 + 8x + 15 = 0$ **(March '19)**

(ii) $x^2 + 15x + 54 = 0$ **(July '23)**

(iii) $x^2 - 15x + 36 = 0$

6. Determine the nature of the roots of the quadratic equation

$2x^2 - 5x + 7 = 0$ using discriminant. **(Sept. '21)**

7. If the sum of the roots of the quadratic equation $x^2 - 4kx + k + 3 = 0$ is double their product, then find the value of k .

Answers :

1. $\boxed{5}, \boxed{2}, \boxed{7}, \boxed{10}.$

2. $\boxed{ax^2 + bx + c = 0}, \boxed{7}, \boxed{12}, \boxed{x^2 - 7x + 12 = 0}.$

3. $\boxed{-9}, \boxed{1}, \boxed{-9}, \boxed{36}.$

4. (i) $k = -6$ (ii) $k = 2.$

5. (i) $-5, -3$ are the roots (ii) $-9, -6$ are the roots.

(iii) $3, 12$ are the roots.

6. The roots are not real.

7. The value of k is 3.

2.4 3 marks each

1. One of the roots of the equation $kx^2 - 10x + 3 = 0$ is 3. Complete the following activity to find the value of k . (March '22)

One of the roots of the equation $kx^2 - 10x + 3 = 0$ is 3.

Putting $x = \boxed{}$ in the above equation.

$$k(\boxed{})^2 - 10 \times \boxed{} + 3 = 0$$

$$\therefore \boxed{} - 30 + 3 = 0$$

$$\therefore 9k = \boxed{} \quad \therefore k = \boxed{}$$

Activity :

One of the roots of the equation $kx^2 - 10x + 3 = 0$ is 3.

Putting $x = \boxed{3}$ in the above equation.

$$k(\boxed{3})^2 - 10 \times \boxed{3} + 3 = 0$$

$$\therefore \boxed{9k} - 30 + 3 = 0$$

$$\therefore 9k = \boxed{27} \quad \therefore k = \boxed{3}$$

2. For a quadratic equation in variable m , the coefficients a , b and c are such that $a = 2$, $b = 4a$, $c = 3a$. Form the quadratic equation and solve it by factorisation method. (Sept. '21)

Solution :

$$a = 2, b = 4a = 4 \times 2 = 8 \text{ and } c = 3a = 3 \times 2 = 6.$$

Taking these values in the quadratic equation $am^2 + bm + c = 0$,

$$2m^2 + 8m + 6 = 0$$

$$\therefore 2m^2 + 2m + 6m + 6 = 0$$

$$\therefore 2m(m + 1) + 6(m + 1) = 0$$

$$\therefore (m + 1)(2m + 6) = 0$$

$$\therefore m + 1 = 0 \quad \text{or} \quad 2m + 6 = 0$$

$$\therefore m = -1 \quad \text{or} \quad 2m = -6$$

$$\text{or} \quad m = -3$$

Ans. The quadratic equation : $2m^2 + 8m + 6 = 0$;

$-1, -3$ are the roots.

3. Solve the following quadratic equations using formula method :

(i) $5x^2 + 13x + 8 = 0$.

(Nov. '20; July '23)

(ii) $x^2 + 10x + 2 = 0$.

(March '22–'23)

Solution :

(i) $5x^2 + 13x + 8 = 0$

Comparing with $ax^2 + bx + c = 0$, $a = 5$, $b = 13$, $c = 8$.

$$b^2 - 4ac = (13)^2 - 4(5)(8)$$

$$= 169 - 160 = 9$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-13 \pm \sqrt{9}}{2 \times 5}$$

$$= \frac{-13 \pm 3}{10}$$

$$\therefore x = \frac{-13 + 3}{10} \quad \text{or} \quad x = \frac{-13 - 3}{10}$$

$$\therefore x = \frac{-10}{10} \quad \text{or} \quad x = \frac{-16}{10}$$

$$\therefore x = -1 \quad \text{or} \quad x = -\frac{8}{5}$$

Ans. $-1, -\frac{8}{5}$ are the roots of the given quadratic equation.

(ii) $x^2 + 10x + 2 = 0$

Comparing with $ax^2 + bx + c = 0$, $a = 1$, $b = 10$, $c = 2$.

$$b^2 - 4ac = (10)^2 - 4(1)(2) = 100 - 8 = 92$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-10 \pm \sqrt{92}}{2 \times 1}$$

$$= \frac{-10 \pm \sqrt{4 \times 23}}{2}$$

$$= \frac{-10 \pm 2\sqrt{23}}{2}$$

$$= \frac{2(-5 \pm \sqrt{23})}{2}$$

$$\therefore x = -5 \pm \sqrt{23}$$

$$\therefore x = -5 + \sqrt{23} \quad \text{or} \quad x = -5 - \sqrt{23}$$

Ans. $-5 + \sqrt{23}$, $-5 - \sqrt{23}$ are the roots of the given quadratic equation.

4. Solve the following problems :

(i) The product of two numbers is 352 and their mean is 19.

Find the numbers.

(July '22)

Solution : Let one number is x .

The product of two numbers is 352.

$\therefore$ the other number is $\frac{352}{x}$.

The mean of the numbers is 19.

$$\therefore \frac{x + \frac{352}{x}}{2} = 19$$

$$\therefore x + \frac{352}{x} = 38$$

$$\therefore x^2 + 352 = 38x$$

... (Multiplying both the sides by x)

$$\therefore x^2 - 38x + 352 = 0$$

$$\therefore x^2 - 16x - 22x + 352 = 0$$

$$\therefore x(x - 16) - 22(x - 16) = 0$$

$$\therefore (x - 16)(x - 22) = 0$$

$$\therefore x - 16 = 0 \quad \text{or} \quad x - 22 = 0$$

$$\therefore x = 16 \quad \text{or} \quad x = 22$$

Ans. The required numbers are **16** and **22**.

$ \begin{aligned} 352 &= 4 \times 88 \\ &= 4 \times 4 \times 22 \\ &= 16 \times 22 \\ &\swarrow \quad \searrow \\ -16 &\quad -22 \end{aligned} $
--

(ii) A dealer sells a toy for ₹ 24 and gains as much per cent as the cost price of the toy. Find the cost price of the toy. (March '20)



Solution : Let the cost price of the toy be ₹ x .

Gain is as much per cent as the cost price of the toy.

$$\therefore \text{gain} = x\% \text{ of } x = \frac{x}{100} \times x = ₹ \frac{x^2}{100}$$

Cost price + Gain = Selling price

$$\therefore x + \frac{x^2}{100} = 24$$

Multiplying both the sides by 100,

$$100x + x^2 = 2400$$

$$\therefore x^2 + 100x - 2400 = 0$$

$$\therefore x^2 + 120x - 20x - 2400 = 0$$

$$\therefore x(x + 120) - 20(x + 120) = 0$$

$$\therefore (x + 120)(x - 20) = 0$$

$$\therefore x + 120 = 0 \quad \text{or} \quad x - 20 = 0$$

$$\therefore x = -120 \quad \text{or} \quad x = 20$$

But the price cannot be negative.

$$\therefore x = -120 \text{ is unacceptable.} \quad \therefore x = 20$$

Ans. The cost price of the toy is ₹ 20.

$$\begin{array}{c} -2400 \\ \swarrow \quad \searrow \\ +120 \quad -20 \end{array}$$

5. Construct a word problem on quadratic equation, such that one of its answers is 20 (years, rupees, centimetres, etc). Also solve it.

(July '19)

Word problem :

The sum of the squares of two consecutive natural numbers is 841.

Find the smaller natural number.

Solution : Let the smaller natural number be x .

From the first condition, the greater consecutive natural number is $(x + 1)$.

From the second condition, $x^2 + (x + 1)^2 = 841$

$$\therefore x^2 + x^2 + 2x + 1 = 841$$

$$\therefore 2x^2 + 2x + 1 - 841 = 0$$

$$\therefore 2x^2 + 2x - 840 = 0$$

$$\therefore x^2 + x - 420 = 0$$

... (Dividing both the sides by 2)

$$\therefore x^2 + 21x - 20x - 420 = 0$$

$$\therefore x(x + 21) - 20(x + 21) = 0$$

$$\therefore (x + 21)(x - 20) = 0$$

$$\therefore x + 21 = 0 \quad \text{or} \quad x - 20 = 0$$

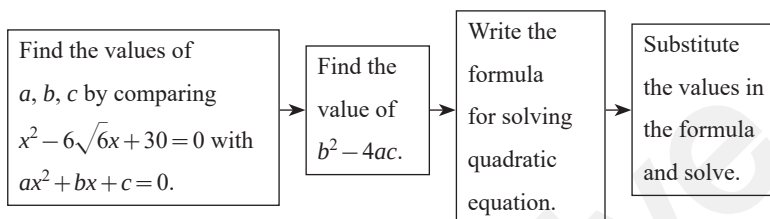
$$\therefore x = -21 \quad \text{or} \quad x = 20$$

$x = -21$ is unacceptable, because -21 is not a natural number.

$$\therefore x = 20 \text{ and } x + 1 = 20 + 1 = 21$$

Ans. The smaller natural number is **20**.

- 6. With the help of flow chart given below, solve the equation $x^2 - 6\sqrt{6}x + 30 = 0$ using the formula :**



Solution :

Step 1 : Comparing $x^2 - 6\sqrt{6}x + 30 = 0$ with $ax^2 + bx + c = 0$,

$$a = 1, b = -6\sqrt{6}, c = 30.$$

Step 2 :

$$b^2 - 4ac = (-6\sqrt{6})^2 - 4(1)(30) = 216 - 120 = 96 = 16 \times 6.$$

$$\text{Step 3 : } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad \dots \text{ (Formula)}$$

$$\begin{aligned} \text{Step 4 : } x &= \frac{-(-6\sqrt{6}) \pm 4\sqrt{6}}{2 \times 1} = \frac{6\sqrt{6} \pm 4\sqrt{6}}{2} \\ &= \frac{2(3\sqrt{6} \pm 2\sqrt{6})}{2} = 3\sqrt{6} \pm 2\sqrt{6} \end{aligned}$$

$$\therefore x = 3\sqrt{6} + 2\sqrt{6} \quad \text{or} \quad x = 3\sqrt{6} - 2\sqrt{6}$$

$$\therefore x = 5\sqrt{6} \quad \text{or} \quad x = \sqrt{6}$$

Ans. $5\sqrt{6}, \sqrt{6}$ are the roots of the given quadratic equation.

1. Complete the following activity to find the roots of the quadratic equation $25x^2 + 30x + 9 = 0$, by formula method. **(July '22)**

$$25x^2 + 30x + 9 = 0.$$

Comparing with $ax^2 + bx + c = 0$, $a = 25$, $b = \boxed{}$, $c = 9$

$$b^2 - 4ac = (30)^2 - 4 \times 25 \times 9$$

$$= 900 - 900 = \boxed{}$$

$$x = \frac{\boxed{} \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{\boxed{} \pm \sqrt{0}}{2 \times 25}$$

$$\therefore x = \frac{-30 + 0}{50} \quad \text{or} \quad x = \frac{\boxed{} - 0}{50}$$

$$\therefore x = -\frac{30}{50} \quad \text{or} \quad x = -\frac{30}{50}$$

$$\therefore x = -\frac{\boxed{}}{5} \quad \text{or} \quad x = -\frac{3}{5}$$

2. If $x = 5$ is a root of the quadratic equation $kx^2 - 14x - 5 = 0$, then find the value of k by completing the following activity : **(Sept. '21)**

One of the roots of the quadratic equation $kx^2 - 14x - 5 = 0$ is 5.

$\therefore$ substituting $x = \boxed{}$ in the above equation,

$$k \boxed{}^2 - 14 \times 5 - 5 = 0$$

$$\therefore \boxed{} k - 70 - 5 = 0$$

$$\therefore 25k = \boxed{} \quad \therefore k = \frac{75}{\boxed{}} \quad \therefore k = \boxed{}$$

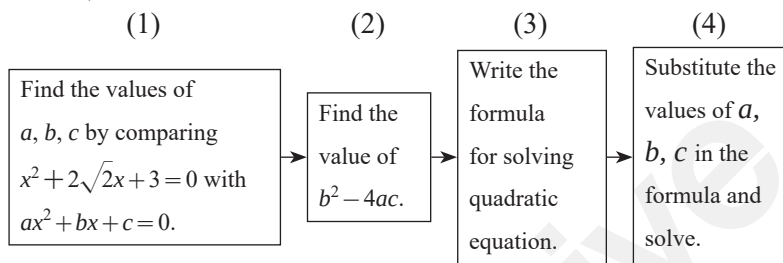
3. Solve the following quadratic equations using formula method :

(i) $3m^2 - m - 10 = 0$. **(March '24)**

(ii) $x^2 - 2x - 3 = 0$. **(March '25)**

4. Solve the following problems :

- (i) In a two-digit natural number, the digit at the tens place is 3 more than the square of the digit at units place. If 45 is subtracted from the number, the digits get interchange. Find the number.
- (ii) The sum of the squares of two consecutive even numbers is 580. Find the numbers.
5. The sum of the two roots of a quadratic equation is 9 and the sum of their cubes is 189. Find the equation.
6. With the help of the flow chart given below, solve the equation $x^2 + 2\sqrt{3}x + 3 = 0$ using the formula :



Answers :

1. $\boxed{30}$, $\boxed{0}$, $\boxed{-b}$, $\boxed{-30}$, $\boxed{-30}$, $\boxed{3}$

2. $\boxed{5}$, $\boxed{5}$, $\boxed{25}$, $\boxed{75}$, $\boxed{25}$, $\boxed{3}$

3. (i) $\left(2, -\frac{5}{3}\right)$ (ii) $(3, -1)$.

4. (i) The required number is 72.

(ii) The required numbers are 16 and 22.

5. The required equation is $x^2 - 9x + 20 = 0$.

6. (1) $a = 1$, $b = 2\sqrt{3}$, $c = 3$ (2) $b^2 - 4ac = 0$

(3) $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

(4) $x = \frac{-2\sqrt{3} \pm 0}{2 \times 1} = \frac{-2\sqrt{3}}{2} = -\sqrt{3}$.

CHALLENGING QUESTIONS

2.5 4 marks each

1. Find the value of m , if the quadratic equation $(m-12)x^2 + 2(m-12)x + 2 = 0$ has real and equal roots.

(March '25)

Solution :

Comparing $(m-12)x^2 + 2(m-12)x + 2 = 0$ with $ax^2 + bx + c = 0$,

$a = (m-12)$, $b = 2(m-12)$, $c = 2$.

$$\begin{aligned} b^2 - 4ac &= [2(m-12)]^2 - 4(m-12)(2) \\ &= 4(m-12)^2 - 8m + 96 \\ &= 4(m^2 - 24m + 144) - 8m + 96 \\ &= 4m^2 - 96m + 576 - 8m + 96 \\ &= 4m^2 - 104m + 672 \end{aligned}$$

The roots of the given quadratic equation are real and equal.

... (Given)

$$\therefore b^2 - 4ac = 0$$

$$\therefore 4m^2 - 104m + 672 = 0$$

$$\therefore m^2 - 26m + 168 = 0$$

... (Dividing by 4)

$$\therefore m^2 - 12m - 14m + 168 = 0$$

$$\therefore m(m-12) - 14(m-12) = 0$$

$$\therefore (m-12)(m-14) = 0$$

$$\therefore m-12=0 \text{ or } m-14=0$$

$$\therefore m=12 \quad \text{or } m=14$$

If $m=12$, $m-12=0$ and $(m-12)x^2=0$

i.e. the quadratic term becomes zero.

$\therefore$ the equation will not be a quadratic one.

$\therefore m=12$ is unacceptable.

$\therefore m=14$

Ans. The value of m is **14**.

2. A train travels 240 km with uniform speed. If the speed of the train is increased by 12 km/h, it takes one hour less to cover the same distance. Find the initial speed of the train. (July '22)

Solution :

Let the initial speed of the train be x km/h.

The increased speed is $(x + 12)$ km/h.

$$\text{Time} = \frac{\text{Distance}}{\text{Speed}}$$

$$\therefore \text{time taken to travel 240 km at the speed of } x \text{ km/h} = \frac{240}{x} \text{ hours}$$

$$\text{and that by the speed of } (x + 12) \text{ km/h} = \frac{240}{x + 12} \text{ hours.}$$

From the given condition,

$$\frac{240}{x} - \frac{240}{x + 12} = 1$$

$$\therefore 240 \left(\frac{1}{x} - \frac{1}{x + 12} \right) = 1$$

$$\therefore 240 \left[\frac{x + 12 - x}{x(x + 12)} \right] = 1$$

$$\therefore 240 \times 12 = 1 \times x(x + 12)$$

$$\therefore 2880 = x^2 + 12x$$

$$\therefore x^2 + 12x = 2880$$

$$\therefore x^2 + 12x - 2880 = 0$$

$$\therefore x^2 + 60x - 48x - 2880 = 0$$

$$\therefore x(x + 60) - 48(x + 60) = 0$$

$$\therefore (x + 60)(x - 48) = 0$$

$$\therefore x + 60 = 0 \quad \text{or} \quad x - 48 = 0$$

$$\therefore x = -60 \quad \text{or} \quad x = 48$$

But the speed cannot be negative.

$$\therefore x = -60 \text{ is unacceptable}$$

$$\therefore x = 48$$

Ans. The initial speed of the train is **48 km/h**.

$$\begin{aligned} 2880 &= 10 \times 288 \\ &= 10 \times 6 \times 48 \\ &= 60 \times 48 \\ &\quad - 2880 \\ &\quad \swarrow \quad \searrow \\ &60 \quad - 48 \end{aligned}$$

3. Out of the total students of class 10th, $\frac{7}{2}$ times the square root of the total number of students are playing on the ground and the remaining 2 students are studying in the classroom. Find the total number of students in class 10th. (Sept. '21)

Solution : Let the total number of students of class 10th be x .

$$\therefore \text{the number of students playing on the ground} = \frac{7}{2}\sqrt{x}.$$

From the given condition,

$$x = \frac{7}{2}\sqrt{x} + 2$$

$$\therefore x - 2 = \frac{7}{2}\sqrt{x}$$

$$\therefore (x - 2)^2 = \left(\frac{7}{2}\right)^2 \times x \quad \dots \text{ (Squaring both the sides)}$$

$$\therefore x^2 - 4x + 4 = \frac{49x}{4}$$

$$\therefore 4x^2 - 16x + 16 = 49x \quad \dots \text{ (Multiplying both the sides by 4)}$$

$$\therefore 4x^2 - 16x - 49x + 16 = 0$$

$$\therefore 4x^2 - 65x + 16 = 0$$

$$\therefore 4x^2 - x - 64x + 16 = 0$$

$$\therefore x(4x - 1) - 16(4x - 1) = 0$$

$$\therefore (4x - 1)(x - 16) = 0$$

$$\therefore 4x - 1 = 0 \quad \text{or} \quad x - 16 = 0$$

$$\therefore 4x = 1 \quad \text{or} \quad x = 16$$

$$\therefore x = \frac{1}{4}$$

But the number of students cannot be a fraction.

$$\therefore x = \frac{1}{4} \text{ is unacceptable.} \quad \therefore x = 16$$

Ans. The total number of students of class 10th is 16.

4. The sum of the squares of five consecutive natural numbers is 1455. Find the numbers. (March '23)

Solution :

Let the five consecutive natural numbers be

$$(x-2), (x-1), x, (x+1) \text{ and } (x+2) \quad \dots (x > 2)$$

From the given condition,

$$(x-2)^2 + (x-1)^2 + x^2 + (x+1)^2 + (x+2)^2 = 1455$$

$$\therefore x^2 - 4x + 4 + x^2 - 2x + 1 + x^2 + x^2 + 2x + 1 + x^2 + 4x + 4 = 1455$$

$$\therefore 5x^2 + 10 = 1455$$

$$\therefore 5x^2 = 1455 - 10$$

$$\therefore 5x^2 = 1445$$

$$\therefore x^2 = 289 \quad \dots \text{(Dividing both the sides by 5)}$$

$$\therefore x = \pm 17 \quad \dots \text{(Taking square roots of both the sides)}$$

$$\therefore x = 17 \quad \text{or} \quad x = -17$$

But $x = -17$ is unacceptable, because -17 is not a natural number.

$$\therefore x = 17$$

$$(x-2) = (17-2) = 15, (x-1) = (17-1) = 16, (x+1) = (17+1) = 18,$$

$$(x+2) = (17+2) = 19.$$

Ans. The required numbers are **15, 16, 17, 18 and 19.**

-
- 5 A tank can be filled up by two taps in 6 hours. The smaller tap alone takes 5 hours more than the bigger tap alone. Find the time required by each tap to fill the tank separately. (March '24)

Solution :

Let the bigger tap alone fill the tank in x hours. Then the smaller tap alone fill the tank in $(x+5)$ hours.

In one hour, the bigger tap will fill $\frac{1}{x}$ part of the tank and the smaller

tap will fill $\frac{1}{x+5}$ part of the tank.

$\therefore$ both the taps together will fill $\left(\frac{1}{x} + \frac{1}{x+5}\right)$ part of the tank.

The tank can be filled up by two taps in 6 hours.



∴ both the taps together fill $\frac{1}{6}$ part of the tank in 1 hour.

$$\therefore \frac{1}{x} + \frac{1}{x+5} = \frac{1}{6}$$

$$\therefore \frac{x+5+x}{x(x+5)} = \frac{1}{6} \quad \therefore \frac{2x+5}{x^2+5x} = \frac{1}{6} \quad \therefore 6(2x+5) = x^2+5x$$

$$\therefore x^2+5x = 6(2x+5) \quad \therefore x^2+5x = 12x+30$$

$$\therefore x^2+5x-12x-30=0 \quad \therefore x^2-7x-30=0$$

$$\therefore x^2-10x+3x-30=0 \quad \therefore x(x-10)+3(x-10)=0$$

$$\therefore (x-10)(x+3)=0$$

$$\therefore x-10=0 \text{ or } x+3=0$$

$$\therefore x=10 \quad \text{or } x=-3$$

∴ $x=-3$ is unacceptable, because the time cannot be negative.

$$\therefore x=10. \quad x+5=10+5=15.$$

Ans. The bigger tap alone can fill the tank in **10 hours** and
the smaller tap alone can fill the tank in **15 hours**.

-
- 6. The speed of a boat in still water is 16 km/h. It can go 60 km upstream and return to the original point in 8 hours. Find the speed of the stream.**

Solution : Let the speed of the stream be x km/h.

The speed of the boat in still water is 16 km/h. ... ($16 > x$)

∴ the speed of the boat upstream = $(16-x)$ km/h and that
downstream = $(16+x)$ km/h.

The time taken by the boat going 60 km upstream is $\frac{60}{16-x}$ hours and

that for going 60 km downstream is $\frac{60}{16+x}$ hours.

From the given condition,

$$\frac{60}{16-x} + \frac{60}{16+x} = 8$$

$$\therefore 60 \left(\frac{1}{16-x} + \frac{1}{16+x} \right) = 8$$

$$\therefore 60 \left[\frac{16+x+16-x}{(16-x)(16+x)} \right] = 8$$

$$\therefore 60 \times 32 = 8 \times (16-x)(16+x)$$

$$\therefore 60 \times 4 = (16-x)(16+x) \quad \dots \text{(Dividing both the sides by 8)}$$

$$\therefore 240 = 256 - x^2$$

$$\therefore x^2 = 256 - 240 \quad \therefore x^2 = 16 \quad \therefore x = \pm 4$$

$$\therefore x = 4 \text{ or } x = -4$$

But the speed cannot be negative.

$\therefore x = -4$ is unacceptable.

$$\therefore x = 4$$

Ans. The speed of the stream is **4 km/h**.

Assignment	2.5
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1. If one root of the quadratic equation $ax^2 + bx + c = 0$ is half of the other root, then show that $b^2 = \frac{9ac}{2}$. **(Nov. '20)**
2. The sum of the areas of two squares is 400 sq m. If the difference between their perimeters is 16 m, find the sides of two squares. **(March '20)**
3. If 460 is divided by a natural number, then the quotient is 2 more than nine times the divisor and the remainder is 5. Find the quotient and the divisor. **(March '22)**
4. A tank can be filled up by two taps in $3\frac{1}{13}$ minutes. The smaller tap alone takes 3 minutes more than the bigger tap alone to fill the tank. Find the time required by each tap to fill the tank separately. **(March '19)**
5. The sum of two roots of a quadratic equation is 5 and the sum of their cubes is 35. Find the equation. **(July '23)**
6. The speed of a boat in still water is 5 km/h. If it takes 1 hour more to go 5.25 km upstream than to return downstream to the same point, find the speed of the stream.



7. A jet plane covers 4500 km in some time. If the regular speed is decreased by 150 km/h, it takes one and a half hours more to complete the journey. Find the original speed of the jet plane.
8. Around a square pool, there is a footpath of width 2 m. If the area of the footpath is 21% that of the pool, find the area of the pool.

Answers :

2. The length of the sides of the squares are 12 m and 16 m respectively.
3. The quotient is 65 and the divisor is 7.
4. The bigger tap alone fills the tank in 5 minutes and the smaller tap alone fills the tank in 8 minutes.
5. The equation is $x^2 - 5x + 6 = 0$.
6. The speed of the stream : 2 km/h.
7. The original speed of the jet plane is 750 km/h.
8. The area of the pool is 1600 m².

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**Question
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3**

ARITHMETIC PROGRESSION

Points To Remember :

- **Sequence :** A sequence is a collection of numbers arranged in a definite order to some definite rule.
(i) 1, 2, 3, 4, 5, ...; (ii) 1, 4, 9, 16, 25, ... are sequences.
- **Terms in a sequence :** The terms in serial order of a sequence are denoted by $t_1, t_2, t_3, \dots, t_n$.
- **Arithmetic Progression :** In the given sequence, if the difference between two consecutive terms ($t_{n+1} - t_n$) is constant, then the sequence is called Arithmetic Progression (A.P.).
- **Common difference :** If the difference between two consecutive terms is constant, then it is called the common difference. It is denoted by d . The value of d may be positive, negative or zero.
- **n th Term of an A.P. :** If the first term (t_1) of an A.P. is a and the common difference is d , then the formula for the n th term of the A.P. is $t_n = a + (n - 1) d$
- $t_{n+1} - t_n = t_n - t_{n-1} = d$.
- **The sum of the first n terms of an A.P. :** The sum of the first n terms of an A.P. whose first term is a and the common difference d is

$$S_n = \frac{n}{2} [2a + (n - 1) d] .$$

If we take t_n as the last term l , then

$$S_n = \frac{n}{2} [a + l] .$$

Short cut method :

- $t_x - t_y = (x - y) \times d$
- $S_n - S_{n-1} = t_n$

• **Particular terms in an A.P. :**

It is convenient to consider :

- (i) Three consecutive terms as $a - d, a, a + d$.
- (ii) Four consecutive terms as $a - 3d, a - d, a + d, a + 3d$.
- (iii) Five consecutive terms as $a - 2d, a - d, a, a + d, a + 2d$.

[Note : In (ii), the common difference is $2d$.]

3.1	1 mark each
------------	--------------------

Q. Choose the correct alternative :

1. The first four terms of an A.P. are whose first term is -2 and the common difference is -2 . **(March '22)**

- (A) $-2, 0, 2, 4$ (B) $-2, 4, -8, 16$
(C) $-2, -4, -6, -8$ (D) $-2, -4, -8, -16$

Ans. (C) $[t_1 = a = -2, t_2 = t_1 + d = -2 + (-2) = -4, \dots]$

2. What is the common difference (d) of the A.P. $2, -2, -6, -10, \dots$?

- (A) -4 (B) 2 (C) -2 (D) 4 **(March '19)**

Ans. (A) $[d = t_2 - t_1.]$

3. The sum of the first five multiples of 3 is **(July '23)**

- (A) 45 (B) 55 (C) 15 (D) 75

Ans. (A) $[S_n = \frac{n}{2}(t_1 + t_n); n = 5, t_1 = 3, t_n = 15.]$

4. For a given A.P., $a = 3.5, d = 0$, then $t_n = \dots$.

(July '19 – '22; March '23)

- (A) 0 (B) 3.5 (C) 103.5 (D) 104.5

Ans. (B) $[t_1 = a = 3.5, d = 0. \therefore t_n = t_1 = 3.5.]$

5. What is the sum of the first 10 natural numbers? **(Nov. '20)**

- (A) 55 (B) 20 (C) 65 (D) 11

Ans. (A) $\left[\text{The sum of natural numbers : } S_n = \frac{n(n+1)}{2}. \right]$

6. Find the common difference of the following A.P. : (March '25)

4, 4, 4, ...

(A) 1 (B) 8 (C) 4 (D) 0

Ans. (D)

$$[d = t_2 - t_1 = 4 - 4 = 0.]$$

Assignment

3.1

Q. Choose the correct alternative :

1. For an A.P., if $t_7 = 4$ and $d = -4$, then what is the value of a ?

(March '24)

(A) 6 (B) 7 (C) 20 (D) 28

2. What is the sum of the first ten terms of the A.P. 15, 10, 5, ...?

(A) -75 (B) -125 (C) 75 (D) 123

3. The first term of an A.P. is 1 and the n th term is 20.

If $S_n = 399$, what is the value of n ?

(A) 42 (B) 38 (C) 21 (D) 19

4. If for an A.P., $d = 5$, then $t_{18} - t_{13} = ?$

(A) 5 (B) 20 (C) 25 (D) 30

5. What is the sum of the first ten terms of the A.P. 20, 16, 12,

(A) 20 (B) 16 (C) -16 (D) -20

6. The first term of an A.P. is 2 and the n th term is 29. If $S_n = 465$, then what is the value of n ?

(A) 15 (B) 16 (C) 30 (D) 32

7. For an A.P., if $S_{10} = 150$ and $S_9 = 126$, find t_{10} .

(A) 20 (B) 22 (C) 24 (D) 26

8. What is the sum of the first 30 natural numbers?

(A) 464 (B) 465 (C) 462 (D) 461

Answers :

1. (D) $[t_n = a + (n-1)d]$. 2. (A) $\left\{ S_n = \frac{n}{2} [2a + (n-1)d] \right\}$.

3. (B) $\left[S_n = \frac{n}{2} (a+l) \right]$. 4. (C) $[t_x - t_y = (x-y) \times d]$.



$$5. (A) \left\{ S_n = \frac{n}{2} [2a + (n-1)d] \right\}. \quad 6. (C) \left\{ S_n = \frac{n}{2} (a+l) \right\}.$$

$$7. (C) [S_n - S_{n-1} = t_n]. \quad 8. (B) \left[S_n = \frac{n(n+1)}{2} \right].$$

3.2 1 mark each

Q. Answer the following questions :

1. Find the second and third terms of an A.P. whose first term is -2 and the common difference is -2 . (March '20)

Ans.

$$a = t_1 = -2, d = -2.$$

$$\begin{aligned} \text{Second term : } t_2 &= t_1 + d \\ &= -2 - 2 = -4 \end{aligned}$$

$$\begin{aligned} \text{Third term : } t_3 &= t_2 + d = -4 - 2 \\ &= -6. \end{aligned}$$

2. Write the second and third terms of an A.P. whose first term is 6 and the common difference is -3 . (March '22)

Ans. $a = t_1 = 6, d = -3.$

$$t_2 = t_1 + d = 6 + (-3) = 3;$$

$$t_3 = t_2 + d = 3 + (-3) = 0$$

$$t_2 = 3; t_3 = 0.$$

3. Find the first term and the common difference for the following A.P. : $127, 135, 143, 151, \dots$ (Nov. '20; July '22)

Ans. Here, $a = t_1 = 127, t_2 = 135, t_3 = 143, \dots$

$$d = t_2 - t_1 = 135 - 127 = 8$$

The first term is **127** and the common difference is **8**.

4. For a given A.P., $a = 3.5, d = 0$, then find t_n . (Sept. '21)

Ans. $a = 3.5, d = 0, t_n = ?$

$$\begin{aligned} t_n &= a + (n-1)d \\ &= 3.5 + (n-1) \times 0 \end{aligned}$$

$$\therefore t_n = 3.5.$$

5. 5, 12, 19, 26, ... is an A.P. Find the next two terms of this A.P.

(July '23)

Ans.

Here, $a = t_1 = 5$, $t_2 = 12$, $t_3 = 19$, $t_4 = 26$, ...

$$d = t_2 - t_1 = 12 - 5 = 7$$

$$t_5 = t_4 + d = 26 + 7 = 33$$

$$t_6 = t_5 + d = 33 + 7 = 40$$

The next two terms are **33** and **40**.

6. Find the first term of the sequence $t_n = 3n - 4$.

(March '25)

Ans.

$$t_n = 3n - 4$$

Substituting $n = 1$,

$$t_1 = 3 \times 1 - 4$$

$$= 3 - 4 = -1$$

The first term is **-1**.

Assignment	3.2
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Q. Answer the following questions :

1. Find the common difference (d) for an A.P., if $t_4 = 12$ and $t_5 = 14$.
2. Find t_2 and t_3 of the A.P., if $a = 10$ and $d = 5$.
3. Find S_3 of the A.P. 13, 10, 7, 4,
4. If for an A.P., $t_{12} = 37$, $t_{18} = 53$, what is the value of t_{15} ?
5. What is the sum of first ten terms of an A.P., if the first term is 7 and the tenth term is 33?
6. Find the first term of the sequence $t_n = 3n - 2$.

(March '24)

Answers :

1. $d = 2$.
2. $t_2 = 15$, $t_3 = 20$.
3. $S_3 = 30$.
4. $t_{15} = 45 \left(t_{15} = \frac{t_{12} + t_{18}}{2} \right)$.
5. $S_{10} = 200$.
6. The first term is 1.



3.3 2 marks each

1. Complete the following activity to find the 27th term of the A.P.
9, 4, -1, -6, -11, (July '22)

Here, $a = 9$, $d = \square$, $n = 27$.

$$t_n = \square + (n - 1)d \quad \dots \text{(Formula)}$$

$$\therefore t_{27} = 9 + \left(\square - 1 \right) (-5) \quad \dots \text{(Substituting the values)}$$

$$\therefore t_{27} = \square.$$

Activity :

Here, $a = 9$, $d = \square - 5$, $n = 27$.

$$t_n = \square a + (n - 1)d \quad \dots \text{(Formula)}$$

$$\therefore t_{27} = 9 + \left(\square 27 - 1 \right) (-5) \quad \dots \text{(Substituting the values)}$$

$$\therefore t_{27} = \square - 121.$$

2. Complete the following activity to find the number of natural numbers from 1 to 171 which are divisible by 5. (July '19)

The natural numbers from 1 to 171 divisible by 5 are

5, 10, 15, ..., 170

This is an A.P. $a = \square$, $d = \square$, $t_n = 170$

$$t_n = a + (n - 1)d$$

$$170 = 5 + (n - 1) \times \square$$

$$n = \square$$

Activity : This is an A.P. $a = \boxed{5}$, $d = \boxed{5}$, $t_n = 170$

$$t_n = a + (n-1)d$$

$$170 = 5 + (n-1) \times \boxed{5}$$

$$n = \boxed{34}$$

$$34 = 1 + (n-1)$$

... (Dividing both the sides by 5)

$$\therefore 34 = n$$

3. The first term and the common difference of an A.P. are 6 and 3 respectively. Complete the following activity to find S_{27} .

(March '20; Sept. '21)

Here, the first term $= a = 6$, common difference $= d = 3$, $S_{27} = ?$

$$S_n = \frac{n}{2} \left[\boxed{} + (n-1)d \right] \quad \dots \text{(Formula)}$$

$$\therefore S_{27} = \frac{27}{2} \left[12 + (27-1) \times \boxed{} \right]$$

$$= \frac{27}{2} \times \boxed{} = 27 \times 45$$

$$\therefore S_{27} = \boxed{}.$$

Activity :

Here, the first term $= a = 6$, common difference $= d = 3$, $S_{27} = ?$

$$S_n = \frac{n}{2} \left[\boxed{2a} + (n-1)d \right] \quad \dots \text{(Formula)}$$

$$\therefore S_{27} = \frac{27}{2} \left[12 + (27-1) \times \boxed{3} \right]$$

$$= \frac{27}{2} \times \boxed{90}$$

$$= 27 \times 45$$

$$\therefore S_{27} = \boxed{1215}.$$

4. Decide whether the following sequence is an A.P. or not. If so, find the 20th term of the progression :

$-12, -5, 2, 9, 16, 23, 30, \dots$

(March '20)



Solution : Here, $a = t_1 = -12$, $t_2 = -5$, $t_3 = 2$, $t_4 = 9$, $t_5 = 16$, ...

$$t_2 - t_1 = -5 - (-12) = -5 + 12 = 7,$$

$$t_3 - t_2 = 2 - (-5) = 2 + 5 = 7,$$

$$t_4 - t_3 = 9 - 2 = 7, \quad t_5 - t_4 = 16 - 9 = 7$$

The common difference $= d = 7$... (A constant number)

$\therefore$ the given sequence is an A.P.

$$t_n = a + (n - 1)d \quad \dots \text{(Formula)}$$

$$\therefore t_{20} = -12 + (20 - 1) \times 7 \quad \dots \text{(Substituting the values)}$$

$$= -12 + 19 \times 7 = -12 + 133$$

$$\therefore t_{20} = 121$$

Ans. The given sequence is an A.P.

The 20th term of the A.P. is **121**.

-
- 5. The first term and the common difference of an A.P. are 12 and 4 respectively. If $t_n = 96$, find n . (March '19)**

Solution : Here, $a = 12$, $d = 4$, $t_n = 96$, $n = ?$

$$t_n = a + (n - 1)d \quad \dots \text{(Formula)}$$

$$\therefore 96 = 12 + (n - 1) \times 4 \quad \dots \text{(Substituting the values)}$$

$$\therefore 96 - 12 = (n - 1) \times 4$$

$$\therefore 84 = (n - 1) \times 4$$

$$\therefore 21 = n - 1 \quad \dots \text{(Dividing both the sides by 4)}$$

$$\therefore n = 21 + 1 \quad \therefore n = 22.$$

Ans. $n = 22$.

-
- 6. The first term $a = 8$ and the common difference $d = 5$ are given. Write an A.P. (July '23)**

Solution : Here, $a = t_1 = 8$ and $d = 5$

$$t_2 = t_1 + d = 8 + 5$$

$$= 13$$

$$t_3 = t_2 + d = 13 + 5$$

$$= 18$$

$$t_4 = t_3 + d = 18 + 5$$

$$= 23$$

Ans. The required A.P. is **8, 13, 18, 23, ...**

7. Find the 24th term of the A.P. 12, 16, 20, 24, (Sept. '21)

Solution : Here, $a = t_1 = 12$, $t_2 = 16$, $t_3 = 20$, ...

$$d = t_2 - t_1 = 16 - 12 = 4$$

$$t_n = a + (n - 1)d \quad \dots \text{(Formula)}$$

$$\therefore t_{24} = 12 + (24 - 1) \times 4 \quad \dots \text{(Substituting the values)}$$

$$= 12 + 23 \times 4$$

$$= 12 + 92$$

$$\therefore t_{24} = 104$$

Ans. The 24th term is **104**.

-
8. Find the sum of the first 21 even natural numbers. (July '22)

Solution :

2, 4, 6, ..., $2n$ are the even natural numbers.

Here, $a = t_1 = 2$, $t_2 = 4$, $t_3 = 6$, ...

$$d = t_2 - t_1 = 4 - 2 = 2.$$

$$S_n = \frac{n}{2}[2a + (n - 1)d] \quad \dots \text{(Formula)}$$

$$\therefore S_{21} = \frac{21}{2}[2 \times 2 + (21 - 1) \times 2] \quad \dots \text{(Substituting the values)}$$

$$= \frac{21}{2}[4 + 20 \times 2]$$

$$= \frac{21}{2}(4 + 40)$$

$$= \frac{21}{2} \times 44$$

$$= 21 \times 22$$

$$\therefore S_{21} = 462$$

Ans. The sum of first 21 even natural numbers is **462**.

Assignment

3.3

1. Find the 23rd term of the A.P., 9, 4, -1, -6, -11, (July '19)
2. Which term of the A.P., 3, 10, 17, 24, ... is 423?



3. For an A.P., $t_1 = 25$ and $t_{20} = 405$. Find the common difference.
4. How many two-digit numbers are divisible by 4?
5. Find the sum of first n even natural numbers. (March '22)
6. Complete the following activity to find the two-digit numbers which are divisible by 6.

The two-digit numbers divisible by 6 are 12, 18, 24, ..., 96.

Here, $a = 12$, $d = 6$, $t_n = 96$.

$$t_n = a + (n - 1)d \quad \dots \text{(Formula)}$$

$$\therefore \boxed{} = 12 + (n - 1) \times \boxed{} \quad \dots \text{(Substituting the values)}$$

$$\therefore \boxed{} = 2 + (n - 1) \quad \dots \text{(Dividing both the sides by 6)}$$

$$\therefore n = \boxed{}.$$

7. The first term of an A.P. is 5 and the common difference is 4. Complete the following activity to find the sum of first 21 terms of this A.P.

Here, $a = 5$, $d = 4$, $S_{21} = ?$

$$S_n = \frac{n}{2} \left[\boxed{} \right] \quad \dots \text{(Formula)}$$

$$\therefore S_{21} = \frac{21}{2} \left[10 + \boxed{} \right] \quad \dots \text{(Substituting the values)}$$

$$= \frac{21}{2} \times \boxed{}$$

$$= \boxed{}.$$

Answers :

1. 23rd term is -101

2. 61st term 3. $d = 20$ 4. 22

5. The sum of first n even natural numbers is $n(n + 1)$

6. $\boxed{96}$, $\boxed{6}$, $\boxed{16}$, $\boxed{15}$.

7. $\boxed{2a + (n - 1)d}$, $\boxed{20 \times 4}$, $\boxed{90}$, $\boxed{945}$.

3.4 3 marks each

1. In an A.P., the first term is -5 and the last term is 45 . If the sum of n terms in the A.P. is 120 , then complete the activity to find n .

(Nov. '20)

$$t_1 = -5, t_n = \boxed{}, S_n = \boxed{}$$

$$\therefore S_n = \frac{n}{2} \left[t_1 + \boxed{} \right] \quad \dots \text{(Formula)}$$

$$\therefore \boxed{} = \frac{n}{2} [-5 + 45] \quad \dots \text{(Substituting the values)}$$

$$\therefore 240 = n \times \boxed{} \quad \therefore n = \boxed{}.$$

Activity :

$$t_1 = -5, t_n = \boxed{45}, S_n = \boxed{120}$$

$$\therefore S_n = \frac{n}{2} \left[t_1 + \boxed{t_n} \right] \quad \dots \text{(Formula)}$$

$$\therefore \boxed{120} = \frac{n}{2} [-5 + 45] \quad \dots \text{(Substituting the values)}$$

$$\therefore 240 = n \times \boxed{40} \quad \therefore n = \boxed{6}.$$

2. There is an auditorium with 27 rows of seats. There are 20 seats in the first row, 22 seats in the second row, 24 seats in the third row, and so on. Find the total number of seats in the auditorium.

(March '22)

Solution : 20 seats in the first row, 22 seats in the second row, 24 seats in the third row.

$\therefore 20, 22, 24, \dots$ is a sequence.

The number of seats increasing in consecutive rows

$$= 24 - 22 = 22 - 20 = 2$$

$$\therefore d = 2$$

$\therefore 20, 22, 24, \dots$ is an A.P.

There are 20 seats in the first row.

$$\therefore a = 20$$

Total number of seats in 27 rows, i.e. we have to find S_{27} .



$$S_n = \frac{n}{2} [2a + (n-1)d] \quad \dots \text{(Formula)}$$

$$\therefore S_{27} = \frac{27}{2} [2 \times 20 + (27-1) \times 2] \quad \dots \text{(Substituting the values)}$$

$$= \frac{27}{2} [40 + 26 \times 2]$$

$$= \frac{27}{2} [40 + 52]$$

$$= \frac{27}{2} \times 92 = 27 \times 46$$

$$\therefore S_{27} = 1242$$

Ans. The total number of seats in the auditorium is **1242**.

- 3. Amit saves certain amount every month in a specific way. In the first month he saves ₹ 200, in the second month ₹ 250, in the third month ₹ 300 and so on. How much will be his total savings in 17 months? (March '19)**

Solution : Savings in the first month ₹ 200,
in the second month ₹ 250,
in the third month ₹ 300 and so on.

200, 250, 300, ... is an A.P.

Here, $a = 200$, $d = 50$, $n = 17$, $S_{17} = ?$

$$S_n = \frac{n}{2} [2a + (n-1)d] \quad \dots \text{(Formula)}$$

$$\therefore S_{17} = \frac{17}{2} [2 \times 200 + (17-1) \times 50] \quad \dots \text{(Substituting the values)}$$

$$= \frac{17}{2} [400 + 16 \times 50]$$

$$= \frac{17}{2} [400 + 800] = \frac{17}{2} \times 1200 = 17 \times 600$$

$$\therefore S_{17} = 10200$$

Ans. Amit's total savings in 17 months will be ₹ **10,200**.

4. Sachin invested ₹ 5000 in the 1st year, ₹ 7000 in the 2nd year, ₹ 9000 in the 3rd year in National Savings Certificate scheme. How much amount did he invest in 12 years? (July '19)

Solution : The amount invested by Sachin :

₹ 5000, ₹ 7000, ₹ 9000.

This is a sequence with common difference $d = 2000$.

∴ this is an A.P.

Here, $a = 5000$, $d = 2000$, $n = 12$.

We want to find S_{12} .

$$S_n = \frac{n}{2}[2a + (n-1)d] \quad \dots \text{(Formula)}$$

$$\begin{aligned} \therefore S_{12} &= \frac{12}{2}[2 \times 5000 + (12-1) \times 2000] \quad \dots \text{(Substituting the values)} \\ &= 6[10000 + 11 \times 2000] \\ &= 6[10000 + 22000] \\ &= 6 \times 32000 \end{aligned}$$

$$\therefore S_{12} = 192000$$

Ans. Sachin invested ₹ 1,92,000 in 12 years.

-
5. In an A.P., the sum of three consecutive terms is 27 and their product is 504. Find the terms. [Assume that three consecutive terms in A.P. are $a-d$, a and $a+d$.] (March '20; Sept. '21)

Solution : Let the three consecutive terms in an A.P. be $a-d$, a and $a+d$.

From the first condition, $(a-d) + a + (a+d) = 27$

$$\therefore a-d+a+a+d=27 \quad \therefore 3a=27 \quad \therefore a=9$$

From the second condition,

$$(a-d) \times a \times (a+d) = 504$$

$$\therefore (9-d) \times 9 \times (9+d) = 504 \quad \dots \text{(Substituting } a=9 \text{)}$$

$$\therefore (9-d)(9+d) = \frac{504}{9} = 56$$

$$\therefore 81 - d^2 = 56 \quad \therefore 81 - 56 = d^2$$

$$\therefore d^2 = 25 \quad \therefore d = \pm 5$$

When $d = 5$, the three consecutive terms are

$$a - d = 9 - 5 = 4, a = 9, a + d = 9 + 5 = 14$$

When $d = -5$, the three consecutive terms are

$$a - d = 9 - (-5) = 9 + 5 = 14, a = 9$$

$$a + d = 9 + (-5) = 9 - 5 = 4.$$

Ans. The three consecutive terms are **4, 9, 14** or **14, 9, 4**.

6. Write any one arithmetic progression with common difference 5.

Find its n th term and the sum of first n terms. (Sept. '21)

(Open Ended Question)

Solution : Here, $d = 5$. Let the first term (a) be 10.

The A.P. is 10, 15, 20, ..., n .

$$n = ?, S_n = ?$$

$$t_n = a + (n - 1)d \quad \dots \text{(Formula)}$$

$$= 10 + (n - 1) \times 5 \quad \dots \text{(Substituting the values)}$$

$$= 10 + 5n - 5$$

$$\therefore t_n = 5 + 5n \quad \dots (1)$$

$$S_n = \frac{n}{2} [t_1 + t_n] \quad \dots \text{(Formula)}$$

$$= \frac{n}{2} [10 + 5 + 5n] \quad \dots \text{(Substituting the values)}$$

$$= \frac{n}{2} (15 + 5n)$$

$$= \frac{n}{2} [5(3 + n)]$$

$$\therefore S_n = \frac{5n(3 + n)}{2} \quad \dots (2)$$

Ans. The n th term is **$5 + 5n$** .

The sum of the first n terms in $\frac{5n(3 + n)}{2}$.

7. Measures of the angles of a quadrilateral are in A.P. The measure of the largest angle is twice the smallest angle. Find the measures of all angles of the quadrilateral. (July '22)

Solution : Let the measures of the angles of the quadrilateral be $(a - 3d)^\circ$, $(a - d)^\circ$, $(a + d)^\circ$ and $(a + 3d)^\circ$.

The common difference is $2d$.

Here, $(a - 3d)^\circ$ is the smallest angle and $(a + 3d)^\circ$ is the largest angle.

$$(a - 3d)^\circ + (a - d)^\circ + (a + d)^\circ + (a + 3d)^\circ = 360^\circ$$

... (The property of the angles of a quadrilateral.)

$$\therefore a - 3d + a - d + a + d + a + 3d = 360$$

$$\therefore 4a = 360 \qquad \therefore a = \frac{360}{4} \qquad \therefore a = 90 \qquad \dots (1)$$

From the given condition,

$$a + 3d = 2(a - 3d)$$

$$\therefore a + 3d = 2a - 6d \qquad \therefore 3d + 6d = 2a - a$$

$$\therefore 9d = a \qquad \therefore 9d = 90 \qquad \dots [\text{From (1)}]$$

$$\therefore d = \frac{90}{9} \qquad \therefore d = 10 \qquad \dots (2)$$

From (1) and (2),

$$a - 3d = 90 - 3(10) = 90 - 30 = 60$$

$$a - d = 90 - 10 = 80$$

$$a + d = 90 + 10 = 100$$

$$a + 3d = 90 + 3(10) = 90 + 30 = 120.$$

Ans. The measures of the angles of the quadrilateral are **60° , 80° , 100° and 120° .**

Assignment

3.4

1. Complete the following activity to find the sum of all two-digit numbers divisible by 8.

Two-digit numbers divisible by 8 are 16, 24, 32, ..., 88, 96.

Here, $a = t_1 = 16$, $d = 8$, $t_n = 96$, $n = ?$, $S_n = ?$

$$t_n = \boxed{} \qquad \dots (\text{Formula})$$

$$\therefore 96 = 16 + \boxed{} \qquad \dots (\text{Substituting the values})$$



$$\therefore 96 = \boxed{}$$

$$\therefore n = 11.$$

$$\text{Now, } S_n = S_{11}$$

$$S_n = \frac{n}{2} \left[\boxed{} \right] \quad \dots \text{ (Formula)}$$

$$\therefore S_{11} = \frac{11}{2} \times \boxed{} \quad \dots \text{ (Substituting the values)}$$

$$\therefore S_{11} = \boxed{}.$$

2. For an A.P., $a = 10$ and $d = 5$. Complete the following activity to find S_{30} of the A.P.

$$\text{Here, } a = 10, d = 5, S_{30} = ?$$

$$S_n = \frac{n}{2} \boxed{} \quad \dots \text{ (Formula)}$$

$$\therefore S_{30} = \frac{30}{2} \left[\boxed{} + (30 - 1) \times \boxed{} \right] \quad \dots \text{ (Substituting the values)}$$

$$= 15 \left[20 + \boxed{} \right]$$

$$= 15 \times \boxed{}$$

$$\therefore S_{30} = \boxed{}.$$

3. The sum of the first 45 terms of an A.P. is 3195. Complete the following activity to find the 23rd term.

$$S_n = S_{45} = 3195$$

Let the first term of the A.P. be a and the common difference d .

$$S_n = \frac{n}{2} \left[2a + \boxed{} \right] \quad \dots \text{ (Formula)}$$

$$\therefore S_{45} = \frac{45}{2} \left[2a + \boxed{} \right] \quad \dots \text{ (Substituting the values)}$$

$$\therefore 3195 = \frac{45}{2} [2a + 44d]$$

$$\therefore 3195 = 45 \times \boxed{}$$

$$\therefore a + 22d = \boxed{}$$

$$\therefore a + 22d = 71 \quad \dots (1)$$

Now, 23rd term is t_{23} .

$$t_n = \boxed{} \quad \dots (\text{Formula})$$

$$\therefore t_{23} = a + (23 - 1)d$$

$$\therefore t_{23} = a + 22d$$

$$\therefore t_{23} = \boxed{}. \quad \dots [\text{From (1)}]$$

4. Divide 207 into three parts such that the numbers are in A.P. and the product of the two smaller parts is 4623.
5. The 11th term and the 21st term of an A.P. are 16 and 29 respectively. Find the 41st term of this A.P.
6. The 9th term of an A.P. is 40 more than the 29th term of the A.P. Find the common difference.
7. Find three consecutive terms in A.P. whose sum is -3 and the product of their cubes is 3375. (Take the value of d positive.)
8. The measure of the angles of a triangle are in A.P. The measure of the smallest angle is five times the common difference. Find the measure of all angles of the triangle. (March '22)

Answers :

$$1. \boxed{a + (n-1)d}, \boxed{(n-1) \times 8}, \boxed{8 + 8n}, \boxed{t_1 + t_n}, \boxed{16 + 96}, \boxed{616}.$$

$$2. \boxed{2a + (n-1)d}, \boxed{20}, \boxed{5}, \boxed{145}, \boxed{165}, \boxed{2475}.$$

$$3. \boxed{(n-1)d}, \boxed{(45-1)d}, \boxed{a + 22d}, \boxed{\frac{3195}{45}}, \boxed{a + (n-1)d}, \boxed{71}.$$

$$4. \text{ The required three parts of 207 are 67, 69 and 71.}$$



5. The 41st term is 55.
6. The common difference is -2 .
7. The three consecutive terms are -5 , -1 and 3 .
8. The measures of all angles of the triangle are 50° , 60° and 70° .

CHALLENGING QUESTIONS

3.5 4 marks each

1. A farmer borrows ₹1000 and agrees to repay with a total interest of ₹140, in 12 instalments. Each instalment being less than the preceding instalment by ₹10. What should be the amount of his first and last instalments? (March '25)

Solution :

The person repays $₹1000 + ₹140 = ₹1140$.

The number of instalments = 12

$$\therefore n = 12, S_n = S_{12} = 1140.$$

Each instalment is ₹10 less than the preceding one.

$$\therefore d = -10. \quad \dots \text{ (A constant number)}$$

This is an A.P.

$$S_n = \frac{n}{2} [2a + (n-1)d] \quad \dots \text{ (Formula)}$$

$$\therefore S_{12} = \frac{12}{2} [2a + (12-1) \times (-10)] \quad \dots \text{ (Substituting the values)}$$

$$\therefore 1140 = 6[2a + 11 \times (-10)]$$

$$\therefore 190 = 2a - 110 \quad \dots \text{ (Dividing both the sides by 6)}$$

$$\therefore 2a - 110 = 190$$

$$\therefore 2a = 190 + 110 \quad \therefore 2a = 300$$

$$\therefore a = \frac{300}{2} \quad \therefore a = 150 \quad \dots \text{ (The first instalment)}$$

t_n = last instalment

$$t_n = a + (n-1)d \quad \dots \text{ (Formula)}$$

$$= 150 + (12 - 1) \times (-10) \quad \dots \text{ (Substituting the values)}$$

$$= 150 + 11 \times (-10) = 150 - 110 = 40$$

Ans. The first instalment is ₹ 150 and the last instalment is ₹ 40.

- 2. How many terms of the A.P. 16, 14, 12, ... are needed to give the sum 60? Write these terms with all possibilities. (Sept. '21)**

Solution : Here, $S_n = 60$, $a = 16$, $d = -2$, $n = ?$

$$S_n = \frac{n}{2} [2a + (n - 1)d] \quad \dots \text{ (Formula)}$$

$$\therefore 60 = \frac{n}{2} [2 \times 16 + (n - 1) \times (-2)] \quad \dots \text{ [Substituting the values]}$$

$$\therefore 60 = \frac{n}{2} (32 - 2n + 2)$$

$$\therefore 60 = \frac{n}{2} (34 - 2n)$$

$$\therefore 60 = \frac{n}{2} \times 2 (17 - n)$$

$$\therefore 60 = n(17 - n)$$

$$\therefore 60 = 17n - n^2$$

$$\therefore n^2 - 17n + 60 = 0$$

$$\therefore n^2 - 5n - 12n + 60 = 0$$

$$\therefore n(n - 5) - 12(n - 5) = 0$$

$$\therefore (n - 5)(n - 12) = 0$$

$$\therefore n - 5 = 0 \quad \text{or} \quad n - 12 = 0$$

$$\therefore n = 5 \quad \text{or} \quad n = 12$$

The required terms are 5 or 12.

Taking $n = 5$, the first 5 terms are 16, 14, 12, 10, 8. [The sum is 60]

Taking $n = 12$, the terms are 16, 14, 12, 10, 8, 6, 4, 2, 0, -2, -4, -6.

[The sum is 60]

Ans. 5 or 12 terms. 5 terms : 16, 14, 12, 10, 8.

12 terms : 16, 14, 12, 10, 8, 6, 4, 2, 0, -2, -4, -6.



3. If the 9th term of an A.P. is zero, then show that the 29th term is twice the 19th term. (March '22)

Proof : Let the first term of the A.P. be a and the common difference d .

$$t_9 = 0 \quad \dots \text{(Given)}$$

$$t_n = a + (n - 1)d \quad \dots \text{(Formula)}$$

$$\therefore t_9 = a + (9 - 1)d$$

$$\therefore 0 = a + 8d \text{ i.e. } a + 8d = 0$$

$$\therefore a = -8d \quad \dots (1)$$

For 19th term,

$$t_{19} = a + (19 - 1)d$$

$$= a + 18d$$

$$= -8d + 18d \quad \dots \text{[From (1)]}$$

$$\therefore t_{19} = 10d \quad \dots (2)$$

For 29th term,

$$t_{29} = a + (29 - 1)d$$

$$= a + 28d$$

$$= -8d + 28d \quad \dots \text{[From (1)]}$$

$$\therefore t_{29} = 20d \quad \dots (3)$$

From (3) and (2),

$$t_{29} = 20d = 2 \times 10d = 2 \times t_{19}$$

$\therefore$ 29th term is twice the 19th term.

-
4. How many two-digit numbers leave remainder 1 when divided by 5.
5. Find the sum of all these numbers.

Solution : The first two-digit number divisible by 5 is $5 \times 2 = 10$

The last two-digit number divisible by 5 is $5 \times 19 = 95$. ($5 \times 20 = 100$ is a three-digit number)

We want to find two-digit numbers which would leave remainder 1 when divided by 5.

$\therefore$ the first two-digit number is $10 + 1 = 11$ and the last one is $95 + 1 = 96$.

Here $t_1 = a = 11$ and $t_n = 96$, $d = 5$.

$$t_n = a + (n - 1)d \quad \dots \text{(Formula)}$$

$$\therefore 96 = 11 + (n - 1) \times 5 \quad \dots \text{(Substituting the values)}$$

$$\therefore 96 - 11 = (n - 1) \times 5$$

$$\therefore 85 = (n - 1) \times 5$$

$$\therefore n - 1 = \frac{85}{5} \quad \therefore n - 1 = 17 \quad \therefore n = 17 + 1 \quad \therefore n = 18.$$

$$\text{Now, } S_n = \frac{n}{2} [t_1 + t_n] \quad \dots \text{(Formula)}$$

$$\therefore S_{18} = \frac{18}{2} [11 + 96] \quad \dots \text{(Substituting the values)}$$

$$= 9 \times 107$$

$$\therefore S_{18} = 963$$

Ans. There are **18** two-digit numbers which leave remainder 1 when divided by 5. The sum of all these numbers is **963**.

-
- 5. If p times the p th term of an A.P. is equal to q times the q th term, then show that $(p + q)$ th term of that A.P. is zero ($p \neq q$).**

(July '23)

Proof :

Let a be the first term and d be the common difference of the given A.P.

Then p times the p th term $= q$ times the q th term ... (Given)

$$\therefore pt_p = qt_q$$

$$\text{Now, } t_p = a + (p - 1)d \text{ and } t_q = a + (q - 1)d$$

$$\therefore p[a + (p - 1)d] = q[a + (q - 1)d]$$

$$\therefore p[a + (p - 1)d] - q[a + (q - 1)d] = 0$$

$$\therefore \underline{ap} + p(p - 1)d - \underline{qa} - q(q - 1)d = 0$$

$$\therefore \underline{ap} - \underline{qa} + p(p-1)d - q(q-1)d = 0$$

$$\therefore a(p-q) + (p^2-p)d - (q^2-q)d = 0$$

$$\therefore a(p-q) + d[p^2-p-q^2+q] = 0$$

$$\therefore a(p-q) + d[p^2-q^2-p+q] = 0$$

$$\therefore a(p-q) + d[(p+q)(p-q) - (p-q)] = 0$$

$$\therefore a(p-q) + d(p-q)(p+q-1) = 0$$

$$\therefore (p-q)[a + d(p+q-1)] = 0$$

$$\therefore a + d(p+q-1) = 0 \dots [\text{Dividing by } (p-q). \because (p-q) \neq 0] \dots (1)$$

$$\text{Now, } (p+q)\text{th term of the A.P.} = a + (p+q-1)d \dots (2)$$

$\therefore$ from (1) and (2),

$(p+q)$ th term of the A.P. is zero.

i.e. $t_{(p+q)} = 0$.

Assignment

3.5

1. In a 'Mahila Bachat Gat', Kavita invested ₹ 20 on the first day of month, ₹ 40 on the second day and ₹ 60 on the third day. If she saves like this, then what would be her total savings in the month of February 2020? (March '24)
2. If the sum of first p terms of an A.P. is equal to the sum of first q terms, then show that the sum of its first $(p+q)$ terms is zero. ($p \neq q$) (July '22)
3. Find the common difference of an A.P. whose first term is 5 and the sum of first four terms is half the sum of the next four terms.
4. One person borrows ₹ 4000 and agrees to repay with a total interest of ₹ 500 in 10 instalments. Each instalment being less than the preceding instalment by ₹ 10. What should be the first and the last instalments? (March '20)
5. Solve : $(-7) + (-4) + (-1) + \dots + (x) = 380$.

Answers :

1. Total savings ₹ 8700
3. $d = 2$
4. The first instalment is ₹ 495 and the last instalment is ₹ 405.
5. $x = 47$ is the solution.

Based on the Board's Evaluation Pattern

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Points To Remember :

- GST stands for **Goods and Service Tax**.
- GST is in effect from 1st July 2017.
- CGST and SGST are the two components of GST.
 - (1) **CGST** (Central Goods and Service Tax) : This is to be paid to the Central government.
 - (2) **SGST** (State Goods and Service Tax) : This is to be paid to the State government.
- **GSTIN** (Goods and Service Tax Identification Number) : It has 15 alphanumerals.
- **PAN** (Permanent Account Number) : It has 10 alphanumerals.
- The value of goods on which GST is levied is called **taxable value**.
- **Input Tax Credit (ITC)** : (1) When a trader pays GST at the time of purchase, it is called **Input tax**. (2) When the trader collects GST at the time of sale, it is called **Output tax**.
- GST payable by the trader = Output tax – ITC.
- **Share** : A share is the smallest unit of the capital.
- **Face Value (FV)** : The value printed on the share certificate is called the face value of the share.
- **Market Value (MV)** : The price at which the shares are sold or purchased in the stock market is called the market value of the share.
- **Dividend** : The part of annual profit of a company which is distributed per share among shareholders is called the dividend.
The dividend is always reckoned on the FV of a share.
 - (1) If $MV > FV$, then the share is at premium.
 - (2) If $MV = FV$, then the share is at par.
 - (3) If $MV < FV$, then the share is at discount.

- Brokerage is paid on the market value of the share.
- At the time of buying shares : Buying price of a share = MV + brokerage.
- At the time of selling shares : Selling price per share = MV – brokerage.

4.1 **1 mark each**

Q. Choose the correct alternative :

1. The tax levied by the Central government for trading within state is
 (A) IGST (B) CGST (C) SGST (D) UTGST **(March '19)**

Ans. (B)

2. How many alphanumerals are there in the GSTIN of a registered dealer?
(March '20–'24; Nov. '20)

(A) 15 (B) 10 (C) 16 (D) 9

Ans. (A)

3. The rate of GST on stainless steel utensils is 18%, then the rate of State GST is

(A) 18% (B) 9% (C) 36% (D) 0.9%

Ans. (B)

$$\left[\text{SGST} = \frac{1}{2} \times \text{GST.} \right]$$

4. What is the trading between GSTIN holder and the consumer known as?

(A) BB (B) B2B (C) BC (D) B2C

Ans. (D)

[B2C means Business to Consumer.]

5. The NAV of a unit in mutual fund scheme is ₹ 10.65. Find the amount required to buy 500 such units.

(A) ₹ 5325 (B) ₹ 5235 (C) ₹ 532500 (D) ₹ 53250

Ans. (A)

[Investment = The number of units × NAV.]

6. A share broker received a brokerage ₹ 7500. He paid ₹ 1350 as GST on brokerage. What is the rate of GST?

(A) 18% (B) 9% (C) 20% (D) 10%

Ans. (A)

$$\left[\text{The rate of GST} = \frac{\text{GST}}{\text{brokerage}} \times 100. \right]$$



Q. Choose the correct alternative :

1. Rate of GST on essential commodities is
(A) 5% (B) 12% (C) 0% (D) 18%
2. What is the amount of dividend received per share of face value ₹ 50 and dividend declared is 25%?
(A) ₹ 25 (B) ₹ 12.50 (C) ₹ 2.50 (D) ₹ 6.25
3. When a registered dealer sells goods to another registered dealer under GST, then this trading is termed as
(A) BB (B) B2B (C) BC (D) B2C
4. Rate of GST on brokerage is
(A) 5% (B) 12% (C) 18% (D) 28%
5. What is the rate of dividend, if a dividend of ₹ 24 is received on 3 shares of face value ₹ 100?
(A) 6% (B) 8% (C) 12% (D) 24%
6. What amount is received by selling 5 shares of FV ₹ 100 at MV ₹ 75?
(A) ₹ 375 (B) ₹ 500 (C) ₹ 125 (D) ₹ 875

Answers :

1. (C) 2. (B) 3. (B) 4. (C) 5. (B)
6. (A) [Amount received = Number of shares × MV]

4.2 1 mark each**Q. Answer the following questions :**

1. 'Pawan Medical' supplies medicines. On some medicines the rate of GST is 12%. What is the rate of CGST and SGST? **(March '20)**

Ans. The rate of SGST = the rate of CGST = half the rate of GST.
Hence, the rate of CGST and SGST is 6%.

2. If the rate of CGST on an article is 9%, what is the rate of SGST and GST on it? (July '23)

Ans. The rate of SGST is equal to the rate of CGST, which is 9%.

$$\begin{aligned}\text{The rate of GST} &= 2 \times \text{the rate of CGST} \\ &= 2 \times 9\% = \mathbf{18\%}.\end{aligned}$$

-
3. The face value of a share is ₹ 100 and the market value is ₹ 150. If the rate of brokerage is 2%, find the brokerage paid on one share. (March '24)

Ans. The brokerage = MV $\times$ the rate of brokerage

$$\begin{aligned}&= ₹ 150 \times \frac{2}{100} \\ &= ₹ 3.\end{aligned}$$

The brokerage paid on one share is ₹ 3.

-
4. How much GST is to be paid on an ice-cream pack of ₹ 200, if the rate of GST is 18%?

Ans. The GST to be paid = $\frac{18}{100} \times 200 = ₹ 36$.

Assignment	4.2
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Q. Answer the following questions :

1. What is the relation between CGST and GST?
2. If the CGST on an article is ₹ 45, what is SGST on it?
3. The taxable value of an article is ₹ 800. How much GST at 5% is to be paid?
4. Ajay Electronics paid GST ₹ 2000 at the time of purchase of an electrical item. He collected ₹ 2500 as GST from the retailer. How much is the output tax?

Answers :

- | | | | |
|--|---------|---------|------------|
| 1. $\text{CGST} = \frac{1}{2} \times \text{GST}$ | 2. ₹ 45 | 3. ₹ 40 | 4. ₹ 2500. |
|--|---------|---------|------------|



4.3 2 marks each

1. Dinesh Chemicals paid ₹ 1,80,000 GST for purchase of some chemicals and collected ₹ 2,50,000 GST at the time of sale. Complete the following activity to find the amount of GST payable by Dinesh Chemicals.

The tax collected at the time of sale is called output tax.

∴ tax in this example is ₹ 2,50,000.

The tax paid at the time of purchase is called input tax.

∴ input tax in this example is ₹ .

∴ ITC = ₹ 1,80,000

GST payable by Dinesh Chemicals = Output tax – ITC

$$= ₹ \text{ } - ₹ 180000 = ₹ \text{ }$$

Activity : The tax collected at the time of sale is called output tax.

∴ **output** tax in this example is ₹ 2,50,000.

The tax paid at the time of purchase is called input tax.

∴ input tax in this example is ₹ **1,80,000**.

∴ ITC = ₹ 1,80,000

GST payable by Dinesh Chemicals = Output tax – ITC

$$= ₹ \text{ **250000** } - ₹ 180000 = ₹ \text{ **70000** }.$$

2. Complete the following table using given information :

(July '23)

Sr. No.	FV	Share is at	MV
1.	₹ 10	Premium of ₹ 7	
2.	₹ 25		₹ 16
3.	₹ 300		₹ 315
4.		At par	₹ 5

Ans.

Sr. No.	FV	Share is at	MV
1.	₹ 10	Premium of ₹ 7	₹ 17
2.	₹ 25	Discount of ₹ 9	₹ 16
3.	₹ 300	Premium of ₹ 5	₹ 315
4.	₹ 5	At par	₹ 5

3. Pushpmala invested ₹ 24,000 and purchased shares of FV ₹ 20 at a premium of ₹ 4. Complete the following activity to find the number of shares she purchased. (March '25)

FV = ₹ 20, Premium = ₹ 4

$$MV = FV + \square$$

$$= 20 + \square = ₹ 24$$

$$\text{Number of shares} = \frac{\text{Total investment}}{MV} = \frac{24000}{\square} = \square \text{ shares.}$$

Activity : FV = ₹ 20, Premium = ₹ 4

$$MV = FV + \text{Premium}$$

$$= 20 + \square 4 = ₹ 24$$

$$\begin{aligned} \text{Number of shares} &= \frac{\text{Total investment}}{MV} = \frac{24000}{\square 24} \\ &= \square 1000 \text{ shares.} \end{aligned}$$

4. Mr Rohit is a retailer. He paid GST of ₹ 6500 at the time of purchase. He collected GST of ₹ 8000 at the time of sale.
- (a) Find the input tax and output tax.
- (b) What is his input tax credit?
- (c) Find his payable GST.
- (d) Hence find the payable CGST and payable SGST. (July '23)

Solution :

(a) Input tax (Tax paid at the time of purchase)

$$= ₹ 6500$$

Output tax (Tax collected at the time of sale)

$$= ₹ 8000$$

(b) Input tax credit (ITC) = ₹ 6500

(c) GST payable = Output tax – ITC

$$= ₹ (8000 - 6500)$$

$$= ₹ 1500$$

(d) Payable CGST = payable SGST

$$= \frac{1}{2} \times \text{GST} = \frac{1}{2} \times 1500$$

$$= ₹ 750.$$

Ans. (a) ₹ 6500; ₹ 8000 (b) ₹ 6500 (c) ₹ 1500 (d) ₹ 750.

5. A share is sold for the market value of ₹ 2000. Brokerage is paid at the rate of 0.5%. What is the amount received after the sale?

(March '25)

Solution :

MV = ₹ 2000, Brokerage = 0.5%.

Brokerage per share = Rate of brokerage × MV

$$= \frac{0.5}{100} \times 2000$$

$$= ₹ 10$$

Amount received after sale = MV – Brokerage

$$= ₹ (2000 - 10)$$

$$= ₹ 1990$$

Ans. The amount received after sale is ₹ 1990.

Assignment	4.3
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1. Complete the following table using given information : (March '23)

Sr. No.	FV	Share is at	MV
1.	₹ 100	At par	
2.		Premium ₹ 500	₹ 575
3.	₹ 10		₹ 5
4.	₹ 200	Discount ₹ 50	

- A shopkeeper sold a mixer costing ₹ 5000 to a consumer giving him 5% discount. What is the taxable value of the mixer?
- On a certain article, if the rate of CGST is 9%, then find the payable GST on article of printed price ₹ 700.
- Prashant purchased a share of FV ₹ 100 when MV was ₹ 150. Company declared a dividend of 12%. What was the rate of return on the investment Prashant made?
- Aseem invested ₹ 13,750 to buy shares of FV ₹ 100 at a premium of ₹ 25. How many shares did he buy?
- What amount is received by selling 150 shares of FV ₹ 100 at a premium of ₹ 25? Brokerage is paid at the rate of 0.3%.
- 25 shares are sold for the market value ₹ 500. Brokerage is paid at 0.1%. What is the amount received?

Answers :

- ₹ 100, ₹ 75, Discount ₹ 5, ₹ 150.
- The taxable value of the mixer is ₹ 4,750.
- The payable GST is ₹ 126.
- The rate of return on the investment made by Prashant is 8%.



5. Aseem purchased 110 shares.
6. The amount received : ₹ 18,693.50.
7. The amount received : ₹ 12,487.50.

4.4 3 marks each

1. The total value (with GST) of a remote controlled toy car is ₹ 2360. The rate of GST is 18% on toys. Complete the following activity to find the taxable value for the toy car. (March '24)

Total value of toy car with GST = ₹ 2360.

The rate of GST = 18%.

Let taxable value for toy car be ₹ x .

$$\therefore \text{GST} = \frac{18}{100} \times x$$

Total value for toy car = $\left(\begin{array}{c} \text{taxable value} \\ \text{for toy car} \end{array} \right) + \boxed{} \dots \text{(Formula)}$

$$\therefore 2360 = \boxed{} + \frac{\boxed{}}{100} \times x$$

$$\therefore 2360 = \frac{\boxed{}}{100} \times x$$

$$\therefore 2360 \times 100 = 118x$$

$$\therefore x = \frac{2360 \times 100}{\boxed{}}$$

$\therefore$ taxable value for toy car is ₹ $\boxed{}$.

Activity :

Total value of toy car with GST = ₹ 2360.

The rate of GST = 18%.

Let taxable value for toy car be ₹ x .

$$\therefore \text{GST} = \frac{18}{100} \times x$$

$$\text{Total value for toy car} = \left(\begin{array}{c} \text{taxable value} \\ \text{for toy car} \end{array} \right) + \boxed{\text{GST}} \quad \dots \text{ (Formula)}$$

$$\therefore 2360 = \boxed{x} + \frac{\boxed{18}}{100} \times x$$

$$\therefore 2360 = \frac{\boxed{118}}{100} \times x$$

$$\therefore 2360 \times 100 = 118x$$

$$\therefore x = \frac{2360 \times 100}{\boxed{118}}$$

$$\therefore \text{taxable value for toy car is ₹ } \boxed{2000}.$$

2. Smita invested ₹ 12,000 and purchased shares of FV ₹ 10 at a premium of ₹ 2. Find the number of shares she purchased. Complete the given activity to get the answer. (July '23)

$$\text{FV} = ₹ 10, \text{ Premium} = ₹ 2$$

$$\text{MV} = \text{FV} + \boxed{} = 10 + \boxed{} = \boxed{}$$

$$\begin{aligned} \therefore \text{Number of shares} &= \frac{\text{Total investment}}{\text{MV}} = \frac{12000}{\boxed{}} \\ &= \boxed{} \text{ shares} \end{aligned}$$

Smita purchased $\boxed{}$ shares.

Activity :

$$\text{FV} = ₹ 10, \text{ Premium} = ₹ 2$$

$$\text{MV} = \text{FV} + \boxed{\text{Premium}} = 10 + \boxed{2} = \boxed{12}$$

$$\begin{aligned} \text{Number of shares} &= \frac{\text{Total investment}}{\text{MV}} = \frac{12000}{\boxed{12}} \\ &= \boxed{1000} \text{ shares} \end{aligned}$$

Smita purchased $\boxed{1000}$ shares.



3. Smt Malhotra purchased solar panels for the taxable value ₹ 85,000. She sold them for ₹ 90,000. The rate of GST is 5%. Find the ITC of Smt Malhotra. What is the amount of GST payable by her? (March '19)

Solution : The tax paid at the time of purchase

$$= ₹ 85000 \times \frac{5}{100} = ₹ 4250$$

$$\therefore \text{input tax} = ₹ 4250. \quad \therefore \text{ITC} = ₹ 4250.$$

$$\text{Output tax} = ₹ 90000 \times \frac{5}{100} = ₹ 4500$$

$$\begin{aligned} \text{Payable GST} &= \text{Output tax} - \text{ITC} \\ &= ₹ (4500 - 4250) = ₹ 250 \end{aligned}$$

Ans. Smt Malhotra will get ₹ 4250 as ITC. She will have to pay ₹ 250 as GST.

-
4. A retailer sold 2 tins of lustre paint and taxable value of each tin is ₹ 2800. If the rate of GST is 28%, then find the amount of CGST and SGST charged in the tax invoice. (Nov. '20)

Solution :

Taxable value of a tin of lustre paint = ₹ 2800

Quantity = 2

$$\therefore \text{taxable amount} = ₹ 2800 \times 2 = ₹ 5600$$

The rate of GST = 28%

The rate of CGST = the rate of SGST

$$= \frac{1}{2} \times \text{the rate of GST}$$

$$= \frac{1}{2} \times 28\% = 14\%$$

The amount of CGST = the rate of CGST $\times$ taxable amount

$$= 14\% \text{ of } ₹ 5600 = \frac{14}{100} \times ₹ 5600 = ₹ 784$$

$\therefore$ the amount of SGST is also ₹ 784

Ans. The amount of CGST and SGST charged in the tax invoice is ₹ 784 each.

5. A ready-made garment shopkeeper gives 5% discount on a dress of ₹ 2000 and charges 5% GST on the remaining amount. What is the purchase price of the dress for the customer? (July '19)

Solution : Discount 5% on ₹ 2000 = $\frac{5}{100} \times 2000 = ₹ 100$

∴ taxable value of the dress = ₹ (2000 – 100) = ₹ 1900.

Rate of GST = 5%

∴ GST = $\frac{5}{100} \times 1900 = ₹ 95$

Purchase price for the customer = Taxable value + GST
= ₹ (1900 + 95)
= ₹ 1995

Ans. The purchase price of the dress for the customer is ₹ 1995.

Assignment	4.4
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1. Fill in the boxes with the help of given information : (March '23)

Tax invoice of services provided (Sample)								
Food Junction, Khed-Shivapur, Pune Invoice No. 58 Mob. No. 7588580000, email – ahar.khed@yahoo.com GSTIN : 27AAAAA5555B1ZA Invoice Date 25 Feb., 2020								
SAC	Food Items	Qty.	Rate (in ₹)	Taxable amount (₹)	CGST		SGST	
9963	Coffee	1	20	20.00	2.5%	₹ 0.50	2.5%	
9963	Masala Tea	1	10	10.00		₹ 0.25	2.5%	
9963	Masala Dosa	2	60		2.5%		2.5%	₹ 3.00
			Total	150.00				₹ 3.75
					Grand Total = ₹ 157.50			



- M/s Jay Chemicals purchased liquid soap having taxable value ₹ 8000 and sold it to the consumers for the taxable value ₹ 10,000. Rate of GST is 18%. Find (i) input tax (ii) output tax (iii) GST payable by M/s Jay Chemicals.
- 'Tip-Top Electronics' supplied 1.5 ton AC to a company. The cost of AC with GST is ₹ 51,200. The rate of GST on AC is 28%. Find (i) the taxable value of AC. (ii) Total amount of GST.
- 50 shares of face value ₹ 10 were purchased for market value of ₹ 25. Company declared 30% dividend on the shares, then find :
(i) Sum invested (ii) Dividend received (iii) Rate of return
(March '24)

Answers :

- ₹ 0.50, 2.5%, ₹ 0.25, 120.00, ₹ 3.00, ₹ 3.75
- (i) ₹ 1440 (ii) ₹ 1800 (iii) ₹ 360.
- (i) ₹ 40,000 (ii) ₹ 11,200.
- (i) ₹ 1250 (ii) ₹ 150 (iii) 12%.

CHALLENGING QUESTIONS

4.5 4 marks each

- Bhujangrao invested ₹ 2,50,590 in shares of FV ₹ 10 when MV is ₹ 250. The rate of brokerage is 0.2% and GST on brokerage is 18%. Find (a) the number of shares purchased, (b) the amount of brokerage paid and (c) GST paid for the trading. (Nov. '20)

Solution : MV = ₹ 250, Brokerage = 0.2%, GST = 18%

Brokerage = MV × Brokerage%

$$= ₹ 250 \times \frac{0.2}{100} = ₹ 0.50$$

GST on brokerage = Brokerage × GST%

$$= ₹ 0.50 \times \frac{18}{100} = ₹ 0.09$$

$$\begin{aligned}\text{Purchase price of 1 share} &= \text{MV} + \text{Brokerage} + \text{GST} \\ &= ₹(250 + 0.50 + 0.09) = ₹ 250.59\end{aligned}$$

$$\begin{aligned}\text{(a) The number of shares purchased} &= \frac{\text{Investment}}{\text{Purchase price of 1 share}} \\ &= \frac{250590}{250.59} = 1000\end{aligned}$$

$$\begin{aligned}\text{(b) Brokerage paid} &= \text{Number of shares} \times \text{Brokerage per share} \\ &= 1000 \times 0.50 = ₹ 500\end{aligned}$$

$$\begin{aligned}\text{(c) GST paid for the training} &= \text{Number of shares} \times \text{GST per share} \\ &= 1000 \times 0.09 = ₹ 90\end{aligned}$$

Ans. (a) 1000 shares (b) ₹ 500 (c) ₹ 90.

- 2. A wholesaler purchased a TV from the manufacturer and sold it to a retailer taking 25% profit. If the purchase price for the retailer is ₹ 52,215, find the cost price for the manufacturer. 18% GST is levied in each transaction.**

Solution : CP → Cost price; SP → Selling price

Let the CP for the manufacturer be ₹ x .

$$\text{GST} = 18\% \quad \therefore \text{GST} = \frac{18}{100} \times x = ₹ \frac{18x}{100} \quad \dots (1)$$

$$\therefore \text{SP for the manufacturer} = ₹ \left(x + \frac{18x}{100} \right) = ₹ \frac{118x}{100}$$

$$\text{i.e. CP for the wholesaler is } ₹ \frac{118x}{100} \quad \dots (2)$$

Wholesaler takes 25% profit while selling to a retailer.

$$\therefore \text{profit} = \frac{25}{100} \times \frac{118x}{100} = ₹ \frac{118x}{400} \quad \dots (3)$$

$$\therefore \text{profit including CP} = ₹ \left(\frac{118x}{100} + \frac{118x}{400} \right) \dots [\text{From (2) and (3)}] \dots (4)$$

GST = 18% (On profit including CP)

$$\therefore \text{GST} = \frac{18}{100} \times \left(\frac{118x}{100} + \frac{118x}{400} \right) \quad \dots (5)$$



$$\therefore \text{SP for the wholesaler} = \left(\frac{118x}{100} + \frac{118x}{400} \right) + \frac{18}{100} \times \left(\frac{118x}{100} + \frac{118x}{400} \right)$$

... [From (4) and (5)]

This is the CP for the retailer which is ₹ 52,215.

$$\begin{aligned} \therefore 52215 &= \left(\frac{118x}{100} + \frac{118x}{400} \right) + \frac{18}{100} \left(\frac{118x}{100} + \frac{118x}{400} \right) \\ &= \left(\frac{118x}{100} + \frac{118x}{400} \right) \left(1 + \frac{18}{100} \right) \\ &= \frac{472x + 118x}{400} \times \frac{118}{100} \end{aligned}$$

$$\therefore 52215 = \frac{590x}{400} \times \frac{118}{100}$$

$$\therefore x = \frac{52215 \times 400 \times 100}{590 \times 118} \quad \dots [52215 = 7.5 \times 118 \times 59]$$

$$\therefore x = 7.5 \times 400 \times 10 \quad \therefore x = 30000$$

Ans. The cost price for the manufacturer is ₹ 30,000.

Assignment

4.5

1. Sandeep Electronics purchased ceiling fans for ₹ 50,176 (with GST) and sold them to the customers for ₹ 56,000 (with GST). The rate of GST is 12%. Find the amount of CGST and SGST to be paid by Sandeep Electronics. Also find the taxable value in each case.
2. Manoj purchased 70 shares at MV ₹ 125. He sold half of the shares at MV ₹ 145 and the remaining shares at MV ₹ 120. He paid brokerage at the rate of 0.2% for each transaction. Find whether he made a profit or incurred a loss in this transaction. How much?

Answers :

1. CGST = SGST = ₹ 312 : Taxable values = ₹ 44,800 and ₹ 50,000
2. Profit ₹ 488.95

Question Set 5

PROBABILITY

Points To Remember :

- **Random Experiment** : An experiment in which all the possible results are known in advance, but none of them can be predicted with certainty is called a *random experiment*.
- **Outcome** : The result of the random experiment is called an *outcome*.
- **Sample Space** : The set of all possible outcomes of a random experiment is called the *sample space*. A sample space is denoted by S . The number of elements in the set S (i.e. number of sample points) is denoted by $n(S)$. The sample space is called a finite sample space, if $n(S)$ is finite.

Sr. No.	Random Experiment	Sample Space	Number of sample points
1.	A coin is tossed	$S = \{H, T\}$	$n(S) = 2$
2.	Two coins are tossed	$S = \{HH, HT, TH, TT\}$	$n(S) = 4$
3.	Three coins are tossed	$S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$	$n(S) = 8$
4.	A die is thrown	$S = \{1, 2, 3, 4, 5, 6\}$	$n(S) = 6$
5.	A card is drawn from the pack of 52 playing cards	$S = \{1, 2, 3, \dots, 51, 52\}$	$n(S) = 52$

- **Event** : A set of all favourable outcomes is an event. e.g. two coins are tossed. A is the event of getting at least one tail. Here, $A = \{HT, TH, TT\}$, $n(A) = 3$.
- **Probability of an Event** : The probability of an event A in a finite sample space S , is written as $P(A)$ and is defined as

$$P(A) = \frac{\text{Number of sample points in event } A}{\text{Number of sample points in sample space}} = \frac{n(A)}{n(S)}.$$

- If E is any event, $0 \leq P(E) \leq 1$ or $0\% \leq P(E) \leq 100\%$.

e.g. the probability $\frac{1}{4}$ is written as 25%.

- Two coins are tossed simultaneously. Write the number of sample points $n(S)$. (July '23)

- Ans. (C)** [A coin has two sample points : H, T.]

- (A) 1 (B) $\frac{1}{3}$ (C) $\frac{1}{2}$ (D) $\frac{1}{6}$

- Ans. (D)** [The number less than 2 is 1. $n(A) = 1$, $n(S) = 6$.]

- (A) 10 (B) $\frac{5}{2}$ (C) $\frac{2}{5}$ (D) $\frac{1}{3}$

- Ans. (A)** $\left[P(A) = \frac{n(A)}{n(S)} \quad \therefore n(S) = \frac{n(A)}{P(A)} \right]$

- (A) $\frac{1}{5}$ (B) $\frac{3}{5}$ (C) $\frac{4}{5}$ (D) $\frac{1}{3}$

- Ans. (A)** [There are 8 cards bearing a number multiple of 5.]

5. A coin is tossed and a die is rolled simultaneously. Write the number of sample points.

(A) 2 (B) 6 (C) 8 (D) 12

Ans. (D) [A coin has 2 sample points. A die has 6 sample points.
 $\therefore$ the number of sample points are $2 \times 6 = 12$.]

6. Two coins are tossed simultaneously. What is the probability of getting at least one tail?

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) $\frac{2}{3}$ (D) $\frac{3}{4}$

Ans. (D) $\left[S = \{HH, HT, TH, TT\}; \text{At least one tail} = \{HT, TH, TT\} \right.$
 $\left. \therefore \text{probability} = \frac{3}{4} \right]$

7. A die is rolled. What is the probability of getting an even number on the upper face?

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) $\frac{1}{3}$ (D) $\frac{3}{4}$

Ans. (B) [2, 4, 6 are even numbers on the die.]

8. A card is drawn at random from a well-shuffled pack of 52 playing cards. What is the probability of getting a black card?

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) $\frac{3}{4}$ (D) $\frac{1}{13}$

Ans. (B) [Out of 52 playing cards, there are 26 black cards.]

Assignment

5.1

Q. Choose the correct alternative :

1. What is the probability of the event that a number chosen from 1 to 100 is a prime number?

(A) $\frac{1}{5}$ (B) $\frac{6}{25}$ (C) $\frac{1}{4}$ (D) $\frac{13}{50}$



2. The probability of failure is 0.25%. What is the probability of success?
(A) 0.9975 (B) 99.75 (C) 0.75 (D) 0.075
3. A die is rolled. What is the probability that the number appearing on the upper face is a multiple of 4?
(A) $\frac{1}{6}$ (B) $\frac{1}{3}$ (C) $\frac{1}{2}$ (D) $\frac{2}{3}$
4. A card is drawn at random from a well-shuffled pack of 52 playing cards. What is the probability of getting **[Each 1 mark]**
- (i) a red card?
(A) $\frac{1}{52}$ (B) $\frac{1}{26}$ (C) $\frac{1}{2}$ (D) $\frac{1}{4}$
- (ii) a spade?
(A) $\frac{1}{52}$ (B) $\frac{1}{26}$ (C) $\frac{1}{2}$ (D) $\frac{1}{4}$
- (iii) a queen or a jack?
(A) $\frac{1}{13}$ (B) $\frac{2}{13}$ (C) $\frac{3}{13}$ (D) $\frac{4}{13}$

Answers :

1. (C) $\left[\begin{array}{l} \text{There are 25 prime numbers from the numbers 1 to 100.} \\ \therefore n(A) = 25, n(S) = 100 \therefore \text{the probability} = \frac{25}{100} = \frac{1}{4}. \end{array} \right]$
2. (A) $[100\% - 0.25\% = 99.75\% = 0.9975$
(Converted into decimal fraction).]
3. (A) [4 is the only multiple of 4.]
4. (i) (C) [There are 26 red cards] (ii) (D) [There are 13 cards of spade.]
(iii) (B) [There are 4 cards each of queen and jack.]

5.2 1 mark each

Q. Answer the following questions :

1. Two coins are tossed simultaneously. Write the sample space S.
(Sept. '21; March '22 – '25)

Ans. Sample space $S = \{HH, HT, TH, TT\}$.

2. If $n(A) = 1$, $n(S) = 2$, what is $P(A)$? (July '23)

Ans. $P(A) = \frac{n(A)}{n(S)} \quad \therefore P(A) = \frac{1}{2}$.

3. A coin is tossed. Write the sample space S . (March '23)

Ans. $S = \{H, T\}$.

4. If $P(A) = \frac{4}{9}$, $n(S) = 36$, what is $n(A)$?

Ans. $P(A) = \frac{n(A)}{n(S)} \quad \therefore \frac{4}{9} = \frac{n(A)}{36} \quad \therefore n(A) = \frac{4 \times 36}{9} \quad \therefore n(A) = 16$.

5. If $P(A) = \frac{3}{4}$, $n(A) = 24$, what is $n(S)$?

Ans. $n(S) = \frac{n(A)}{P(A)} \quad \therefore n(S) = 24 \times \frac{4}{3} \quad \therefore n(S) = 32$.

Assignment

5.2

Q. Answer the following questions :

1. A die is rolled. A is the event of getting a number more than 3. Write event A .
2. A die is rolled. What is the probability of getting a number less than 4 on the upper face?
3. A card is drawn at random from a well-shuffled pack of 52 playing cards. What is the probability of getting (i) a king (ii) a black card?
4. Write the number of sample points $[n(S)]$, when **[Each 1 mark]**
 - (i) three coins are tossed
 - (ii) a coin and a die are tossed. **[Each 1 mark]**
5. Two-digit numbers are formed using the digits 2, 3 and 5 without repeating a digit. Write the sample space. **(Nov. '20; March '24)**

Answers :

1. $A = \{4, 5, 6\}$ 2. $\frac{1}{2}$ 3. (i) $\frac{1}{13}$ (ii) $\frac{1}{2}$
4. (i) $n(S) = 8$ (ii) $n(S) = 12$ 5. $S = \{23, 25, 32, 35, 52, 53\}$.
-



5.3 2 marks each

1. A die is rolled. Complete the following activity to find the probability of getting a prime number on the upper face of the die.

(Sept. '21; March '22; July '22)

A die is rolled. S is the sample space.

$$\therefore S = \left\{ \boxed{} \right\}, \quad \therefore n(S) = 6$$

Event A : To get prime number on the upper face.

$$\therefore A = \left\{ \boxed{} \right\}, \quad \therefore n(A) = 3$$

$$P(A) = \frac{\boxed{}}{n(S)} \quad \dots \text{(Formula)}$$

$$\therefore P(A) = \boxed{}$$

Activity :

A die is rolled. S is the sample space.

$$\therefore S = \left\{ \boxed{1, 2, 3, 4, 5, 6} \right\}, \quad \therefore n(S) = 6$$

Event A : To get prime number on the upper face.

$$\therefore A = \left\{ \boxed{2, 3, 5} \right\}, \quad \therefore n(A) = 3$$

$$P(A) = \frac{n(A)}{n(S)} \quad \dots \text{(Formula)}$$

$$\therefore P(A) = \boxed{\frac{1}{2}}$$

2. Two coins are tossed simultaneously. Complete the following activity of writing the sample space (S) and the given events A and B in the set form :

(i) Event A : to get at least one head

(ii) Event B : to get no head.

(March '19 – '24)

If two coins are tossed simultaneously,

$$S = \left\{ \boxed{}, HT, TH, \boxed{} \right\}$$

(i) Event A : to get at least one head $\therefore A = \left\{ HH, \boxed{}, TH \right\}$

(ii) Event B : to get no head $\therefore B = \left\{ \boxed{} \right\}$

Activity : If two coins are tossed simultaneously.

$$S = \left\{ \boxed{HH}, HT, TH, \boxed{TT} \right\}$$

(i) Event $A : A = \left\{ HH, \boxed{HT}, TH \right\}$ (ii) Event $B : B = \left\{ \boxed{TT} \right\}$.

-
- 3. A box contains 5 red, 8 blue and 3 green pens. Rutuja wants to pick a pen at random. What is the probability that the pen picked up is blue? (March '22)**

Solution : $S = \{ 5 \text{ red} + 8 \text{ blue} + 3 \text{ green pens} \}$

$$\therefore n(S) = 16$$

Let A be the event of picking up a blue pen.

$$\text{Then } n(A) = 8$$

$$P(A) = \frac{n(A)}{n(S)} \quad \therefore P(A) = \frac{8}{16} \quad \therefore P(A) = \frac{1}{2}$$

Ans. The probability of picking up a blue pen is $\frac{1}{2}$.

-
- 4. Two coins are tossed simultaneously. Find the probability of the event of getting no head. (July '22)**

Solution : Let S be the sample space.

$$\text{Then } S = \{ HH, HT, TH, TT \} \quad \therefore n(S) = 4$$

Let A be the event of getting no head.

$$\therefore A = \{ TT \}. \quad \therefore n(A) = 1.$$

$$P(A) = \frac{n(A)}{n(S)} \quad \therefore P(A) = \frac{1}{4}.$$

Ans. The probability of getting no head is $\frac{1}{4}$.



- There are 9 tickets in a box, each bearing one of the numbers from 1 to 9. One ticket is drawn at random from the box.

Event A : Ticket shows an even number.

(July '19)

Complete the following activity :

$$S = \left\{ \boxed{} \right\}, n(S) = \boxed{}, A = \left\{ \boxed{} \right\}, n(A) = \boxed{}$$

- Fifty cards bearing numbers 1 to 50 are placed in a box. One card is drawn at random. Complete the following activity to find the probability of the event A that the number on the card is divisible by 11.

$$S = \{1, 2, 3, \dots, 49, 50\}; n(S) = \boxed{}$$

$$A = \left\{ \boxed{} \right\}; \quad n(A) = \boxed{}$$

$$P(A) = \frac{n(A)}{n(S)} = \boxed{}$$

- In each of the following experiments, write the sample space (S), the number of sample points $n(S)$, events R and T using set form, $n(R)$ and $n(T)$:

Two-digit numbers, using the digits 6, 7, 8 are formed without repeating the digits.

(i) R is the event that the number is divisible by 2.

(ii) T is the event that the number is a prime number. **[Each 2 marks]**

- The faces of a die bear the numbers 1, 3, 5, 7, 9, 11. The die is rolled. Find the probability of getting a perfect square number on the upper face of the die.
- A box contains 5 strawberry chocolates, 6 coffee chocolates and 2 peppermint chocolates. Find the probability that the chocolate picked up at random from the box is a coffee chocolate. **(Sept. '21)**

Answers :

$$1. \left\{ \boxed{1, 2, 3, 4, 5, 6, 7, 8, 9} \right\}, \boxed{9}, \left\{ \boxed{2, 4, 6, 8} \right\}, \boxed{4}.$$

$$2. \boxed{50} \left\{ \boxed{11, 22, 33, 44} \right\}; \boxed{4} \boxed{\frac{2}{25}}.$$

$$3. S = \{67, 68, 76, 78, 86, 87\}; n(S) = 6$$

$$(i) R = \{68, 76, 78, 86\}; n(R) = 4 \quad (ii) T = \{67\}; n(T) = 1.$$

$$4. \text{Probability } \frac{1}{3}. \quad 5. \text{Probability } \frac{6}{13}.$$

5.4 3 marks each

1. A two-digit number is to be formed from the digits 2, 3, 5 without repetition of the digits. Complete the following activity to find the probability that the number so formed is an odd number.

(Sept. '21; March '25)

Let S be the sample space.

$$\therefore S = \left\{ 23, 25, 32, \boxed{}, 52, 53 \right\} \quad \therefore n(S) = \boxed{}$$

Now condition of event A is that number so formed is an odd number.

$$\therefore A = \left\{ 23, 25, \boxed{}, 53 \right\} \quad \therefore n(A) = 4$$

$$\therefore P(A) = \frac{\boxed{}}{n(S)}$$

... (Formula)

$$\therefore P(A) = \frac{\boxed{}}{6} \quad \therefore P(A) = \frac{\boxed{}}{3}$$

Activity : Let S be the sample space.

$$\therefore S = \left\{ 23, 25, 32, \boxed{35}, 52, 53 \right\}$$

$$\therefore n(S) = \boxed{6}$$

Now condition of event A is that number so formed is an odd number.

$$\therefore A = \left\{ 23, 25, \boxed{35}, 53 \right\} \quad \therefore n(A) = 4$$



$$P(A) = \frac{n(A)}{n(S)} \quad \dots \text{(Formula)}$$

$$\therefore P(A) = \frac{4}{6} \quad \therefore P(A) = \frac{2}{3}.$$

2. If one die is rolled once, then find the probability of each of the following events :

(a) Number on the upper face is prime.

(b) Number on the upper face is even. (July '23)

S is the sample space.

$$S = \{1, 2, 3, 4, 5, 6\} \therefore n(S) = \boxed{}$$

(a) Event A : Prime number on the upper face.

$$A = \{2, 3, 5\} \therefore n(A) = \boxed{}$$

$$P(A) = \frac{n(A)}{n(S)} \quad \dots \text{(Formula)}$$

$$\therefore P(A) = \frac{3}{\boxed{}} = \boxed{}.$$

(b) Event B : Even number on the upper face.

$$B = \{2, 4, 6\} \therefore n(B) = \boxed{}$$

$$P(B) = \frac{n(B)}{n(S)} \quad \dots \text{(Formula)}$$

$$\therefore P(B) = \frac{\boxed{}}{\boxed{}} = \frac{1}{2}.$$

Activity :

S is the sample space.

$$S = \{1, 2, 3, 4, 5, 6\} \therefore n(S) = \boxed{6}$$

(a) Event A : Prime number on the upper face.

$$A = \{2, 3, 5\} \therefore n(A) = \boxed{3}$$

$$P(A) = \frac{n(A)}{n(S)} \quad \dots \text{(Formula)}$$

$$\therefore P(A) = \frac{3}{\boxed{6}} = \boxed{\frac{1}{2}}.$$

(b) Event B : Even number on the upper face.

$$B = \{2, 4, 6\}$$

$$\therefore n(B) = \boxed{3}$$

$$P(B) = \frac{n(B)}{n(S)}$$

... (Formula)

$$\therefore P(B) = \frac{\boxed{3}}{\boxed{6}} = \frac{1}{2}.$$

3. A card is drawn at random from a pack of well-shuffled 52 playing cards. Complete the following activity to find the probability that the card drawn is

(i) an ace (Event A) (ii) a spade (Event B). (March '22)

S is the sample space.

$$\therefore n(S) = 52$$

(i) Event A : The card drawn is an ace.

$$\therefore n(A) = \boxed{}$$

$$P(A) = \frac{\boxed{}}{\boxed{}}$$

... (Formula)

$$\therefore P(A) = \frac{\boxed{}}{52} \quad \therefore P(A) = \frac{\boxed{}}{13}$$

(ii) Event B : The card drawn is a spade.

$$\therefore n(B) = \boxed{}$$

$$P(B) = \frac{n(B)}{n(S)} \quad \therefore P(B) = \frac{\boxed{}}{4}$$

Activity :

S is the sample space.

$$\therefore n(S) = 52$$

(i) Event A : The card drawn is an ace.

$$\therefore n(A) = \boxed{4}$$

$$P(A) = \frac{n(A)}{n(S)}$$

... (Formula)

$$\therefore P(A) = \frac{\boxed{4}}{52} \quad \therefore P(A) = \frac{\boxed{1}}{13}$$



(ii) Event B : The card drawn is a spade.

$$\therefore n(B) = \boxed{13}$$

$$P(B) = \frac{n(B)}{n(S)} \therefore P(B) = \frac{\boxed{1}}{4}.$$

4. A coin is tossed and a die is rolled simultaneously. Find the probability of the following events :

(i) Event A : To get a head and a prime number.

(ii) Event B : To get a tail and an odd number. (March '24)

Solution : A coin is tossed and a die is rolled simultaneously.

$\therefore$ the sample space $S = \{H1, H2, H3, H4, H5, H6, T1, T2, T3, T4, T5, T6\}$

$$\therefore n(S) = 12.$$

Event A : To get a head and a prime number.

$$\therefore A = \{H2, H3, H5\} \therefore n(A) = 3$$

$$P(A) = \frac{n(A)}{n(S)} \therefore P(A) = \frac{3}{12} \therefore P(A) = \frac{1}{4}.$$

Event B : To get a tail and an odd number.

$$\therefore B = \{T1, T3, T5\} \therefore n(B) = 3$$

$$P(B) = \frac{n(B)}{n(S)} \therefore P(B) = \frac{3}{12} \therefore P(B) = \frac{1}{4}.$$

Ans. The probability of event A is $\frac{1}{4}$ and the probability of event B is $\frac{1}{4}$.

5. Two-digit numbers are formed using digits 2, 3, 5, 7, 9 without repetition. What is the probability of the following events?

Event A : The number formed is an odd number.

(2 marks) (March '20)

Event B : The number formed is a multiple of 5. (For both events.)

(March '23)

Solution : The sample space is

$$S = \{23, 25, 27, 29, 32, 35, 37, 39, 52, 53, 57, 59, 72, 73, 75, 79, 92, 93, 95, 97\}$$

$$\therefore n(S) = 20.$$

Event A : The number formed is an odd number.

Then $A = \{23, 25, 27, 29, 35, 37, 39, 53, 57, 59, 73, 75, 79, 93, 95, 97\}$

$$\therefore n(A) = 16$$

$$P(A) = \frac{n(A)}{n(S)} \quad \therefore P(A) = \frac{16}{20} \quad \therefore P(A) = \frac{4}{5}$$

Event B : The number formed is a multiple of 5.

Then $B = \{25, 35, 75, 95\}$

$$\therefore n(B) = 4$$

$$P(B) = \frac{n(B)}{n(S)} \quad \therefore P(B) = \frac{4}{20} \quad \therefore P(B) = \frac{1}{5}$$

Ans. The probability of event A is $\frac{4}{5}$. The probability of event B is $\frac{1}{5}$.

6. A bag contains 8 red and some blue balls. One ball is drawn at random from the bag. If the ratio of the probability of getting red ball and blue ball is 2 : 5, then find the number of blue balls.

(March '22)

Solution : Let the number of blue balls be x .

$$\therefore n(B) = x.$$

The number of red balls = 8 ... (Given)

$$\therefore n(R) = 8.$$

Total number of balls = $(x + 8)$ $\therefore n(S) = (x + 8)$.

The probability of getting a red ball,

$$P(R) = \frac{n(R)}{n(S)} \quad \therefore P(R) = \frac{8}{x+8} \quad \dots (1)$$

The probability of getting a blue ball,

$$P(B) = \frac{n(B)}{n(S)} = \frac{x}{x+8} \quad \dots (2)$$

$$\text{From the given condition, } \frac{P(R)}{P(B)} = \frac{2}{5} \quad \dots (3)$$

$$\therefore \frac{8}{x+8} \div \frac{x}{x+8} = 2 \div 5 \quad \dots [\text{From (1), (2) and (3)}]$$

$$\therefore \frac{8}{x} = \frac{2}{5} \quad \therefore 2x = 8 \times 5 \quad x = 4 \times 5 \quad \therefore x = 20$$

Ans. The number of blue balls is 20.



7. Six faces of a die are as shown below :

A **B** **C** **D** **E** **O**.

If the die is rolled once, find the probability of event M that the English vowel appears on the upper face. (Nov. '20)

Solution : $S = \{\text{A} \text{ B} \text{ C} \text{ D} \text{ E} \text{ O}\}$. $\therefore n(S) = 6$

$M = \{A, E, O\}$ $\therefore n(M) = 3$

$$P(M) = \frac{n(M)}{n(S)} \quad \therefore P(M) = \frac{3}{6} = \frac{1}{2}$$

Ans. The probability is $\frac{1}{2}$.

Assignment

5.4

1. A two-digit number is to be formed from the digits 2, 3, 5 without repetition of the digits. Complete the following activity to find the probability that the number so formed is an odd number. (March '25)
Let S be the sample space.

$$\therefore S = \{23, 25, 32, \boxed{}, 52, 53\}$$

$$\therefore n(S) = \boxed{}$$

Event A : The number so formed is an odd number.

$$\therefore A = \{23, 25, \boxed{}, 53\} \quad \therefore n(A) = 4$$

$$P(A) = \frac{\boxed{}}{n(S)} \quad \text{..... (Formula)}$$

$$\therefore P(A) = \frac{\boxed{}}{6} \quad \therefore P(A) = \frac{\boxed{}}{3}.$$

2. A card is drawn at random from a well-shuffled pack of 52 playing cards. Complete the following activity to find the probability of the events that the card drawn is (i) a heart (ii) not a face card.

The sample space S contains 52 sample points.

$$\therefore n(S) = 52.$$

(i) Let A be the event of getting a heart card.

There are $\boxed{}$ cards of heart. $\therefore n(A) = \boxed{}$

$$P(A) = \frac{n(A)}{n(S)} \quad \therefore P(A) = \frac{\boxed{}}{\boxed{}}$$

(ii) Let B be the event of not getting a face card.

There are cards which are not face cards. $\therefore n(B) =$

$$P(B) = \frac{n(B)}{n(S)} \quad \therefore P(B) = \frac{\text{---}}{\text{---}}$$

3. A committee of two members is to be formed from three boys and two girls. Find the probability of the following events : **(March '25)**
Event A : At least one girl must be a member of the committee.
Event B : Committee must be of one boy and one girl.
4. A die is rolled and a coin is tossed simultaneously. Write the sample space S and the number of sample points $n(S)$. Write the events A and B in set form and their number of sample points according to the given condition.
(a) Condition for event A : To get a head or tail on the coin and a number divisible by 3 on the upper face of the die.
(b) Condition for event B : To get a number on the upper face of the die greater than 7 and a head on the coin. **(Sept. '21)**
5. Two-digit numbers are formed from the digits 0, 1, 2, 3. Repetition of the digits are allowed. Find the probability that the number so formed is a prime number. **(March '19)**
6. A student made a cube-shaped die from a card sheet. Instead of writing numbers 1, 2, 3, 4, 5, 6 on its faces, he wrote letters a, b, c, d, e, f one on each face. He rolled the die twice. Find the probability that he got a vowel on the upper faces of both the times. **(July '19)**
7. A bag contains 3 red, 3 white, 3 green and 3 black balls. One ball is taken out of the bag at random. What is the probability that the ball picked up is (i) white (ii) not white? **(July '19)**

Answers :

1. 35, 6, 35, $n(A)$, 4, 2.

2. (i) 13, 13, $\frac{1}{4}$. (ii) 40, 40, $\frac{10}{13}$.

3. The probability of event A is $\frac{7}{10}$ and the probability of event B is $\frac{3}{5}$.



$$4. S = \{H1, H2, H3, H4, H5, H6, T1, T2, T3, T4, T5, T6\}$$

$$\therefore n(S) = 12.$$

(a) Event A : To get a head or a tail on the coin and a number divisible by 3 on the upper face of the die.

$$A = \{H3, H6, T3, T6\}; \quad \therefore n(A) = 4$$

(b) Event B : To get a number on the upper face of the die greater than 7 and head on the coin.

There is no number greater than 7 on a die.

$$\therefore B = \{ \quad \} \quad \therefore n(B) = 0.$$

$$5. \text{ The probability is } \frac{1}{3}.$$

$$6. \frac{1}{9}.$$

$$7. \text{ (i) } \frac{1}{4} \quad \text{(ii) } \frac{3}{4}.$$

CHALLENGING QUESTIONS

5.5 4 marks each

1. A game of chance consist of spinning an arrow which comes to rest pointing at one of the numbers 15, 20, 25, 30, 35, 40, 45, 50 and these are equally likely outcomes. What is the probability that it will point at

(i) an even number ?

(ii) at a number multiple of 5

(iii) at a number divisible by 3

(iv) at a prime number.

Solution : The sample space is

$$S = \{15, 20, 25, 30, 35, 40, 45, 50\}$$

$$\therefore n(S) = 8.$$

- (i) Let A be the event that the arrow points at an even number.

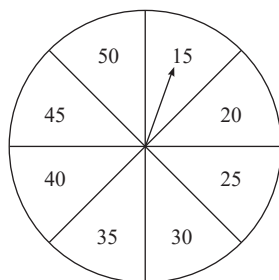
$$\text{Then } A = \{20, 30, 40, 50\} \quad \therefore n(A) = 4$$

$$P(A) = \frac{n(A)}{n(S)} \quad \therefore P(A) = \frac{4}{8} = \frac{1}{2}$$

- (ii) Let B be the event that the arrow points at a number multiple of 5.

$$\text{Then } B = \{15, 20, 25, 30, 35, 40, 45, 50\} \quad \therefore n(B) = 8$$

$$P(B) = \frac{n(B)}{n(S)} \quad \therefore P(B) = \frac{8}{8} = 1$$



- (iii) Let C be the event that the arrow points at a number divisible by 3.

$$\text{Then } C = \{15, 30, 45\} \quad \therefore n(C) = 3$$

$$P(C) = \frac{n(C)}{n(S)} \quad \therefore P(C) = \frac{3}{8}$$

- (iv) Let D be the event that the arrow points at a prime number.

$$\text{Then } D = \{\quad\}, \quad \therefore n(D) = 0$$

$$P(D) = \frac{n(D)}{n(S)} \quad \therefore P(D) = \frac{0}{8} = 0$$

Ans. (i) $\frac{1}{2}$ (ii) 1 (iii) $\frac{3}{8}$ (iv) 0.

- 2. There are 60 balls, all of the same size, in a bag. Some of them are green, some red and some are blue. The number of blue balls are 13 times the number of green balls. The number of red balls is less than the number of blue balls but more than the number of green balls. A ball is picked up at random from the bag. What is the probability that it is blue?**

Solution : The number of green balls < the number of red balls < the number of blue balls.

Green balls (number)	Blue balls (number)	Red balls (number)	Possibility
1	13	$60 - 14 = 46$	Not possible
2	26	$60 - 28 = 32$	Not possible
3	39	$60 - 42 = 18$	Possible
4	52	$60 - 56 = 4$	Not possible

So the number of green balls = 3,
the number of blue balls = 39,
the number of red balls = 18.



The probability of picking up a blue ball,

$$P(B) = \frac{n(B)}{n(S)}$$

$$\therefore P(B) = \frac{39}{60}$$

$$= \frac{13}{20}$$

... [$n(S)$ = total number of balls]

Ans. The probability of picking up a blue ball is $\frac{13}{20}$.

Assignment

5.5

- Two-digit numbers are formed from the digits 0, 1, 2, 3, 4 where repetition of the digits are allowed. Find the probability of the following events : (a) Event A : The number is composite, (b) Event B : The number is a multiple of 7, (c) Event C : The number is odd.
- A coin is tossed and a die is rolled simultaneously.
 - A is the event of getting a head and an odd number.
 - B is the event of getting either H or T and an even number.
 - C is the event of getting a tail and a prime number.

Find the probability for each event.

Answers :

- $n(S) = 20$; (a) 14 composite numbers. $\therefore n(A) = 14$; $P(A) = \frac{7}{10}$.
 (b) Multiples of 7 are 3. $\therefore n(B) = 3$; $P(B) = \frac{3}{20}$.
 (c) Odd numbers are 8. $\therefore n(C) = 8$; $P(C) = \frac{2}{5}$.
- $n(S) = 12$; (i) $n(A) = 3$; $P(A) = \frac{1}{4}$.
 (ii) $n(B) = 6$; $P(B) = \frac{1}{2}$. (iii) $n(C) = 3$; $P(C) = \frac{1}{4}$.

**Question
Set
6**

STATISTICS

Points To Remember :

- Mean
The mean of the statistical data = $\frac{\text{The sum of all scores}}{\text{Total number of scores}} = \frac{\sum_{i=1}^N x_i}{N}$

- Mean from classified frequency distribution :

(A) Direct method : Mean = $\bar{X} = \frac{\sum x_i f_i}{N}$

- (B) Assumed mean method :

$$d_i = x_i - A.$$

(d_i : deviation; x_i : class mark; A : assumed mean)

$$\bar{d} = \frac{\sum f_i d_i}{\sum f_i} \quad \bar{X} = A + \bar{d}$$

- (C) Step deviation method :

$$d_i = x_i - A$$

g , the GCD of all d_i

$$u_i = \frac{x_i - A}{g} = \frac{d_i}{g}$$

$$\bar{u} = \frac{\sum f_i u_i}{\sum f_i} \quad \bar{X} = A + \bar{u}g$$

- Median from grouped frequency distribution

$$\text{Median} = L + \left[\frac{\frac{N}{2} - cf}{f} \right] \times h$$

L = lower class limit of the median class.

N = sum of frequencies.

cf = cumulative frequency of the class preceding the median class.

f = frequency of the median class.

h = class interval of the median class.

- Mode from grouped frequency distribution :

$$\text{Mode} = L + \left[\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right] \times h$$

L = lower class limit of the modal class.

f_1 = frequency of the modal class.

f_0 = frequency of the class preceding the modal class.

f_2 = frequency of the class succeeding the modal class.

h = class interval of the modal class.

- Histogram and frequency polygon are the graphical methods of data presentation.
- Pie diagram is the pictorial presentation of the numerical data.
- For drawing a histogram the classes must be continuous (exclusive type).
- For drawing a frequency polygon, plot two additional points, one preceding the first class with frequency zero and the other succeeding the last class with frequency zero.
- For drawing a pie diagram, first find the measure of the central angle for each item or component using the following formula :

$$\text{Central angle} = \frac{\text{Value of the item or component}}{\text{Total value of all the items or components}} \times 360^\circ.$$

6.1 1 mark each

Q. Choose the correct alternative :

1. The formula to find mean from a grouped frequency table is

$$\bar{X} = A + \frac{\sum f_i u_i}{\sum f_i} \times g. \text{ In the formula, } u_i = \dots$$

- (A) $\frac{x_i + A}{g}$ (B) $(x_i - A)$ (C) $\frac{x_i - A}{g}$ (D) $\frac{A - x_i}{g}$

Ans. (C)

2. Cumulative frequencies in a grouped frequency table are useful to find

(A) mean (B) median (C) mode (D) all of these

Ans. (B)

3. The persons of O-blood group are 40%. The classification of persons based on blood groups is to be shown by a pie diagram. What should be the measure of central angle for the persons of O-blood group?

(A) 114° (B) 140° (C) 104° (D) 144°

Ans. (D)

$$\left[\frac{40\%}{100\%} \times 360^\circ \right]$$

Assignment

6.1

Q. Choose the correct alternative :

1. Find the value of the mean $\bar{X}$, if $\Sigma f_i x_i = 30$ and $\Sigma f_i = 6$.
- (A) $\frac{1}{5}$ (B) 5 (C) 36 (D) 24
2. The class mark of the class 2000–2500 is taken as the assumed mean. What is the valuation of the deviation (d_i) for the class 2500–3000?
- (A) – 500 (B) 500 (C) – 1000 (D) 1000

Answers :

1. (B) $\left[\bar{X} = \frac{\Sigma f_i x_i}{\Sigma f_i} \right]$

2. (B) [2250 → A. Width of the class 500. Class mark of the class 2500–3000 is 2750. $d_i = x_i$ (class mark) – A]

6.2 1 mark each

Q. Answer the following questions :

1. Find the mean ($\bar{X}$) in the following :
- $\Sigma f_i = 20$, $\Sigma f_i x_i = 50$.



$$\text{Ans. } \bar{X} = \frac{\sum f_i x_i}{\sum f_i} = \frac{50}{20} = \frac{5}{2}.$$

2. Find $\bar{d}$, if $\sum f_i d_i = 108$ and $\sum f_i = 100$. (Nov. '20)

$$\text{Ans. } \bar{d} = \frac{\sum f_i d_i}{\sum f_i} = \frac{108}{100} \quad \therefore \bar{d} = \mathbf{1.08}.$$

3. Find $\sum f_i$, if $\bar{u} = -0.1$ and $\sum f_i u_i = -5$.

$$\text{Ans. } \bar{u} = \frac{\sum f_i u_i}{\sum f_i} \quad \therefore -0.1 = \frac{-5}{\sum f_i} \quad \therefore \sum f_i = \frac{-5}{-0.1} \quad \therefore \sum f_i = \mathbf{50}.$$

4. The classes are 5–9, 10–14, 15–19, ... (i) Find the lower boundary of the class 10–14. (ii) Find the upper boundary of the class 15–19.

Ans. (i) The lower boundary of the class 10–14 is **9.5**.

(ii) The upper boundary of the class 15–19 is **19.5**.

Assignment	6.2
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1. Find the mean ($\bar{X}$), if $\sum f_i x_i = 100$ and $\sum f_i = 25$.
2. If $\sum f_i x_i = 3880$ and mean ($\bar{X}$) = 388, find $\sum f_i$.
3. Find $\bar{u}$, if $\sum f_i u_i = 66$ and $\sum f_i = 100$.
4. If $\sum f_i = 100$ and $\sum f_i u_i = -28$, find $\bar{u}$.
5. Write the (i) lower boundary and (ii) upper boundary of the class 20–30, if the classes are 10–20, 20–30, 30–40,

Answers :

1. $\bar{X} = 4$ 2. $\sum f_i = 10$ 3. $\bar{u} = 0.66$ 4. $\bar{u} = -0.28$
 5. (i) 20 (ii) 30.

6.3	2 marks each
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1. Complete the following activity to prepare a table of coordinates of points to draw a frequency polygon : (Nov '20)

Class	18–19	19–20	20–21	
Class mark	18.5	19.5		21.5
Frequency	4		15	19
Coordinates of points		(19.5, 13)	(20.5, 15)	(21.5, 19)

Activity :

Class	18–19	19–20	20–21	21–22
Class mark	18.5	19.5	20.5	21.5
Frequency	4	13	15	19
Coordinates of points	(18.5, 4)	(19.5, 13)	(20.5, 15)	(21.5, 19)

2. The following table shows the daily supply of electricity to different places in a town. To show the information by a pie diagram, measures of central angles of sectors are to be decided. Complete the following activity to find the measures : *(March '19)*

Places	Supply of electricity (Thousand units)	Measure of the central angle
Roads	4	$\frac{4}{30} \times 360^\circ = 48^\circ$
Factories	12	$\times 360^\circ = 144^\circ$
Shops	6	$\frac{6}{30} \times 360^\circ =$
Houses	8	$\times 360^\circ =$
Total	30	

Activity :

Places	Supply of electricity (Thousand units)	Measure of the central angle
Roads	4	$\frac{4}{30} \times 360^\circ = 48^\circ$
Factories	12	$\frac{12}{30} \times 360^\circ = 144^\circ$
Shops	6	$\frac{6}{30} \times 360^\circ = 72^\circ$
Houses	8	$\frac{8}{30} \times 360^\circ = 96^\circ$
Total	30	

3. Find the mean from the given values :

$$\Sigma x_i f_i = 1265; N = 50.$$

*(July '23)***Solution :**

$$\text{Mean } (\bar{X}) = \frac{\Sigma x_i f_i}{N} \quad \dots \text{(Formula)}$$

$$= \frac{1265}{50}$$

$$= 25.3$$

Ans. The mean from the given values is **25.3**.**4. If $L = 10$, $f_1 = 70$, $f_0 = 58$, $f_2 = 42$, $h = 2$, then find the mode using formula.***(July '19; March '20)***Solution :**

$$\text{Mode} = L + \left[\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right] \times h \quad \dots \text{(Formula)}$$

$$= 10 + \left[\frac{70 - 58}{2(70) - 58 - 42} \right] \times 2 \quad \dots \text{(Substituting the values)}$$

$$\begin{aligned}
 &= 10 + \left(\frac{12}{140 - 100} \right) \times 2 \\
 &= 10 + \frac{12 \times 2}{40} = 10 + \frac{24}{40} \\
 &= 10 + 0.6 = 10.6
 \end{aligned}$$

Ans. The mode is **10.6**.

- 5. The following table shows the classification of number of workers and number of hours they work in software company. Prepare less than upper limit type cumulative frequency distribution table : (March '24)**

Daily number of hours	Number of workers
8–10	150
10–12	500
12–14	300
14–16	50

Ans.

Daily number of hours	Number of workers	Cumulative frequency less than upper limit type
8–10	150	150
10–12	500	650
12–14	300	950
14–16	50	1000



6. Observe the following table and find Mean :

Assumed mean $A = 300$.

(March '23)

Class	Class mark (x_i)	$d_i = x_i - A$ $d_i = x_i - 300$	Frequency (f_i)	Frequency $\times$ deviation ($f_i d_i$)
200–240	230	– 80	5	– 400
240–280	260	– 40	10	– 400
280–320	300 $\rightarrow A$	0	15	0
320–360	340	40	12	480
360–400	380	80	8	640
Total			$\Sigma f_i = 50$	$\Sigma f_i d_i = 320$

Solution : Here, $A = 300$, $\Sigma f_i d_i = 320$, $\Sigma f_i = 50$

$$\bar{d} = \frac{\Sigma f_i d_i}{\Sigma f_i} = \frac{320}{50} = 6.4$$

$$\begin{aligned}\text{Mean} = \bar{X} &= A + \bar{d} \\ &= 300 + 6.4 = 306.4\end{aligned}$$

Ans. Mean = **306.4**.

7. The following table shows causes of noise pollution. Find the measures of the central angles for each to draw a pie diagram :

Construction	Traffic	Aircraft take off	Industry
10%	50%	15%	25%

(Nov '20)

Solution :

Measures of the central angles for

(i) Construction : $\frac{10\%}{100\%} \times 360^\circ = 36^\circ$

(ii) Traffic : $\frac{50\%}{100\%} \times 360^\circ = 180^\circ$

(iii) Aircraft take off : $\frac{15\%}{100\%} \times 360^\circ = 54^\circ$

$$(iv) \text{ Industry : } \frac{25\%}{100\%} \times 360^\circ = 90^\circ.$$

Ans. Measures of the central angles for (i) Construction : 36°

(ii) Traffic : 180° (iii) Aircraft take off : 54° (iv) Industry : 90° .

Assignment

6.3

1. The following table shows the number of students and the time they utilized daily for their studies. Find the mean time spent by students for their studies : **(July '19; March '25)**

Time (hours)	0–2	2–4	4–6	6–8	8–10
Number of students	8	14	18	10	10

2. Find the mean of the data given in the following table :

Class	20–40	40–60	60–80	80–100
Frequency	4	5	7	4

3. For the frequency distribution of time (in minutes) a worker takes to complete the work, $\Sigma f_i = 100$, $\Sigma f_i d_i = 185$ and mean $(\bar{X}) = 38.85$. Find the assumed mean (A).
4. Draw the histogram for the following frequency distribution :

House rent per month (in ₹)	4000–6000	6000–8000	8000–10000	10000–12000
Number of families	200	240	30	150

5. The age group and the number of persons in the age group who donated blood is given below. Find the measures of the central angles to show the information by a pie diagram :

Age group	20–25	25–30	30–35	35–40
Number of persons	75	60	40	25



Answers :

1. The mean time spent by a student is 5 hours.
2. The mean of the data is 61.
3. The assumed mean $(A) = 37$. 5. $135^\circ; 108^\circ, 72^\circ; 45^\circ$.

6.4 3 marks each

1. The following frequency distribution table shows the classification of the number of vehicles and the volume of petrol filled in them. Complete the following activity, to find the mode of the volume of petrol filled : (March '24)

Class (Petrol filled in litres)	Frequency (Number of vehicles)
0.5–3.5	33
3.5–6.5	40
6.5–9.5	27
9.5–12.5	18
12.5–15.5	12

From the given table,

Modal class =

$$\text{Mode} = \boxed{} + \left[\frac{f_1 - f_0}{2f_1 - f_0 - \boxed{}} \right] \times h$$

$$\therefore \text{mode} = 3.5 + \left[\frac{40 - 33}{2(40) - 33 - 27} \right] \times \boxed{}$$

$$\therefore \text{mode} = 3.5 + \left[\frac{7}{80 - 60} \right] \times 3$$

$$\therefore \text{mode} = \boxed{}$$

$\therefore$ the mode of the volume of petrol filled is .

Activity :

From the given table, [the maximum frequency (40) is in the class 3.5 – 6.5 ...]

$$\text{Modal class} = \boxed{3.5-6.5}$$

$$\text{Mode} = \boxed{L} + \left[\frac{f_1 - f_0}{2f_1 - f_0 - \boxed{f_2}} \right] \times h$$

$$\therefore \text{mode} = 3.5 + \left[\frac{40 - 33}{2(40) - 33 - 27} \right] \times \boxed{3}$$

$$\therefore \text{mode} = 3.5 + \left[\frac{7}{80 - 60} \right] \times 3$$

$$\therefore \text{mode} = \boxed{4.55}$$

$\therefore$ the mode of the volume of petrol filled is $\boxed{4.55 \text{ litres}}$.

2. Complete the activity by fill in the boxes : (March '20)

Age group (in years)	Number of persons	Measure of the central angle
20–25	80	$\frac{\boxed{}}{200} \times 360^\circ = \boxed{}$
25–30	60	$\frac{60}{200} \times 360^\circ = \boxed{}$
30–35	35	$\frac{35}{200} \times \boxed{} = 63^\circ$
35–40	25	$\frac{25}{200} \times 360^\circ = \boxed{}$
Total	200	$\boxed{}$

Activity :

Age group (in years)	Number of persons	Measure of the central angle
20–25	80	$\frac{80}{200} \times 360^\circ = 144^\circ$
25–30	60	$\frac{60}{200} \times 360^\circ = 108^\circ$
30–35	35	$\frac{35}{200} \times 360^\circ = 63^\circ$
35–40	25	$\frac{25}{200} \times 360^\circ = 45^\circ$
Total	200	360°

3. A milk centre sold milk to 50 customers. The table below gives the number of customers and the milk they purchased. Find the mean of the milk they sold by direct method : (March '20)

Milk sold (Litres)	1–2	2–3	3–4	4–5	5–6
Number of customers	17	13	10	7	3

Solution :

Class Milk sold (Litres)	Class mark (x_i)	Frequency (Number of customers) (f_i)	Class mark $\times$ frequency ($x_i f_i$)
1–2	1.5	17	25.5
2–3	2.5	13	32.5
3–4	3.5	10	35.0
4–5	4.5	7	31.5
5–6	5.5	3	16.5
Total		$N = \Sigma f_i = 50$	$\Sigma x_i f_i = 141$

Here, $\sum x_i f_i = 141$, $\sum f_i = 50$

$$\text{Mean} = \bar{X} = \frac{\sum x_i f_i}{\sum f_i} = \frac{141}{50} = 2.82$$

Ans. The mean of milk sold is **2.82 litres**.

4. In a general store the prices of different articles and its demand is shown in the following frequency distribution table. Find the Median of the prices. (March '25)

Price in Rupees	Number of Articles
Less than 20	140
20–40	100
40–60	80
60–80	60
80–100	20

Solution :

Class (Price in Rupees)	Frequency (Number of Articles)	Cumulative Frequency (Less than type)
Less than 20	140	140 $\rightarrow cf$
20–40 Median class	100 $\rightarrow f$	240
40–60	80	320
60–80	60	380
80–100	20	400
Total	$\sum f_i = 400$	

Here, $N = \sum f_i = 400$

$$\frac{N}{2} = \frac{400}{2} = 200$$

200th score is in the class 20–40

$\therefore$ 20–40 is the median class.



$$L = 20, cf = 140, f = 100, h = 20$$

$$\text{Median} = L + \left[\frac{\frac{N}{2} - cf}{f} \right] \times h \quad \dots (\text{Formula})$$

$$= 20 + \left[\frac{200 - 140}{100} \right] \times 20 \quad \dots (\text{Substituting the values})$$

$$= 20 + \frac{60}{100} \times 20 = 20 + 12 = 32$$

Ans. The median of the price is ₹ 32.

5. The following frequency distribution table shows the distances travelled by some rickshaws in a day. Observe the table and answer the following questions : (March '19)

Class (Daily distance travelled in km)	Continuous classes	Frequency (Number of rickshaws)	Cumulative frequency (less than type)
60–64	59.5–64.5	10	10
65–69	64.5–69.5	34	10 + 34 = 44
70–74	69.5–74.5	58	44 + 58 = 102
75–79	74.5–79.5	82	102 + 82 = 184
80–84	79.5–84.5	10	184 + 10 = 194
85–89	84.5–89.5	6	194 + 6 = 200

- (1) Which is the modal class? Why?
- (2) Which is the median class? Why?
- (3) Write the cumulative frequency (cf) of the class preceding the median class.
- (4) What is the class interval (h) to calculate median?

Ans. (1) The maximum frequency is **82**.

$\therefore$ the modal class is **74.5 – 79.5**

(2) $N = 200 \quad \therefore \frac{N}{2} = 100$

Cumulative frequency which is just greater than 100 is **102**.

$\therefore$ the corresponding class **69.5 – 74.5** is the median class.

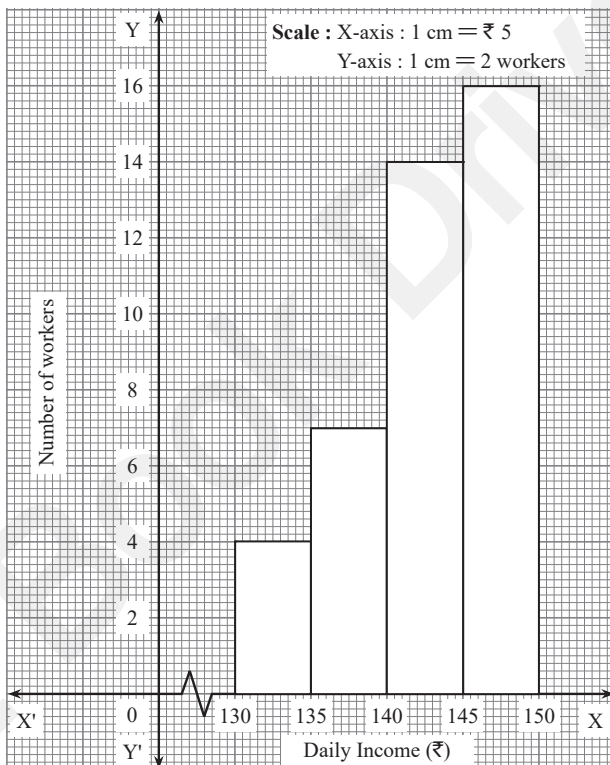
(3) The cumulative frequency (cf) of the class preceding the median class is **44**.

(4) Class interval (h) to calculate the median is **5**.

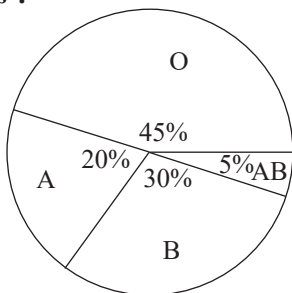
6. Represent the following data using histogram : (July '23)

Daily Income (₹)	130–135	135–140	140–145	145–150
Number of Workers	4	7	14	16

Solution :



7. The following pie diagram shows percentage of persons according to blood group in a blood group checking camp. Answer the following questions :



- (a) Find the measure of central angle for each blood group.
 (b) Find the total number of persons, if there are 600 persons of blood group B. (March '25)

Solution :

(a)

Blood Group	Measure of the central angle
O	$\frac{45\%}{100\%} \times 360^\circ = 162^\circ$
A	$\frac{20\%}{100\%} \times 360^\circ = 72^\circ$
B	$\frac{30\%}{100\%} \times 360^\circ = 108^\circ$
AB	$\frac{5\%}{100\%} \times 360^\circ = 18^\circ$
Total	360°

- (b) Let the total number of persons be x .

The central angle for blood group B

$$= \frac{\text{The number of persons in blood group B}}{\text{The total number of persons}} \times 360^\circ$$

$$\therefore 108^\circ = \frac{600}{x} \times 360^\circ \quad \therefore x = \frac{600 \times 360^\circ}{108^\circ}$$

$$\therefore x = 2000$$

The total number of persons is **2000**.

Assignment	6.4
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1. The following table shows the number of students of class X and the time they utilized daily for their studies. Find the mean time spent by 50 students for their studies by direct method : **(July '23)**

Time	0–2	2–4	4–6	6–8	8–10
Number of Students	7	18	12	10	3

2. The frequency distribution table shows the number of mango trees in a grove and their yield of mangoes. Find the median of data. **(March '23)**

Number of Mangoes	50–100	100–150	150–200	200–250	250–300
Number of trees	33	30	90	80	17

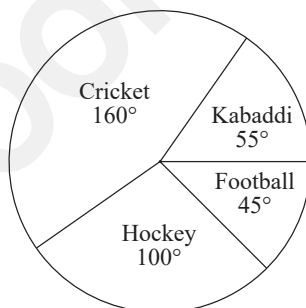
3. Time allotted for the preparation of an examination by some students is shown in the table. Draw a histogram to show the information :

Time (minutes)	60–80	80–100	100–120	120–140
Number of students	14	20	24	22

(Nov. '20)

4. In the following figure, the pie diagram represents the amount spent on different sports by a school administration in a year. If the money spent on football is ₹ 9,000, answer the following questions :

(March '24)



- (a) What is the total amount spent on sports?
 (b) What is the amount spent on cricket?



Answers :

1. The mean time spent by students is 4.36 hours.
2. The median of the data is 184.4 mangoes.
4. (a) The total amount spent on sports is ₹ 72,000.
(b) The amount spent on cricket is ₹ 32,000.

CHALLENGING QUESTIONS**6.5** 4 marks each

1. The following frequency distribution table shows marks obtained by 180 students in Mathematics examination : (March '19-'25)

Marks	0–10	10–20	20–30	30–40	40–50
Number of students	25	x	30	$2x$	65

Find the value of x .

Also draw a histogram representing the above information.

Solution :

First we find the value of x .

$$25 + x + 30 + 2x + 65 = 180$$

$$\therefore 3x + 120 = 180$$

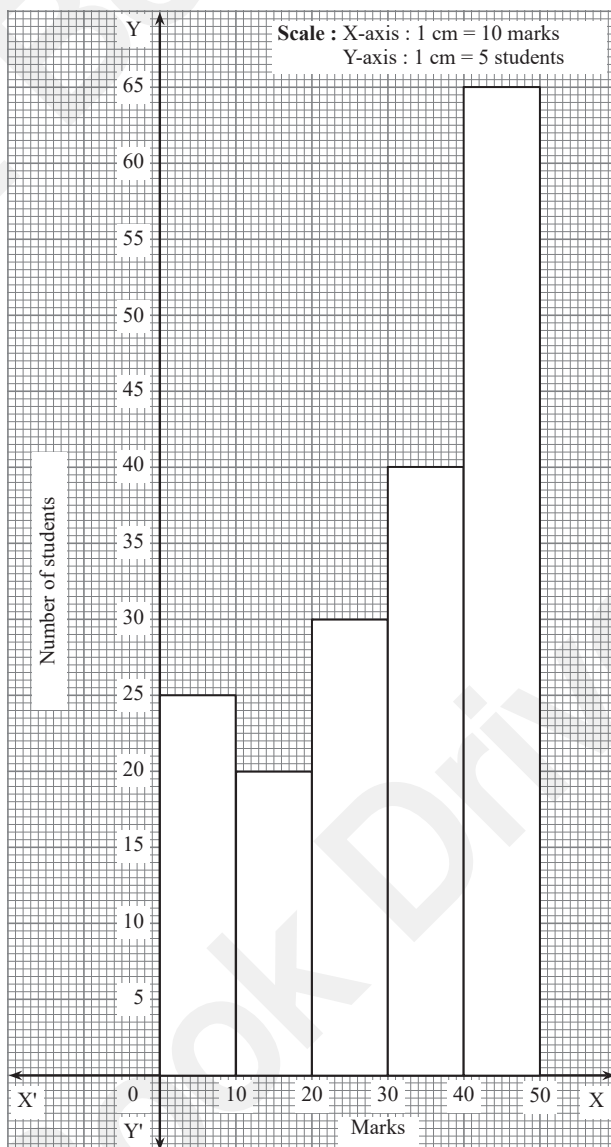
$$\therefore 3x = 180 - 120$$

$$\therefore 3x = 60$$

$$\therefore x = 20$$

The value of x is **20**.

Marks	0–10	10–20	20–30	30–40	40–50
Number of students	25	20	30	40	65



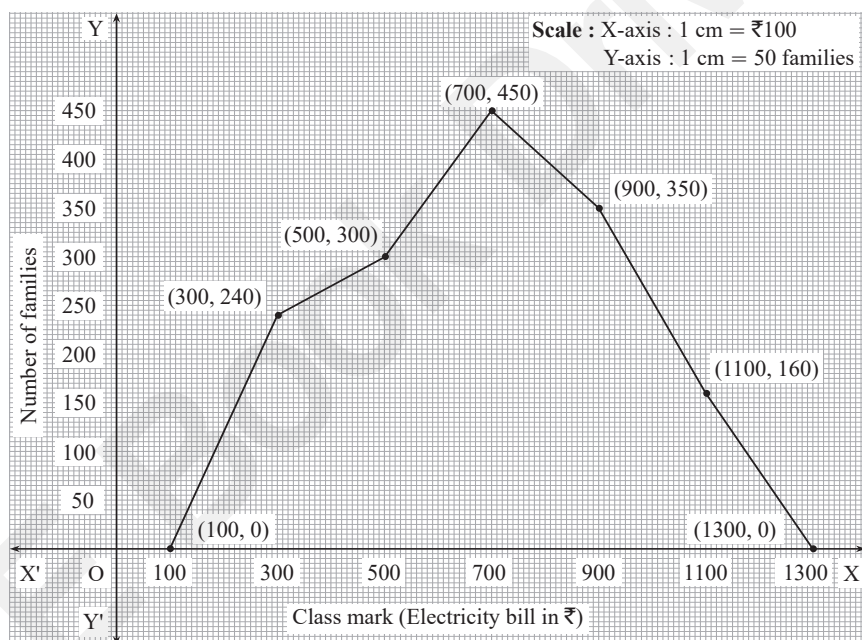
2. Show the following data by a frequency polygon. (March '23)

Electricity bill (₹)	200–400	400–600	600–800	800–1000	1000–1200
Families	240	300	450	350	160

Solution :

To draw the frequency polygon, we take two more classes. The class preceding the first class and the class succeeding the last class, each with frequency zero. The table to draw the frequency polygon is as follow :

Class [Electricity bill (₹)]	Class mark	Frequency (Number of families)	Coordinates of the points
0–200	100	0	(100, 0)
200–400	300	240	(300, 240)
400–600	500	300	(500, 300)
600–800	700	450	(700, 450)
800–1000	900	350	(900, 350)
1000–1200	1100	160	(1100, 160)
1200–1400	1300	0	(1300, 0)



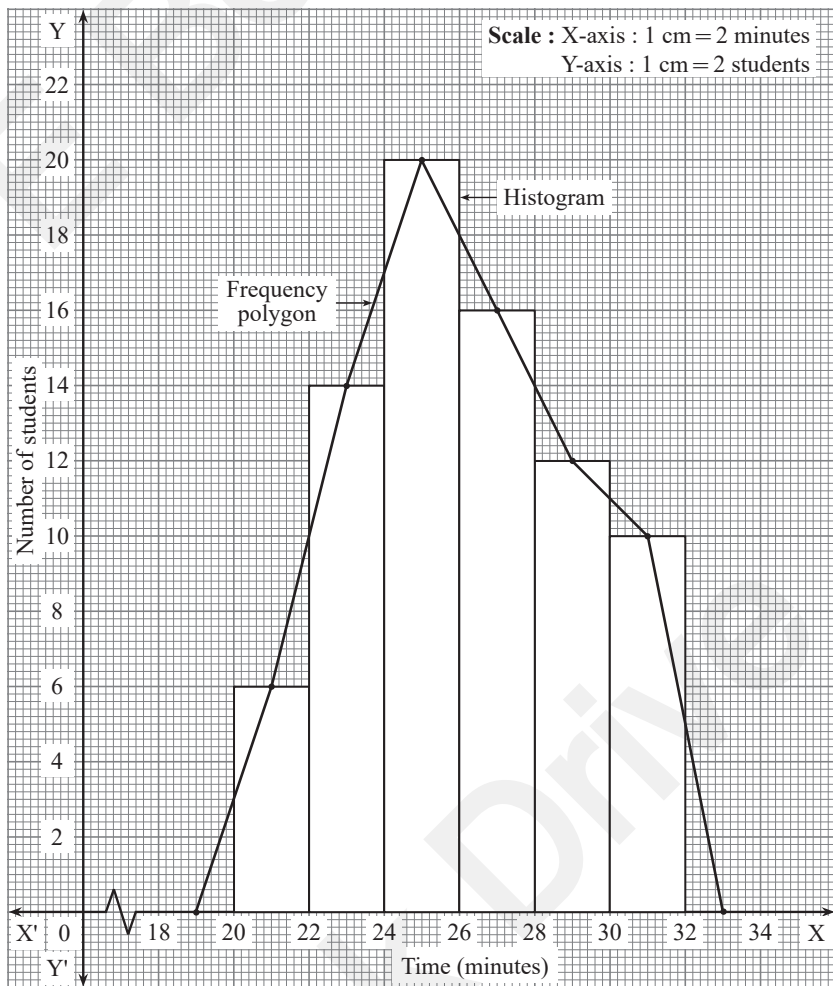
3. The time required for some students to complete a science experiment and the number of students is shown in the following grouped frequency distribution table. Draw a frequency polygon with the help of histogram using given information.

Time required for experiment (minutes)	20–22	22–24	24–26	26–28	28–30	30–32
Number of students	6	14	20	16	12	10

(July '19)

Solution : For drawing a frequency polygon, we take two additional classes. 18–20 with frequency zero and 32–34 with frequency zero.

Class (Time in minutes)	Class mark	Frequency	Coordinates of the points
18–20	19	0	(19, 0)
20–22	21	6	(21, 6)
22–24	23	14	(23, 14)
24–26	25	20	(25, 20)
26–28	27	16	(27, 16)
28–30	29	12	(29, 12)
30–32	31	10	(31, 10)
32–34	33	0	(33, 0)



4. The following table shows frequency distribution of number of trees planted by students in the school :

Number of trees planted	0-10	10-20	20-30	30-40	40-50
Number of students	30	70	100	70	40

Find the mode of trees planted.

(Nov. '20)

Solution :

Number of trees planted	0-10	10-20	20-30	30-40	40-50
Number of students	30	$70 \rightarrow f_0$	$100 \rightarrow f_1$	$70 \rightarrow f_2$	40

Here, the maximum frequency (100) is in the class 20–30.

∴ the modal class is 20–30.

In the given example,

$$L = 20, f_1 = 100, f_0 = 70, f_2 = 70, h = 10.$$

$$\begin{aligned}\text{Mode} &= L + \left[\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right] \times h \\ &= 20 + \left[\frac{100 - 70}{2(100) - 70 - 70} \right] \times 10 \\ &= 20 + \left(\frac{30}{200 - 140} \right) \times 10 \\ &= 20 + \frac{30}{60} \times 10 \\ &= 20 + \frac{1}{2} \times 10 \\ &= 20 + 5 = 25\end{aligned}$$

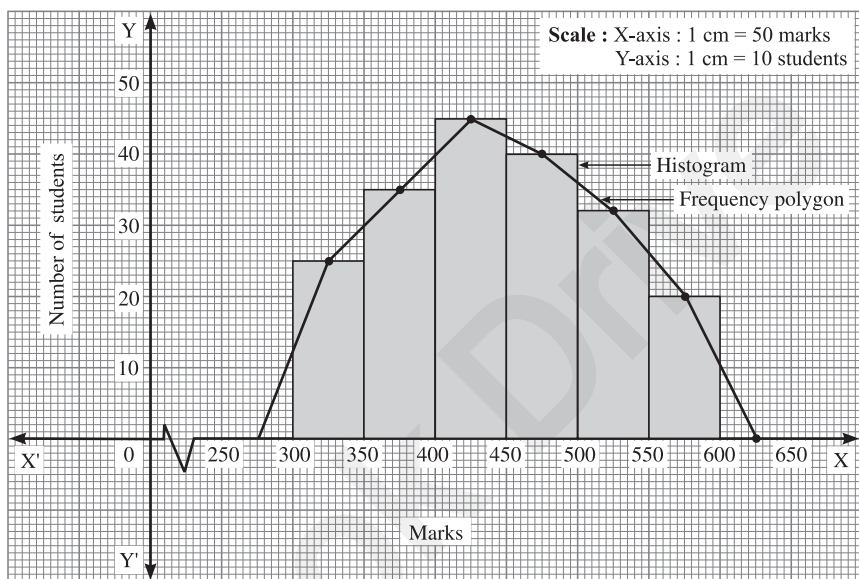
Ans. The mode of trees planted is 25.

5. The following table is based on the marks of the first term examination of 10th class student. Show the information by a histogram. Also, draw a frequency polygon with the help of the histogram :

Class mark of marks	325	375	425	475	525	575
Number of students	25	35	45	40	32	20

Solution : For drawing a frequency polygon take a class preceding the lowest class with frequency zero and a class succeeding the highest class with frequency zero.

Class	Class mark of class	Frequency (Number of students)	Coordinates of points
250–300	275	0	(275, 0)
300–350	325	25	(325, 25)
350–400	375	35	(375, 35)
400–450	425	45	(425, 45)
450–500	475	40	(475, 40)
500–550	525	32	(525, 32)
550–600	575	20	(575, 20)
600–650	625	0	(625, 0)



Assignment 6.5

- From the following frequency distribution table, find the values of f_1 and f_2 . The mean of the data is 166.4 and the sum of the observation is 50. Take assumed mean 165 :

Class	140–150	150–160	160–170	170–180	180–190	190–200
Frequency	5	f_1	20	f_2	6	2

2. Represent the following data by histogram. (March '20)

Price of sugar (per kg in ₹)	18–20	20–22	22–24	24–26	26–28	28–30
Number of weeks	4	8	22	12	6	8

3. The following table shows the classification of percentage of marks of students and the number of students. Draw frequency polygon from that table without drawing histogram :

Result (Percentage)	20–40	40–60	60–80	80–100
Number of students	25	65	80	15

(March '24)

4. Draw a pie diagram to represent the world population given in the following table :

Country	Japan	England	India	China
Percentages of Population	20	10	40	30

(July '23)

Answers :

1. The values of f_1 and f_2 are 9 and 8 respectively.

4. The measures of central angles :

Japan : 72° , England : 36° , India : 144° , China : 108° .



Time : 2 Hours]

Solution on QR Code

[Max. Marks : 40

- Note :** (i) *All questions are compulsory.*
(ii) *Use of a calculator is **not** allowed.*
(iii) *The numbers to the right of the questions indicate full marks.*
(iv) *In case of MCQs [Q. No. 1(A)], only the first attempt will be evaluated and will be given credit.*

- 1. (A) Four alternative answers are given for every subquestion. Choose the correct alternative and write its alphabet with subquestion number :**

4

- (i) The value of the determinant $\begin{vmatrix} 5 & 3 \\ -7 & -4 \end{vmatrix}$ is
(A) -1 (B) -41 (C) 41 (D) 1
- (ii) Out of the following equations, which one is not a quadratic equation?
(A) $x^2 + 4x = 11 + x^2$ (B) $x^2 = 4x$
(C) $5x^2 = 90$ (D) $2x - x^2 = x^2 + 5$
- (iii) For given A.P., $a = 3.5$, $d = 0$, then $t_2 = ?$
(A) 0 (B) 3.5 (C) 7 (D) 10.5
- (iv) From the following numbers, which number cannot represent a probability?
(A) 0.6 (B) 2.0 (C) 0.15 (D) 0.75

- 1. (B) Solve the following subquestions :**

4

- (i) For simultaneous equations in variables x and y , $D_x = 49$, $D_y = -63$, $D = 7$, then find the value of x .
- (ii) Write the quadratic equation $2y = 10 - y^2$ in standard form.
- (iii) Face value of one share is ₹ 100 and premium is ₹ 10. Find the market value of the share.
- (iv) Find the class mark of the class 6–10.

2. (A) Complete the following activities and rewrite them :

(Any two) 4

- (i) Complete the following activity to solve the quadratic equation

$$x^2 + 8x - 48 = 0 \text{ by completing square method.}$$

Activity :

$$x^2 + 8x - 48 = 0$$

$$\therefore x^2 + 8x + 16 - \boxed{} - 48 = 0$$

$$\therefore (x+4)^2 - \boxed{} = 0$$

$$\therefore (x+4)^2 = 64$$

$$\therefore x+4 = \boxed{} \text{ or } x+4 = -8$$

$$\therefore x = 4 \text{ or } x = \boxed{}.$$

- (ii) Courier service agent charged total ₹ 590 to courier a parcel from Nashik to Nagpur. In the tax invoice, taxable value is ₹ 500 on which CGST is ₹ 45 and SGST is ₹ 45. Complete the following activity to find the rate of GST charged for this service :

Activity :

$$\text{Total GST} = \text{CGST} + \text{SGST}$$

$$= ₹ \left(\boxed{} + 45 \right)$$

$$= ₹ \boxed{}$$

$$\text{Rate of GST} = \frac{90}{500} \times \boxed{}$$

$$\therefore \text{rate of GST charged by agent is } \boxed{}\%.$$

- (iii) The monthly expenditure of a family on different items is shown in the following table. Complete the following activity to calculate the related central angles.



Activity :

Different items	Percentage of expenditure	Measure of the central angle
Food	40	$\frac{40}{100} \times 360^\circ = \square^\circ$
Clothing	20	$\frac{20}{100} \times 360^\circ = \square^\circ$
Education	30	$\frac{30}{100} \times 360^\circ = \square^\circ$
Other expenditure	10	$\frac{10}{100} \times 360^\circ = \square^\circ$
Total	100	360°

2. (B) Solve the following subquestions : (Any four)**8**

- Find the values of $(x + y)$ and $(x - y)$ for simultaneous equations $101x + 99y = 501$, $99x + 101y = 499$.
- Solve the quadratic equation $x^2 - 15x + 54 = 0$ by factorisation method.
- Which term of the A.P., 2, 11, 20, 29, ... is 560?
- Market value of a share is ₹ 200. If the brokerage rate is 0.3%, then find the purchase value of the share.
- If $\sum f_i d_i = 10,000$, $\sum f_i = 100$ and $A = 2000$, then find mean $(\bar{X})$.

3. (A) Complete the following activities and rewrite it :**(Any one) 3**

- In an A.P., the sum of three consecutive terms is 27 and their product is 504. Complete the following activity to find the terms.

Activity :

Let the three consecutive terms be $a - d$, a , $a + d$.

$$\therefore a - d + a + a + d = \square$$

$$\therefore a = \square$$

Similarly :

$$(a-d) \times a \times (a+d) = \square$$

$$\therefore [(9)^2 - d^2] \times 9 = 504$$

$$\therefore (81 - d^2) \times 9 = 504$$

$$\therefore d^2 = 81 - \square$$

$$\therefore d = \pm 5$$

Thus by putting $a=9$ and $d=5$, we get, three consecutive terms = $\square$

Or by putting $a=9$ and $d=-5$, we get, three consecutive terms = $\square$.

- (ii) A card is drawn from a well-shuffled pack of 52 playing cards. Complete the following activity to find the probability of the following events :

(a) A red card (b) A face card

Activity :

Let S is the sample space

$$\therefore n(S) = \square$$

Event A : Card drawn is a red card.

$$\text{Total red cards} = \square$$

$$\therefore n(A) = \square$$

$$P(A) = \frac{n(A)}{n(S)} \quad \dots \text{(Formula)}$$

$$\therefore P(A) = \frac{26}{52}$$

$$\therefore P(A) = \square$$

Event B : Card drawn is a face card.

$$\text{Total face cards} = 12$$

$$\therefore n(B) = \square$$

$$P(B) = \frac{n(B)}{n(S)} \quad \dots \text{(Formula)}$$

$$\therefore P(B) = \frac{12}{52} \quad \therefore P(B) = \boxed{\quad}$$

3. (B) Solve the following subquestions : (Any two)

6

- (i) In the following table the yearly investment made by 210 families is given. From this information draw the histogram.

Investment (Thousand rupees)	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60
Number of families	30	50	60	55	15

- (ii) Shri Shivajirao purchased 150 shares of FV ₹ 100, for MV of ₹ 120. Company paid 7% dividend. Find the rate of return on his investment.
- (iii) The number denoting the product of Shraddha's age 2 years ago and 3 years hence is 84. Find her present age.
- (iv) Solve the following simultaneous equations graphically :
 $x + y = 6$, $x - y = 4$.

4. Solve the following subquestions : (Any two)

8

- (i) In an agriculture field, the ratio of the wages of skilled and unskilled workers is 5 : 4. Total wages of one day of both of them is ₹ 900. Find daily wages of skilled and unskilled workers.
- (ii) Two dice are thrown. Find the probability of the following events :
- (a) Event A : The sum of the numbers on their upper faces is at least 9.
- (b) Event B : The sum of the numbers on their upper faces is divisible by 5.
- (c) Event C : The number on the upper face of the first die is greater than the number on the upper face of the second die.

- (iii) In the following frequency distribution table, the ages of 300 patients and the number of patients in a hospital is given. Find the median age of the patients :

Age (in years)	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60	60 – 70
Number of patients	60	42	55	70	53	20

5. Solve the following subquestion : (Any one)

3

- (i) Three conditions are given below. Write the quadratic equation for each of them to satisfy that condition :
- If discriminant $\Delta = 0$
 - If discriminant $\Delta > 0$
 - If discriminant $\Delta < 0$.
- (ii) If first term of an A.P. is p , second term is q and last term is r , then show that the sum of all terms is $\frac{(p+r)(q+r-2p)}{2(q-p)}$.

Scan  this **QR Code** for the
Model Answers to
Board's July 2025 Question Paper

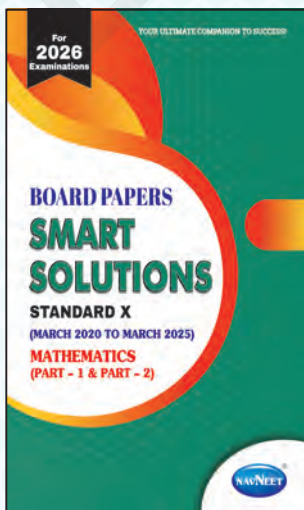


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